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Instrumentation
Chromatography-IV

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Chromatography

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Thin layer Chromatography

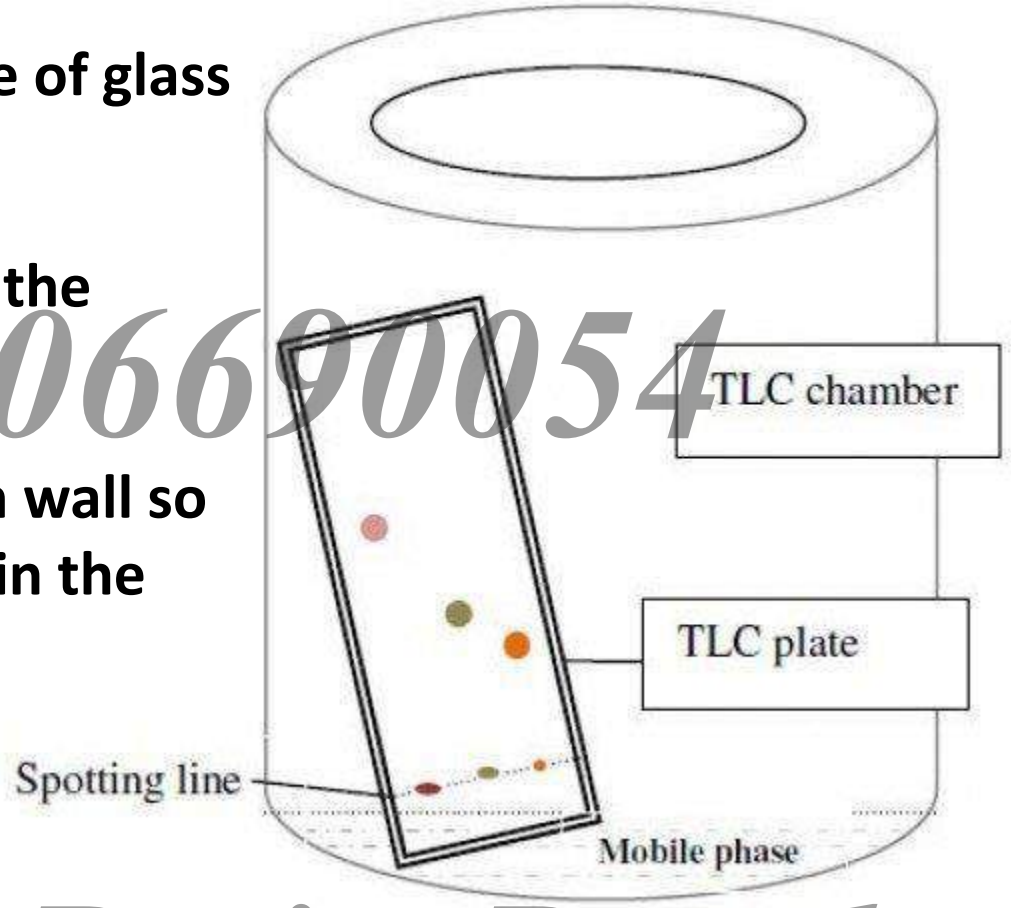
- Stationary Phase used: adsorbent silica pasted on glass or aluminum plates
- Mobile Phase used: usually organic solvents (pure or mixed in variable ratios)

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Chromatography

Thin layer Chromatography

- TLC plates are made of glass or aluminum
- Solvent equilibrate the environment
- May be placed with wall so that it may not fall in the chamber



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Chromatography

- Separations in TLC involve distributing a mixture of two or more substances between a stationary phase and a mobile phase.
- The stationary phase is a thin layer of adsorbent (usually silica gel or alumina) coated on a plate.
- The mobile phase is a developing liquid which travels up the stationary phase, carrying the samples with it.
- Components of the samples will separate on the stationary phase according to “how much they adsorb on the stationary phase versus how much they dissolve in the mobile phase” (affinity).

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Chromatography





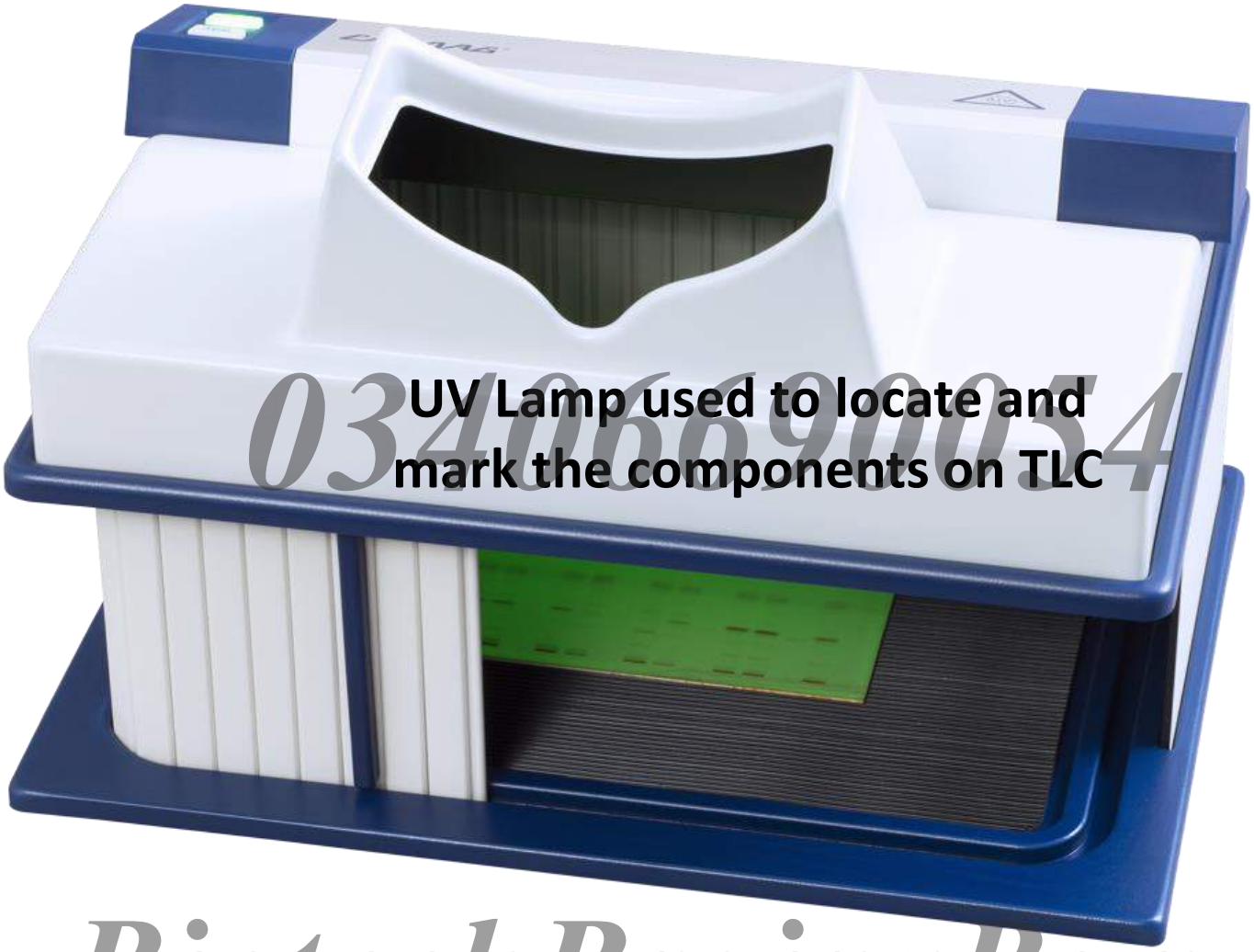
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Glass and Aluminum Plates cut to the required sizes

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Chromatography



UV Lamp used to locate and mark the components on TLC

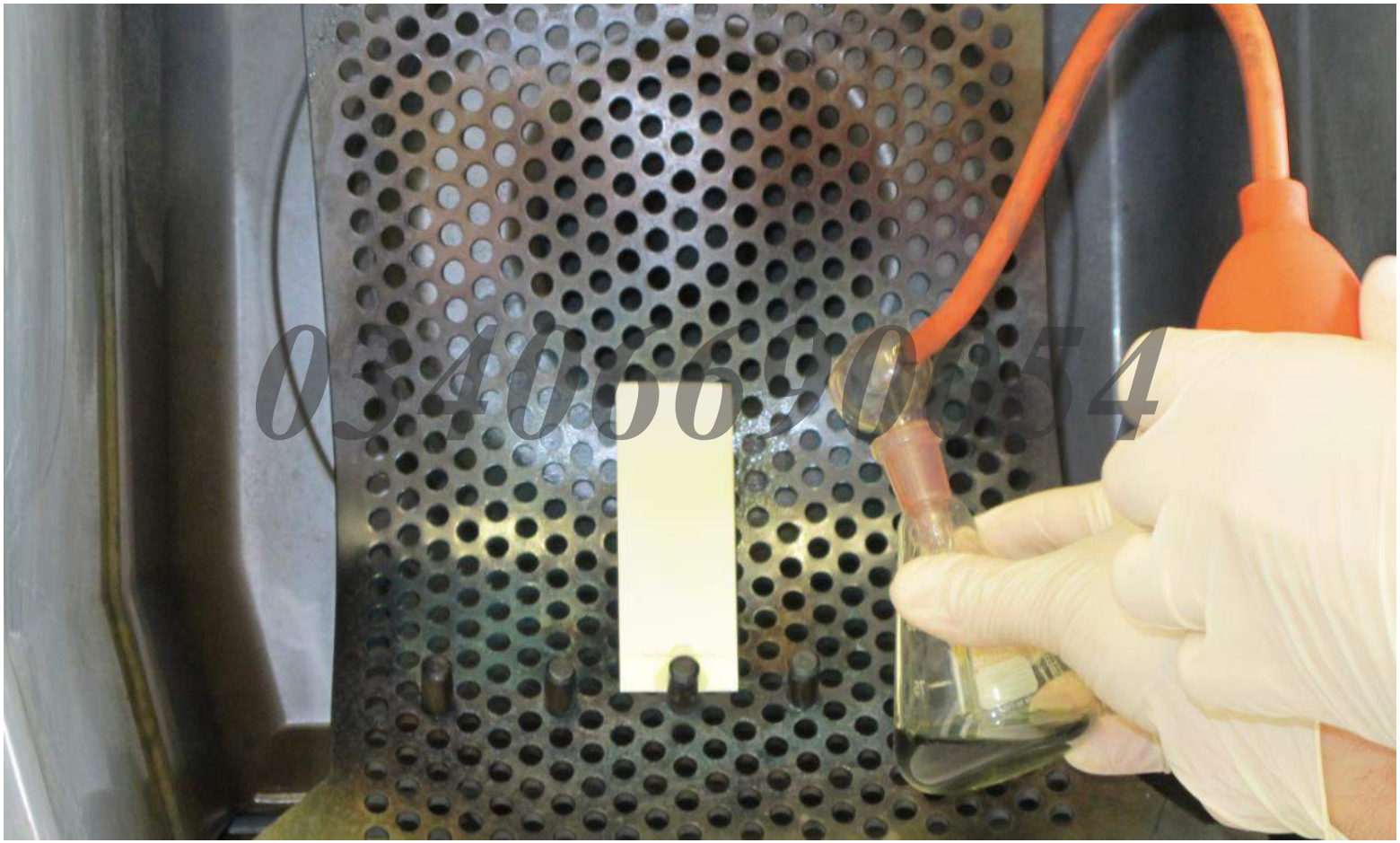
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Chromatography

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Spraying Chamber for locating agent spray

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Chromatography

Thin layer Chromatography

- Developing agents:
- Ninhydrin (amino acids), dragondorf reagent (for alkaloides), ceric sulphate (for terpenes)

- Qualitative technique when compared with standard
- Quantitative technique when scratched and quantified the analyte

END

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Instrumentation
Chromatography-V

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Chromatography

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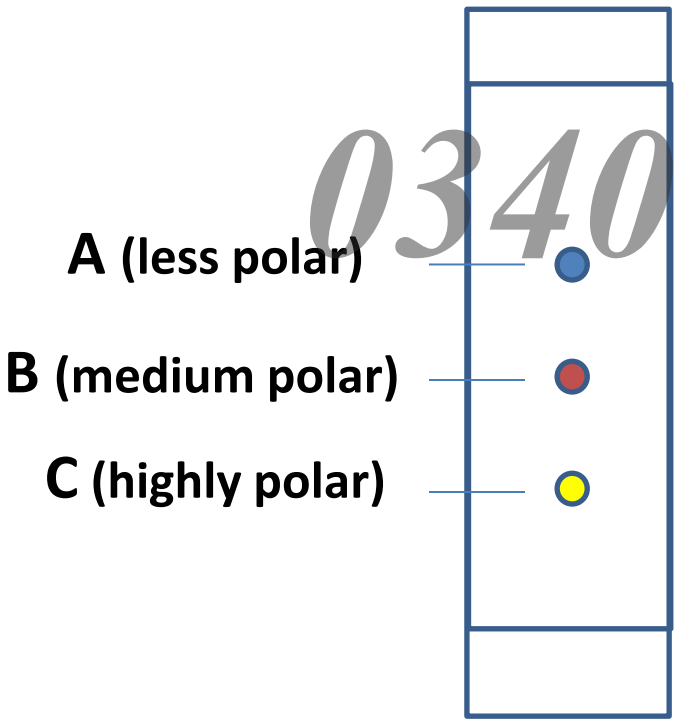
Column Chromatography

- Stationary Phase used: Silica gel, flash silica, alumina
- Mobile Phase used: organic solvents (starting from less polar to high polarity)

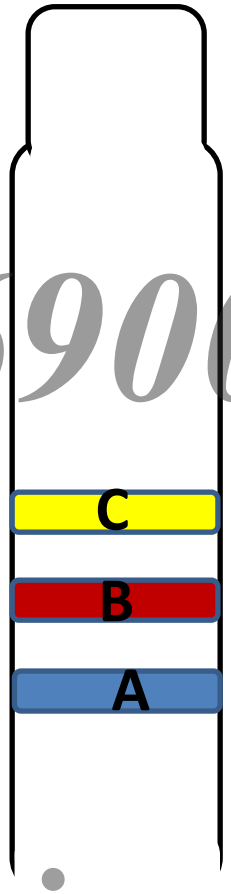
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Chromatography

Thin Layer Chromatography



Column Chromatography

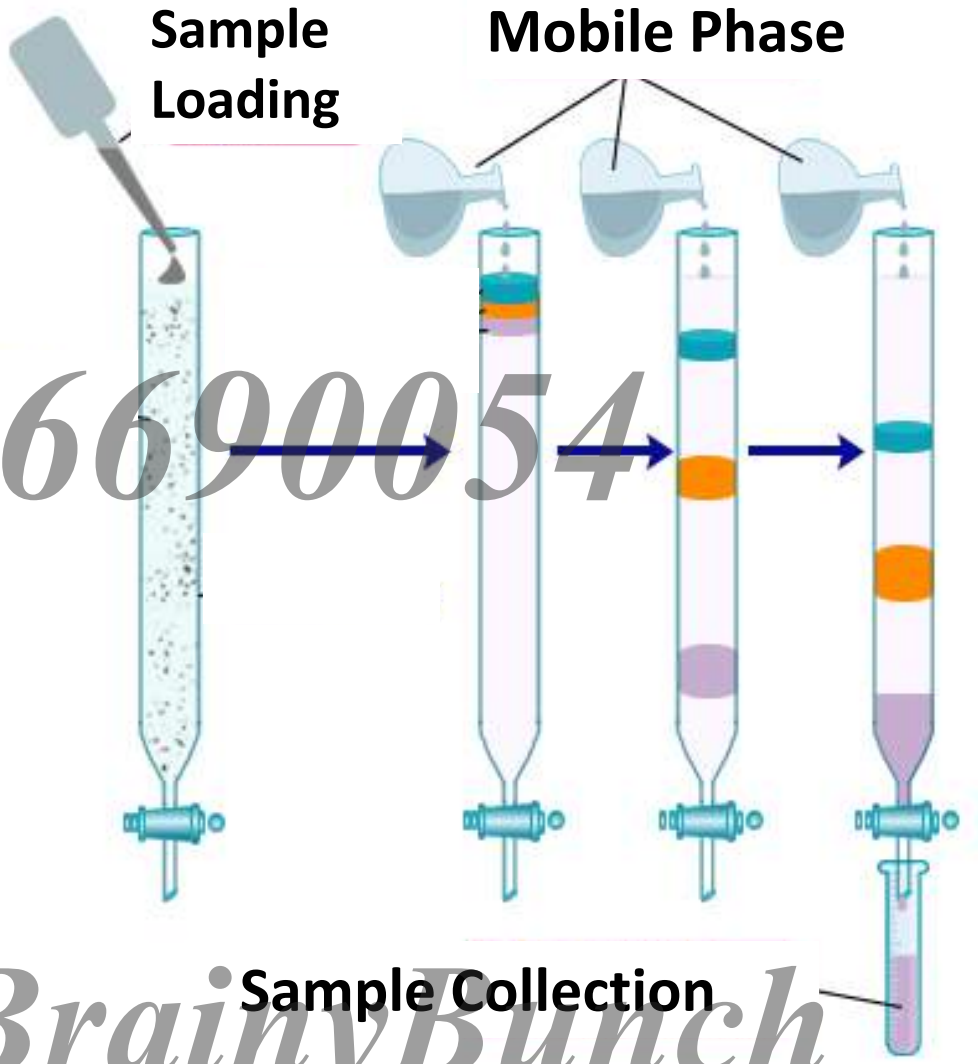


- Choose of glass column based on amount of sample and difference in R_f values) of component.

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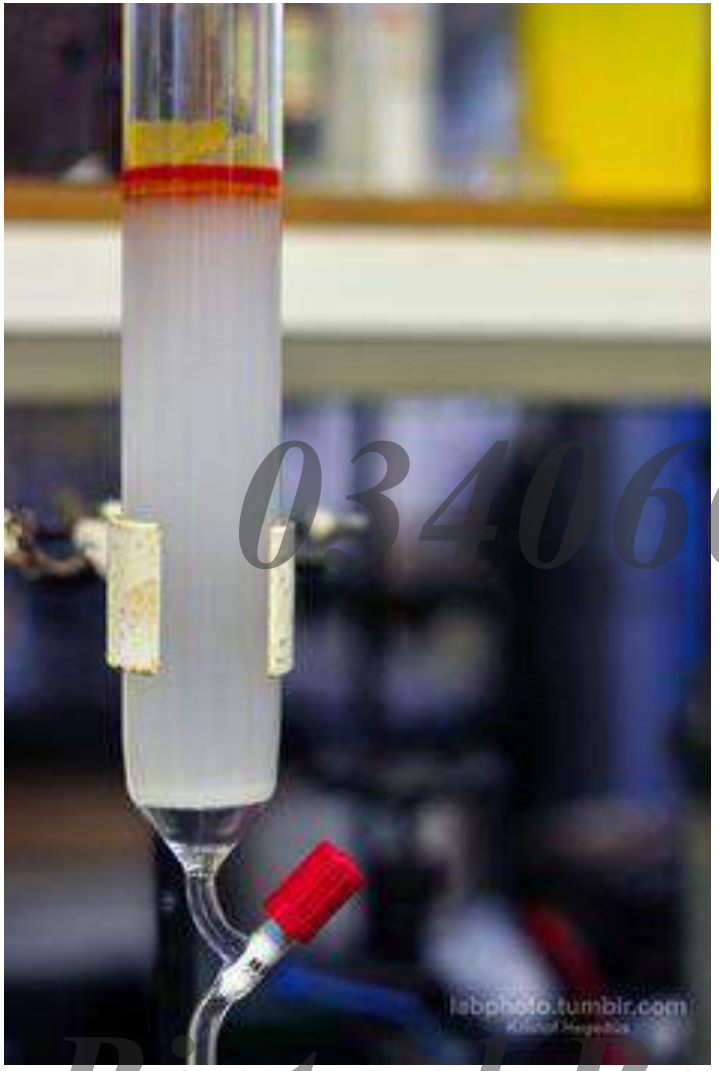
Chromatography

- Stationary phase is held in a narrow tube through which the mobile phase is forced under pressure or under the effect of gravity.
- Choice of stationary phase depends upon chemistry of analytes.

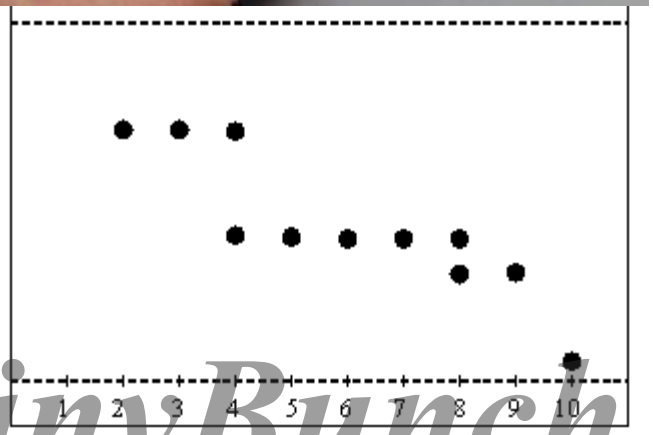


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Chromatography







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Chromatography

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Flash Chromatography:

In flash column chromatography the sample is pumped through the packed glass column.

Flash silica has quite smaller mesh size.

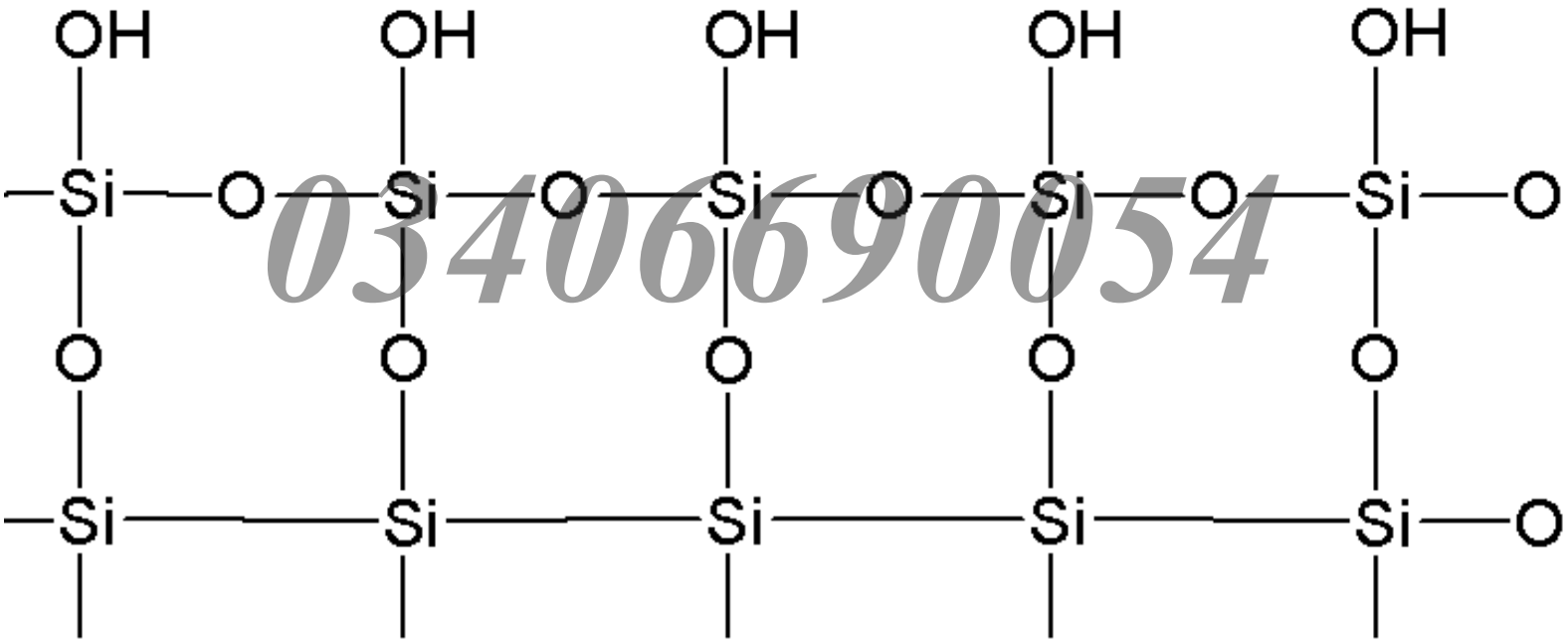


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Chromatography

Silica Gel Chemistry:



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Column Chromatography

- Silica Gel used, is available in varied mesh sizes starting from 60 to 800 Mesh.
- Alumina (Al_2O_3) used as solid support, available in acidic basic and neutral phases.

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Instrumentation
Chromatography-VI

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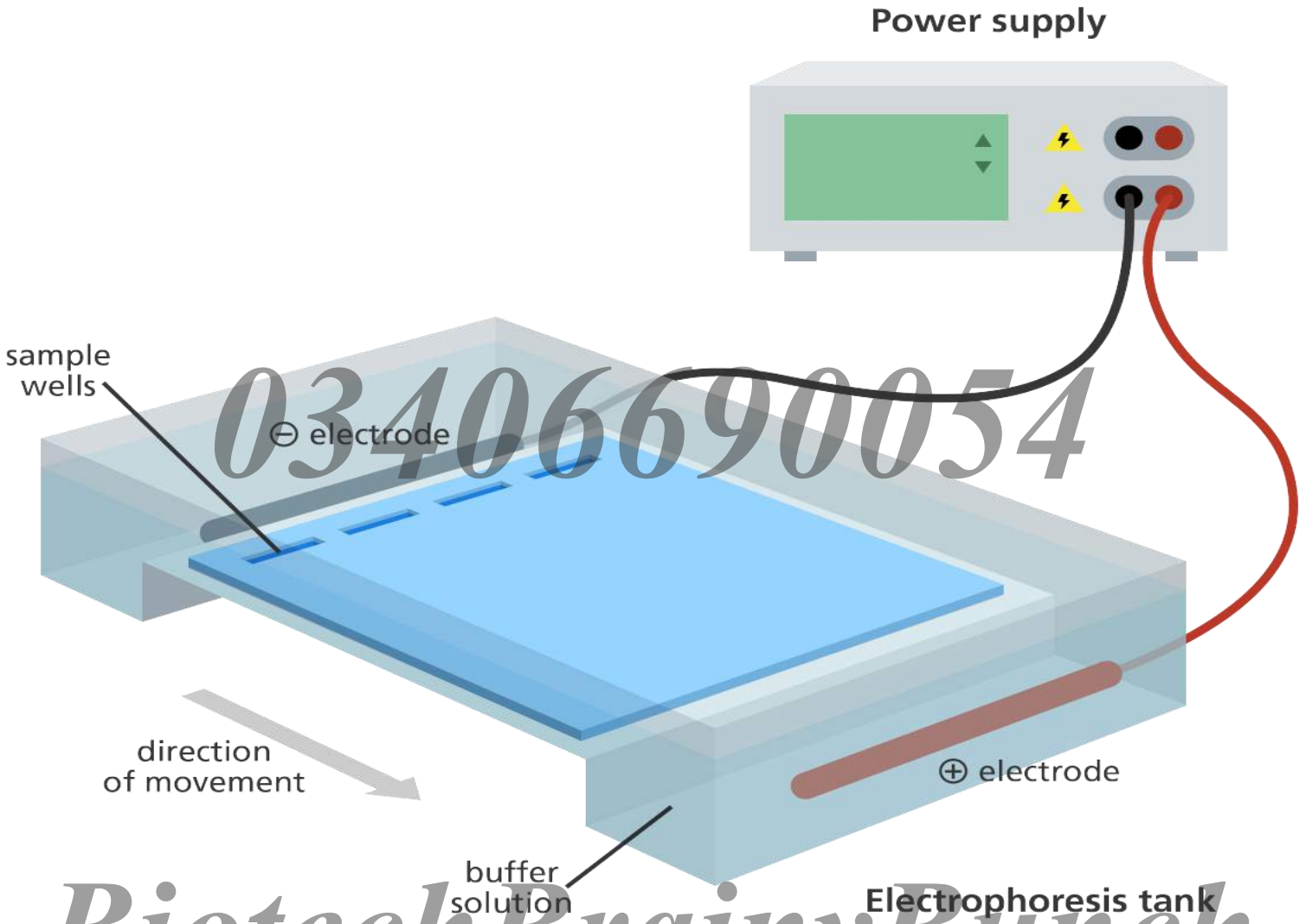
Chromatography

Electrophoresis

- Electrophoresis is a technique used in laboratories in order to separate macromolecules based on size or charge.
- The technique applies a charge so proteins move towards a positive charge electrode.

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Chromatography



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Chromatography

- Electrophoresis may be of two types;
  - Slab electrophoresis-TLC
  - Capillary electrophoresis-Column

- Velocity (v) of charged molecule in an electric field can be assessed with formula;

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$$v = Eq/F$$

- where E is applied electric charge, q is net charge on a molecule, and F is frictional coefficient depends upon mass and shape of molecule.

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Chromatography

- Stationary phases: Starch gel, PAGE (poly acrylamide gel), Agarose, Cellulose acetate
  - SDS-PAGE (polyacrylamide gel as a support medium and sodium dodecyl sulfate to denature the proteins)
  - IEF: Isoelectric focusing using agarose or polyacrylamide
- Mobile phase: Buffer system
  - Phosphate buffer (pH 7.0), Tris-Borate-EDTA buffer (TBE, pH 8.0), Tris-Cotrate EDTA buffer (TCE, pH 7.0), Tris-Maleic acid-EDTA buffer (TME, pH 7.5), Tris-Acetate EDTA buffer (TAE, pH 8.0), Tris glycerin buffer (TG, pH 8.5), Lithium Borate buffer (LB, pH 8.6)

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Chromatography

Identification through staining systems

- For Proteins
  - General: Coomassie brillient blue R, Kenacid blue, Amido black
  - Specific: Oil red O, PAS (Periodic acid Schiff), Rubeanic acid, Transferrin-specific & for calcium binding proteins
- Allozyme: Histochemical staining
- DNA: EtBr (Ethedium bromide), SyBR green, Propidium iodide and silver staining

- Steps: fixing, staining, destaining

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Chromatography

Two dimensional Electrophoresis

- This technique combines two techniques; one separate the mixture of proteins according to charges (IEF) and second with the size (SDS-PAGE).

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Instrumentation
Chromatography-VII

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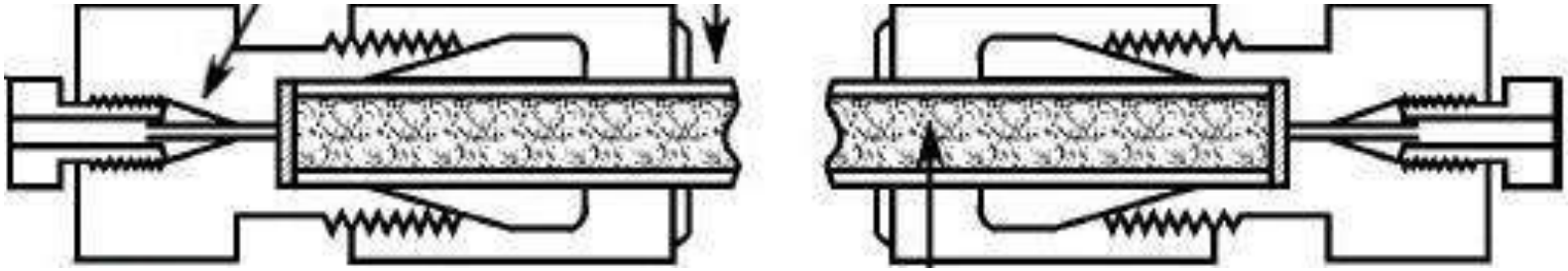
Chromatography

High Performance Liquid Chromatograph

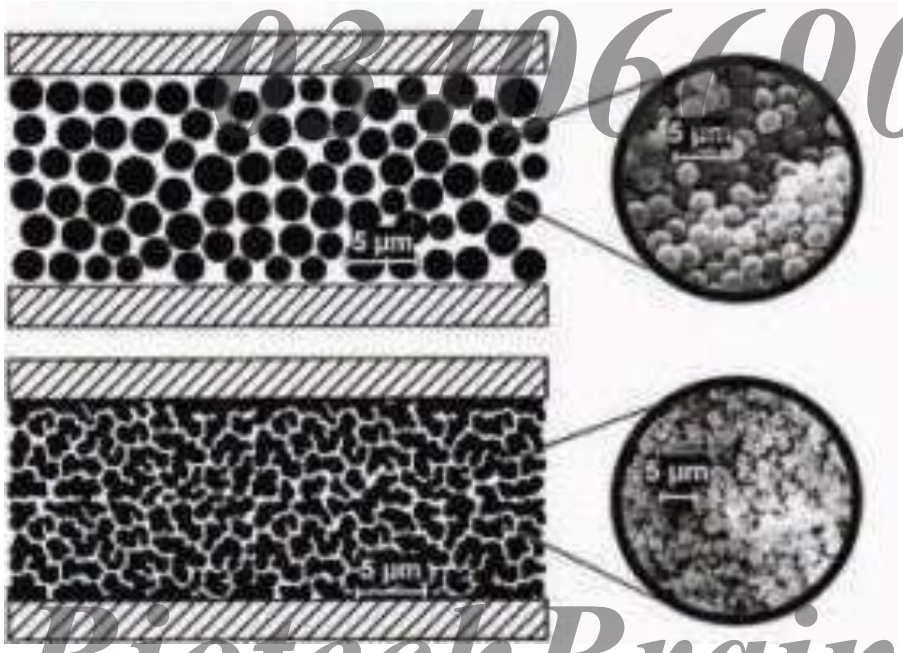
- It is normally used as reversed phase chromatographic technique
- Mobile phase is pumped into the column with high pressure
- Separation occur based on size, charge, affinity etc

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Chromatography



Conventional end capped column



Monolithic column

Chromatographic column

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Chromatography

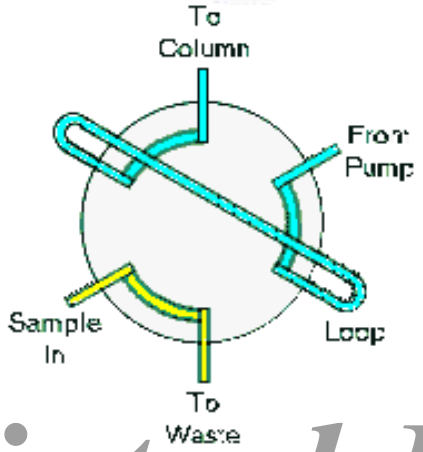
Gradient and Isocratic HPLC

HPLC Pump

High pressure gradient and
Low pressure gradient

Injector

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Detector

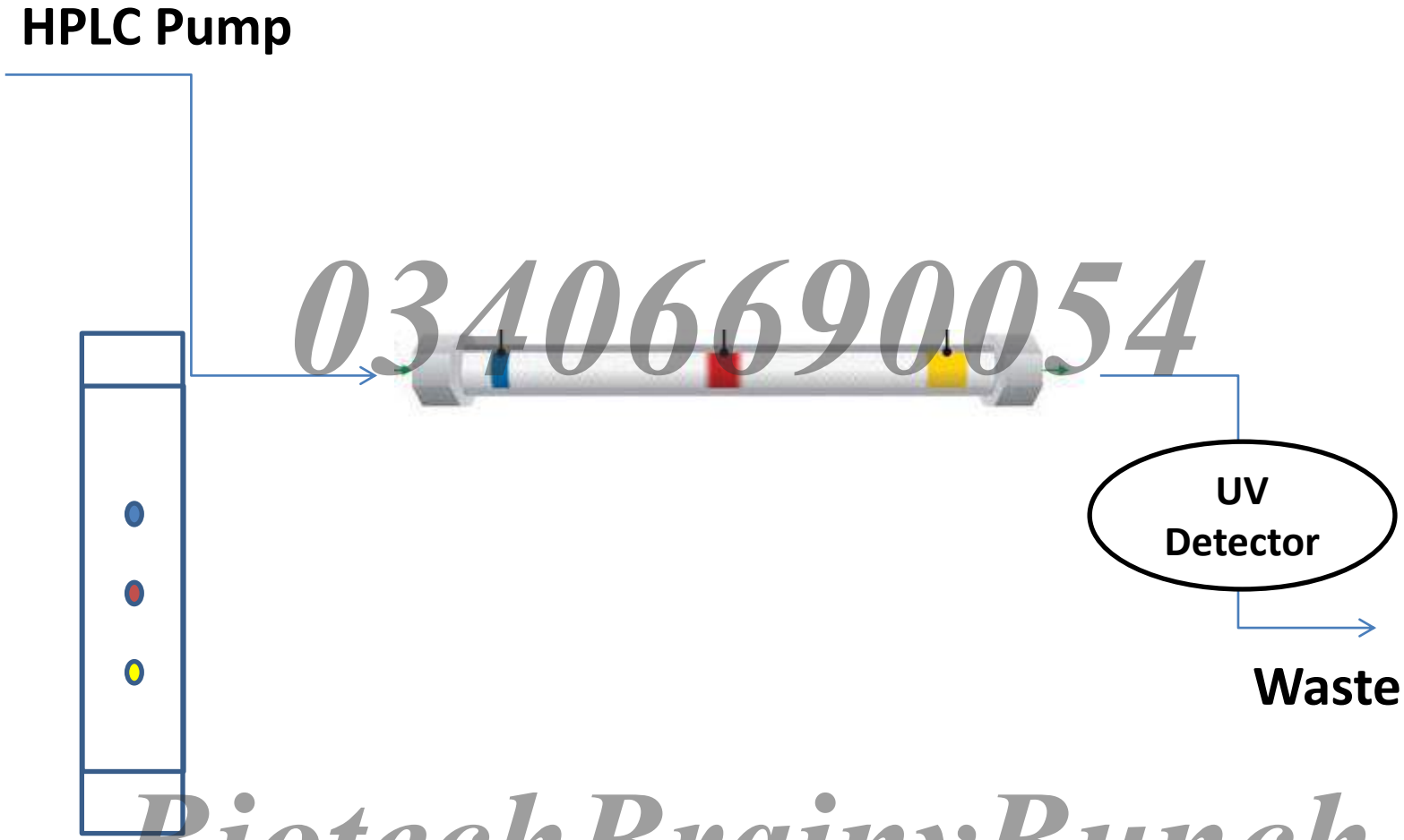
Waste

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Chromatography

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Chromatography

Detectors:

- **UV Detector: for coloured or UV active compounds**
- **PDA detector: UV based scanning**
- **RI: Refractive Index detector for polysacharides**
- **ELSD: Evaporating light scattering detector**
- **Conductivity: for charged containing molecules**

END

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Instrumentation
Chromatography-VIII

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Chromatography

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HPLC Columns

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- Columns, made of stainless steel of varying length and internal diameter (Ø) are used for analytical chromatography.

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Chromatography

Classification of Column Dimensions:

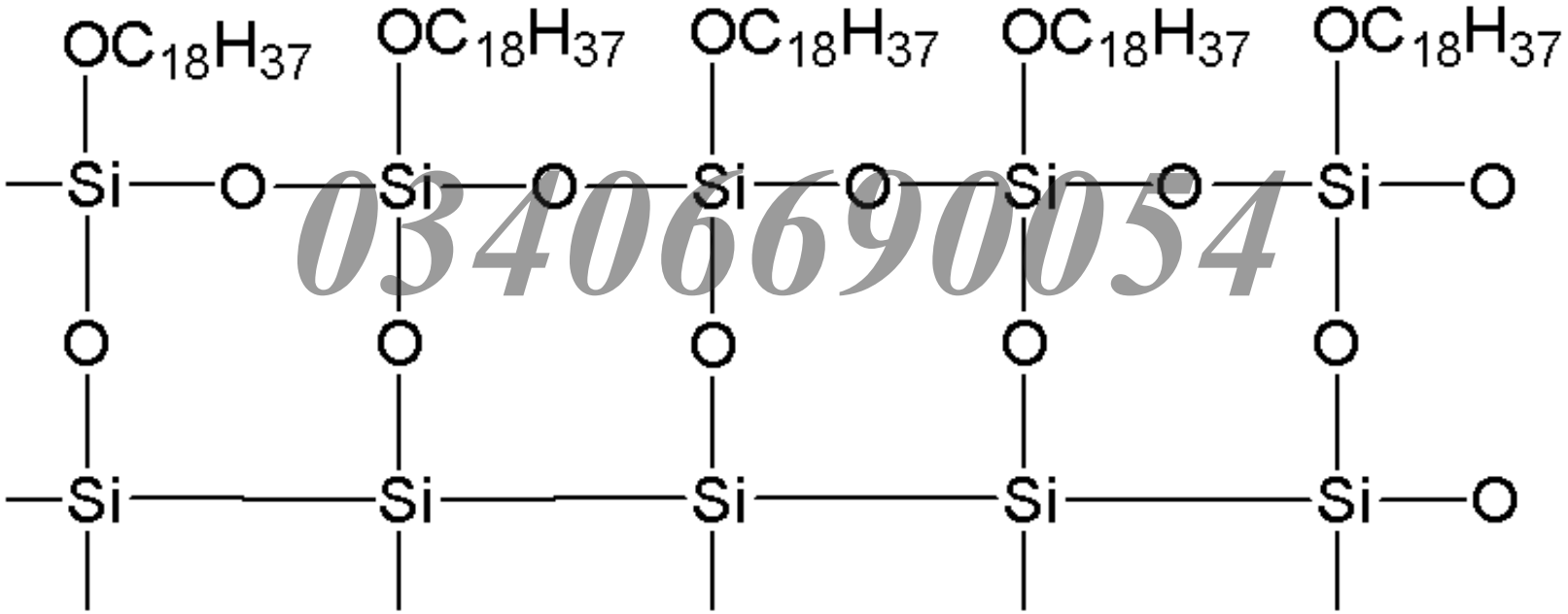
- **Standard column: Diameter 4.6 mm, length 15 to 25 cm**
- **Micro Bore Columns (MBC): Columns with internal diameters of less than 2 mm are often referred to as MBC. Mobile phase consumption 1/20 compared to that in 4.6 mm diameter. Small injection volume required.**
- **Small Bore Column (SBC): Columns with internal diameter around 2.5 mm referred to as SBC. Mobile phase consumption 1/3 compared to that in 4.6 mm diameter.**
- **Fast-LC columns: These columns packed with 3µm dia, column packing is just 5 cm long permit very fast analysis.**

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Chromatography

Commonly used bonded phases

Octyl = Si–$(CH_2)_7$–CH_3 **C_8**

Octadecyl = Si–$(CH_2)_{17}$–CH_3 **C_{18}**

Phenyl = Si–(CH_2)-*n*-(C_6H_5) **C_6H_5**

Cyanopropyl = Si–$(CH_2)_3$–CN **CN**

Aminopropyl = Si–$(CH_2)_3$–NH_2 **NH_2**

Diol = Si–$(CH_2)_3$–OCH(OH)–CH_2–OH

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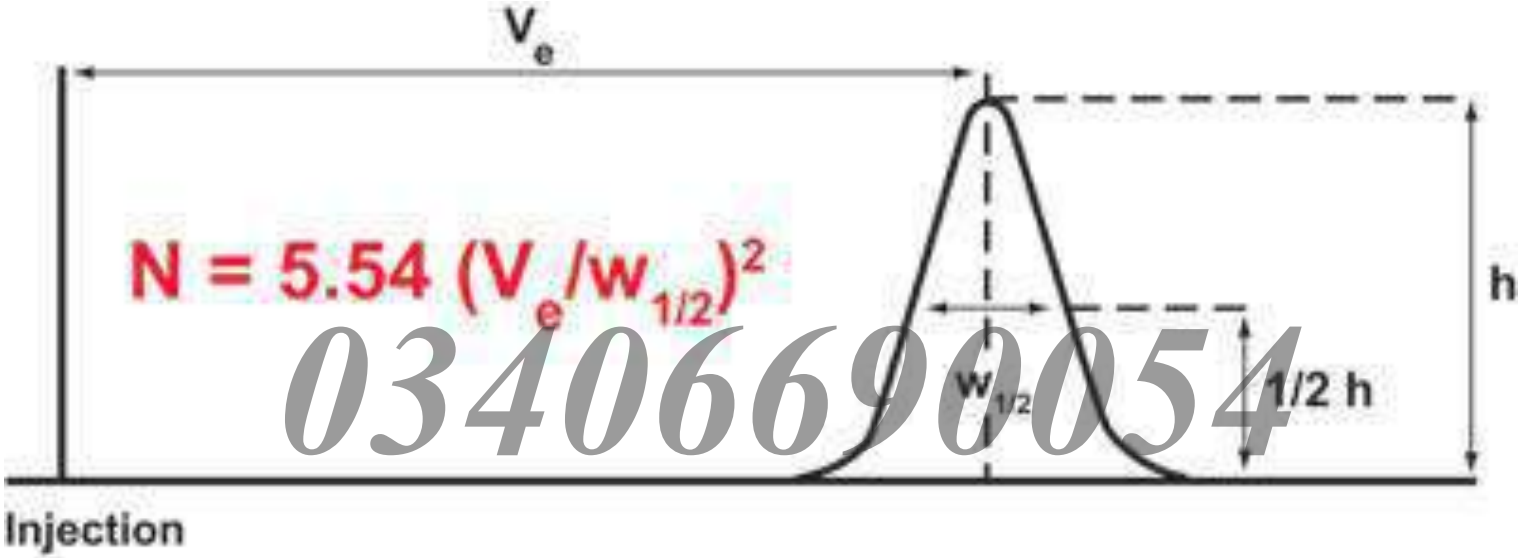
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Chromatography

- A theoretical plate in many separation processes is a hypothetical zone or stage in which two phases, such as the liquid and vapor phases of a substance, establish an equilibrium with each other.
- Such equilibrium stages may also be referred to as an equilibrium stage, ideal stage, or a theoretical tray.
- No. of theoretical plates = $5.54*(R_t / \text{Peak } 1/2 \text{ width})^2$

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Chromatography



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HPLC Columns:

- **Silica-based reversed-phase columns are stable in pH range 2-8.**
- **Columns containing porous graphite or particles of polymeric materials such as styrene–divinylbenzene copolymer are stable over a wider pH range.**
- **Columns may be heated to give higher efficiency. However, not to be heated above 60°**

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Instrumentation
Chromatography-IX

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Chromatography

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Types of Chromatography

1. Adsorption
2. Partition
3. Ion-Exchange
4. Molecular Exclusion
5. Affinity

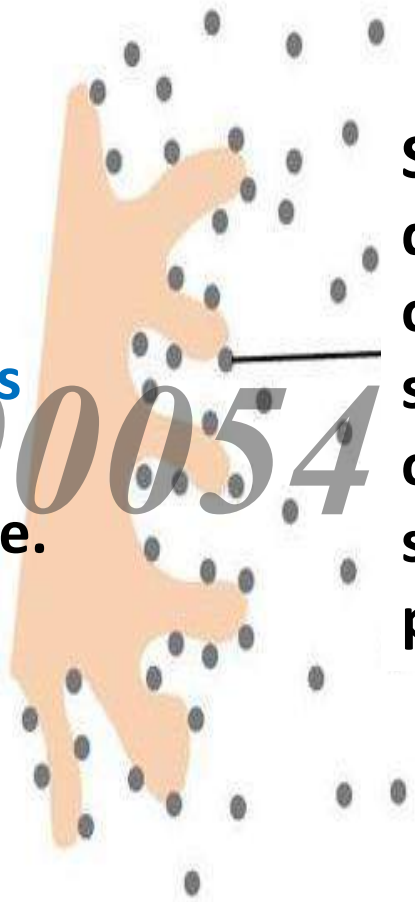
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Chromatography

Adsorption Chromatography

- Adsorption chromatography is probably one of the oldest types of chromatography around. It utilizes a mobile liquid or gaseous phase that is adsorbed onto the surface of a stationary solid phase. The equilibration between the mobile and stationary phase accounts for the separation of different solutes.



Solute dissolved on surface of stationary phase

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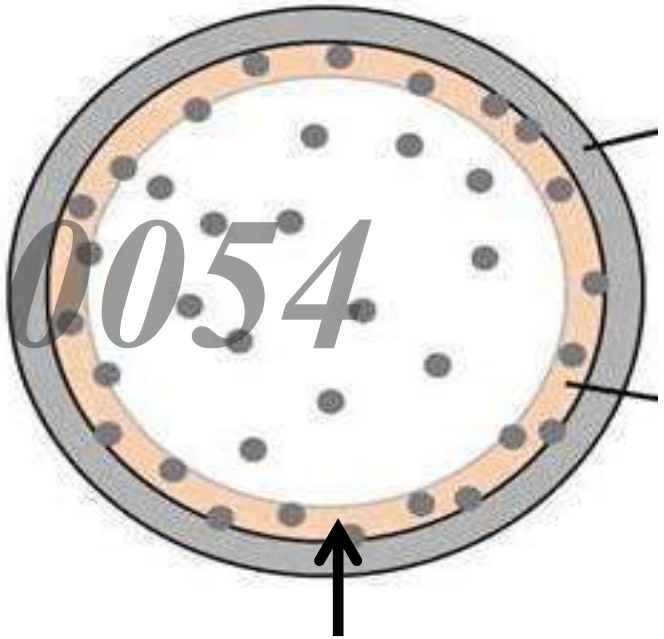
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Partition Chromatography

- This form of chromatography is based on a thin film formed on the surface of a solid support by a liquid stationary phase. Solute equilibrates between the mobile phase and the stationary liquid.

Cross section of open tubular column



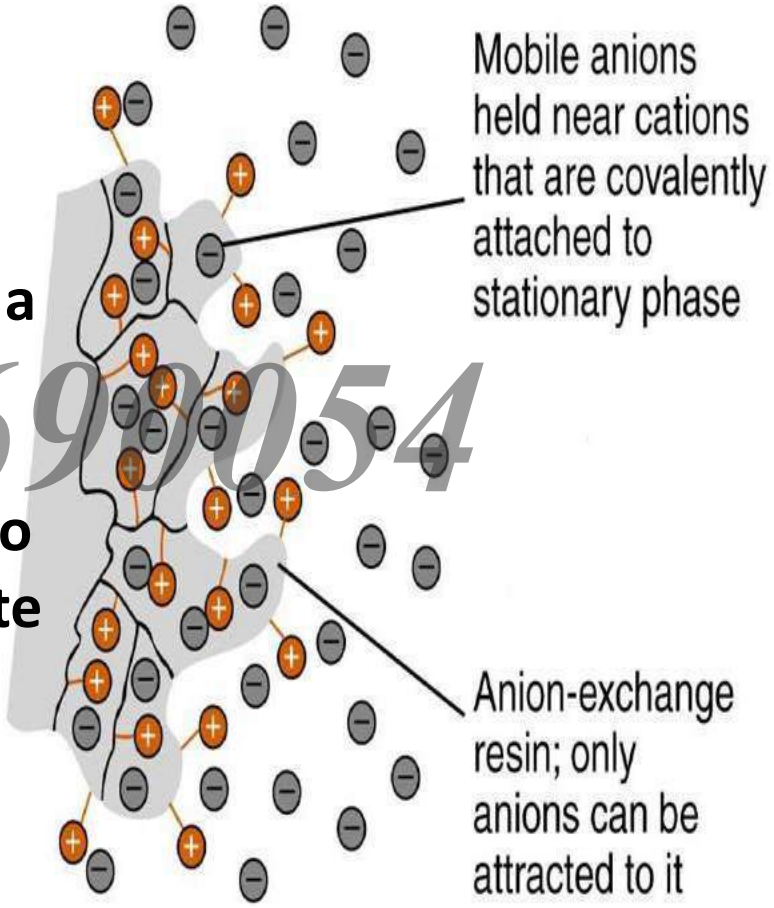
Solute dissolved in liquid phase bounded to the surface of column

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Ion Exchange Chromatography

- In this type of chromatography, the use of a resin (the stationary solid phase) is used to covalently attach anions or cations onto it. Solute ions of the opposite charge in the mobile liquid phase are attracted to the resin by electrostatic forces.

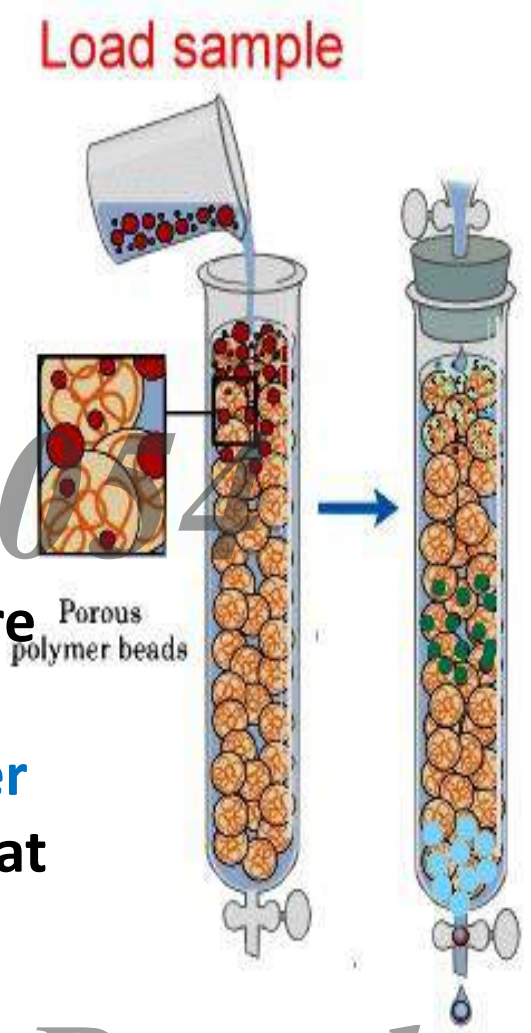


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Chromatography

Molecular Exclusion Chromatography

- Also known as gel permeation or gel filtration, based on interaction between the stationary phase and solute. While passing through a porous gel molecules separate according to their sizes. The pores are normally small and allows smaller molecules to enter the gel. The larger molecules pass through the column at a faster rate.



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Chromatography

Affinity Chromatography

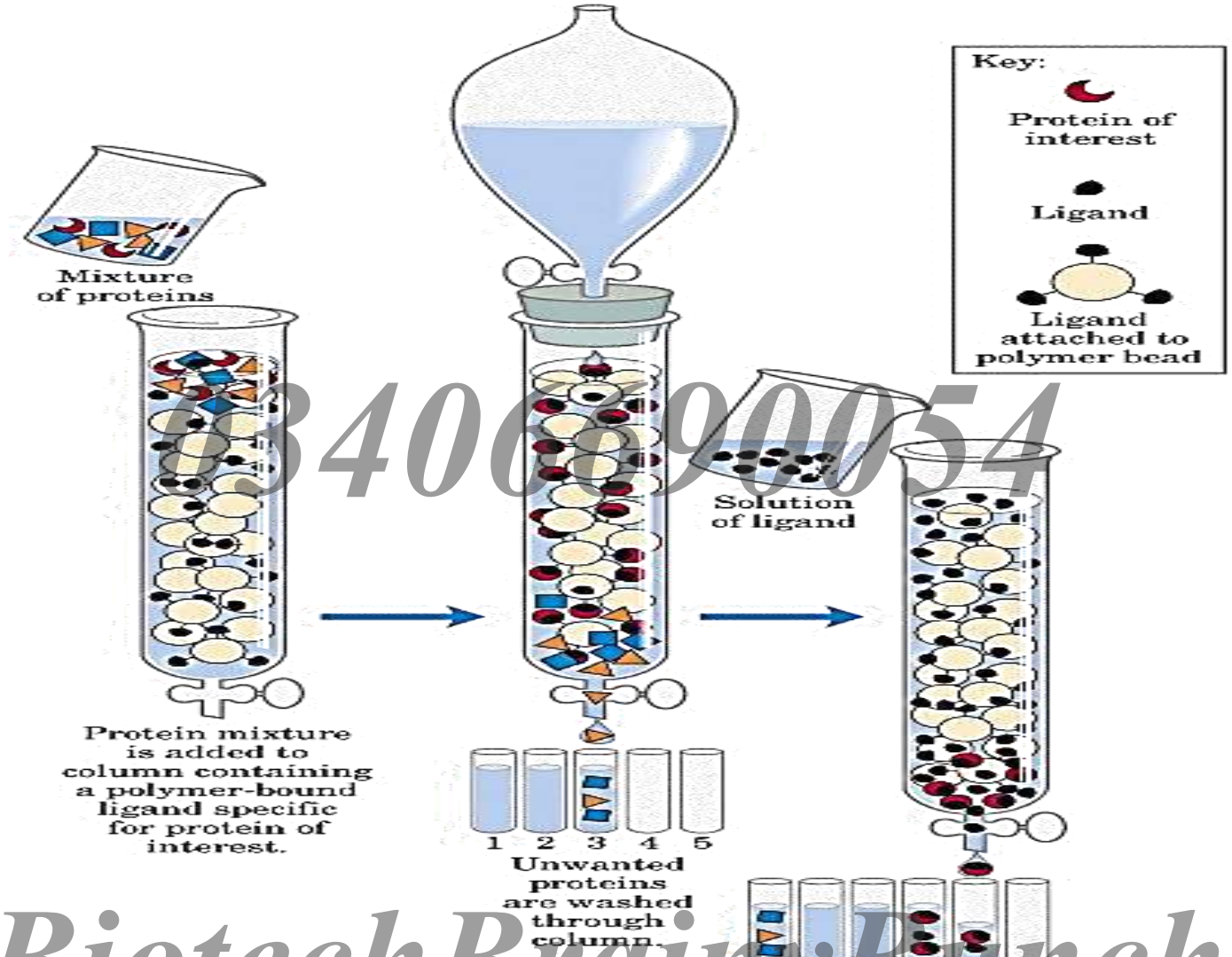
- This is the most selective type of chromatography employed. It utilizes the specific interaction between one kind of solute molecule and a second molecule that is immobilized on a stationary phase. For example, the immobilized molecule may be an antibody to some specific protein. When solute containing a mixture of proteins are passed by this molecule, only the specific protein is reacted to this antibody, binding it to the stationary phase. This protein is later extracted by changing the ionic strength or pH.

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Chromatography



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Choice of Column:

- **Based on chemistry of analytes and related compounds (impurities) or excipients etc.**

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END

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Instrumentation
Chromatography-X

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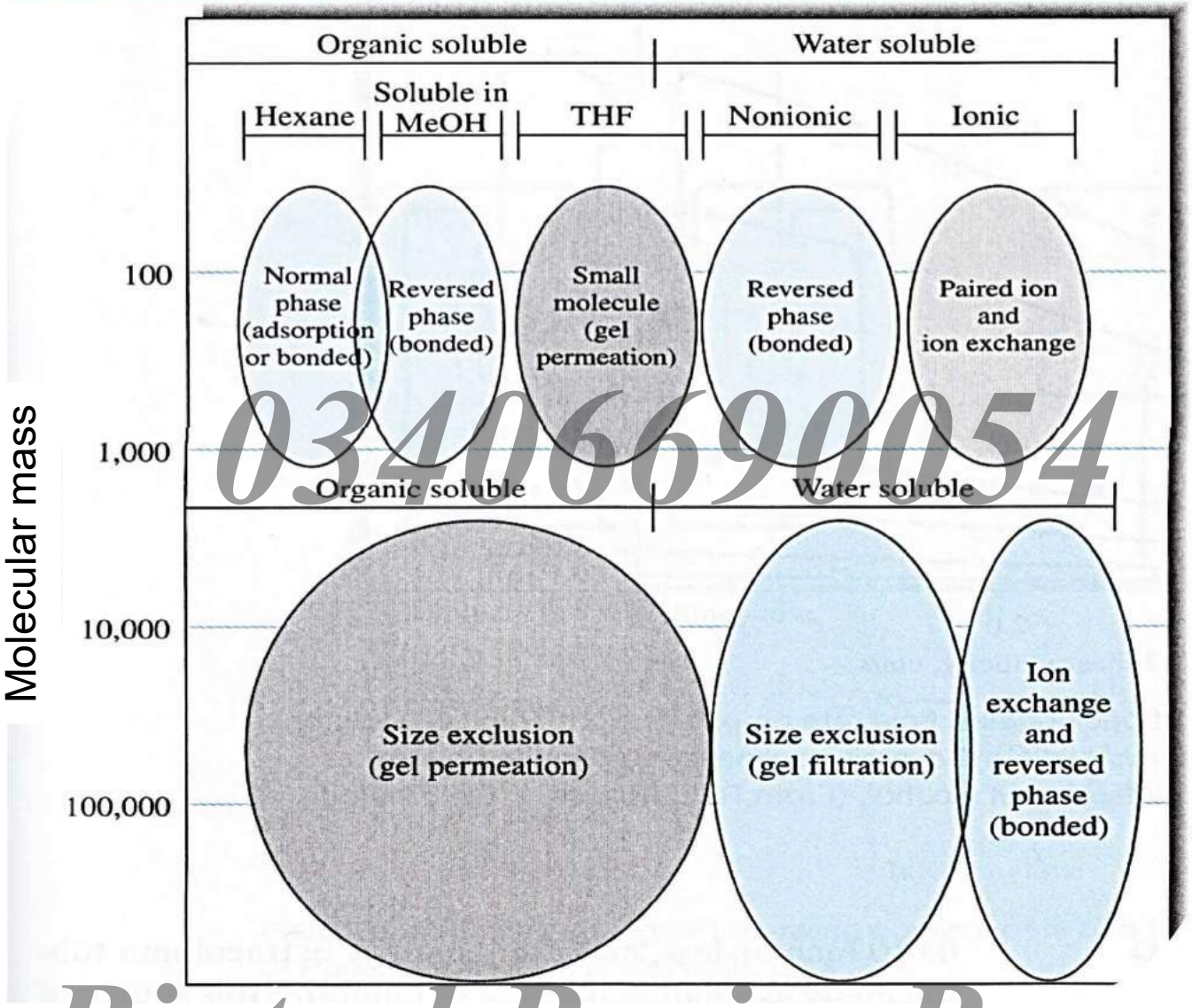
Chromatography

Method development on HPLC

1. Chose appropriate column
2. Adjust mobile phase (pH or combination)
3. Isocratic or gradient system
4. Choice of detector

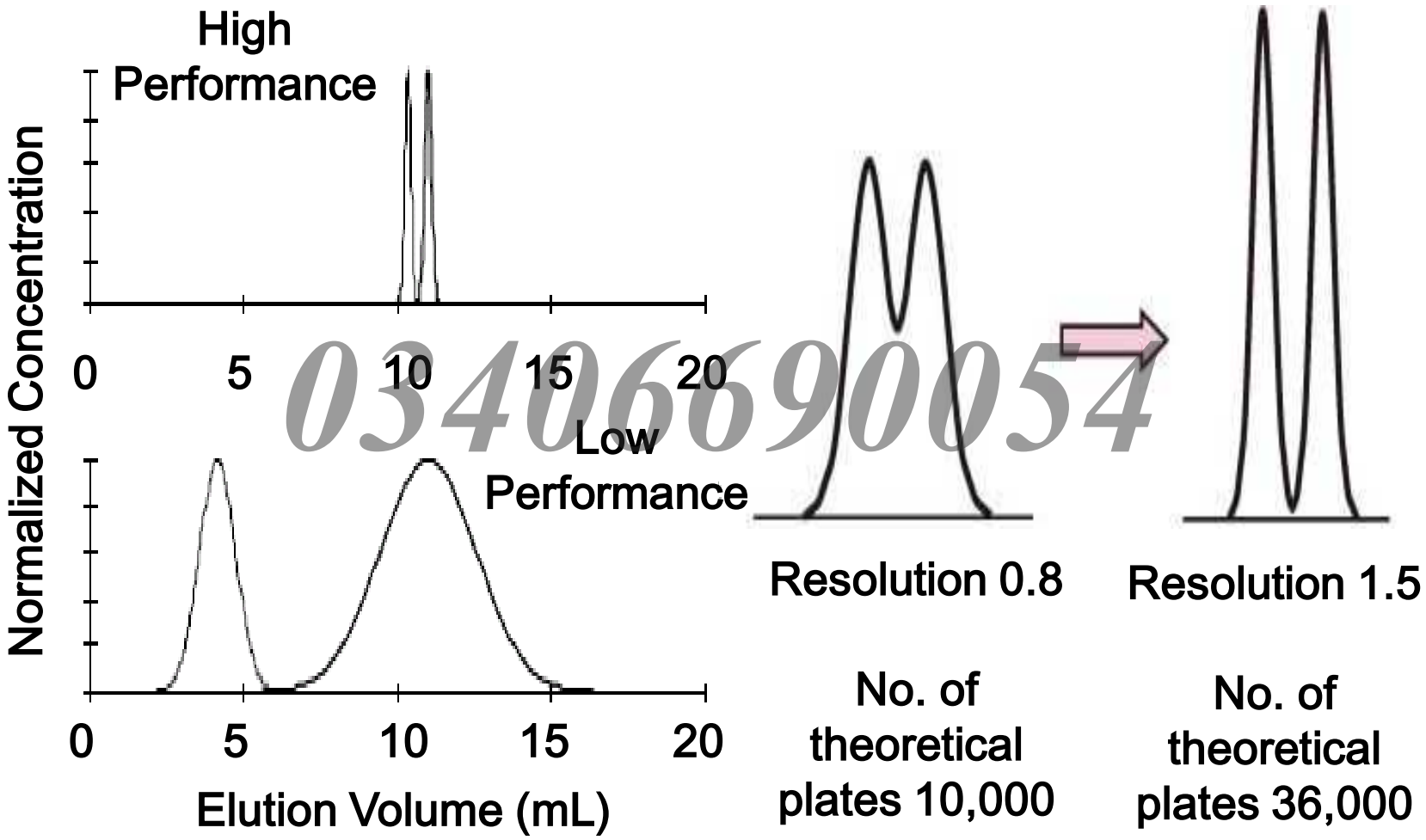
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Chromatography



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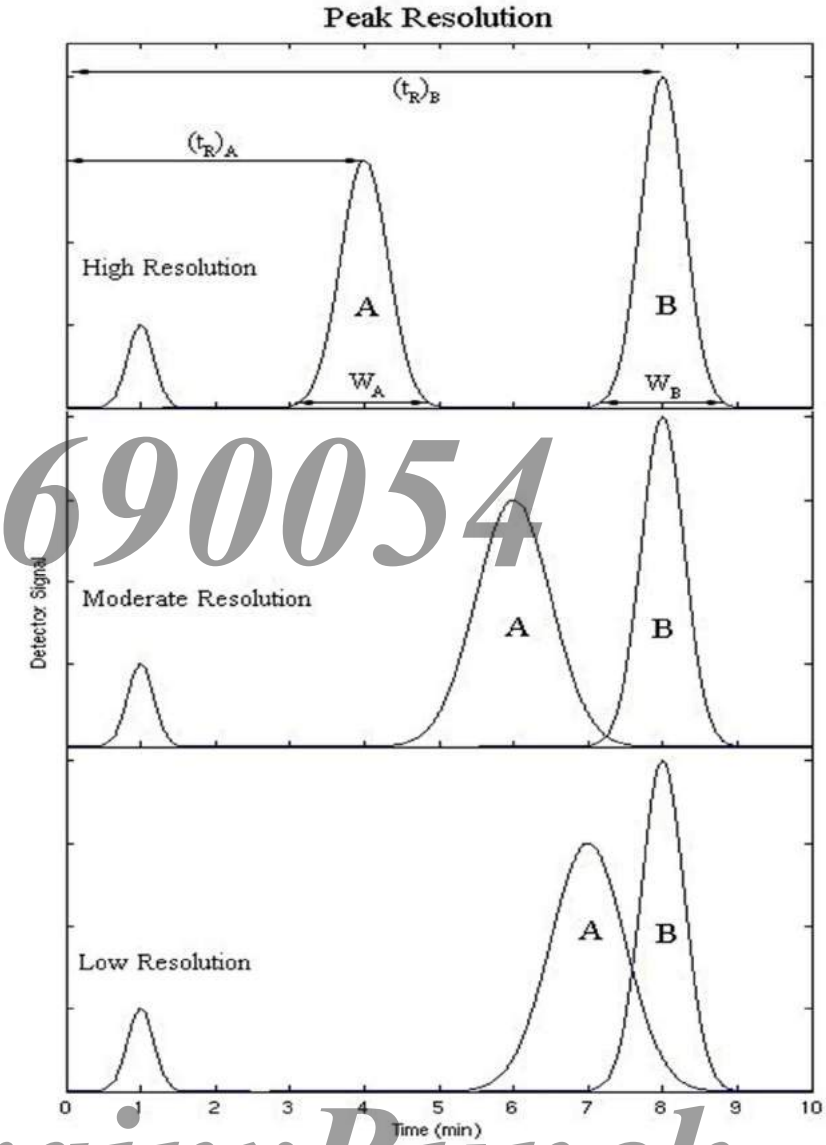
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Chromatography

Peak Resolution

$$R = \frac{2(t_2 - t_1)}{W_1 + W_2}$$

- t_1, W_1 = retention time and baseline width for first eluting peak
- t_2, W_2 = retention time and baseline width for first eluting peak
- R ~ 1.5 represent complete separation
- R ~ 1.0 consider adequate in most cases

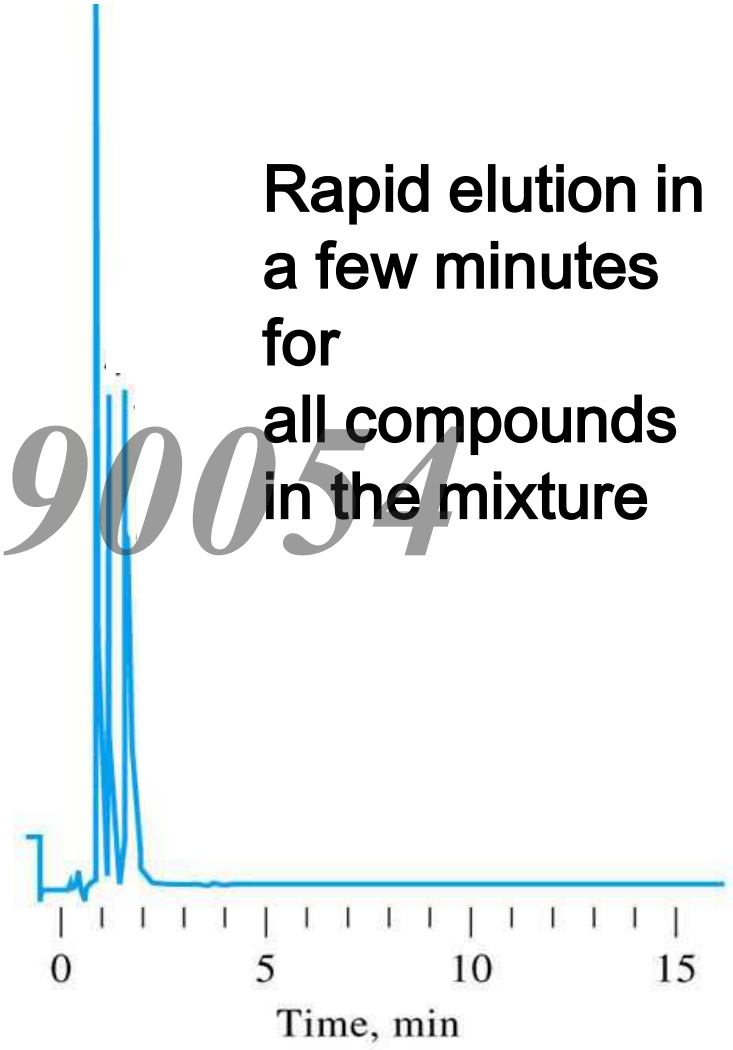


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Chromatography

Peak Resolution

- Shows that column has little interaction with the compounds.
- Analytes has greater affinity with mobile phase.
- May be elution speed is too high.
- No separation at all



Rapid elution in a few minutes for all compounds in the mixture

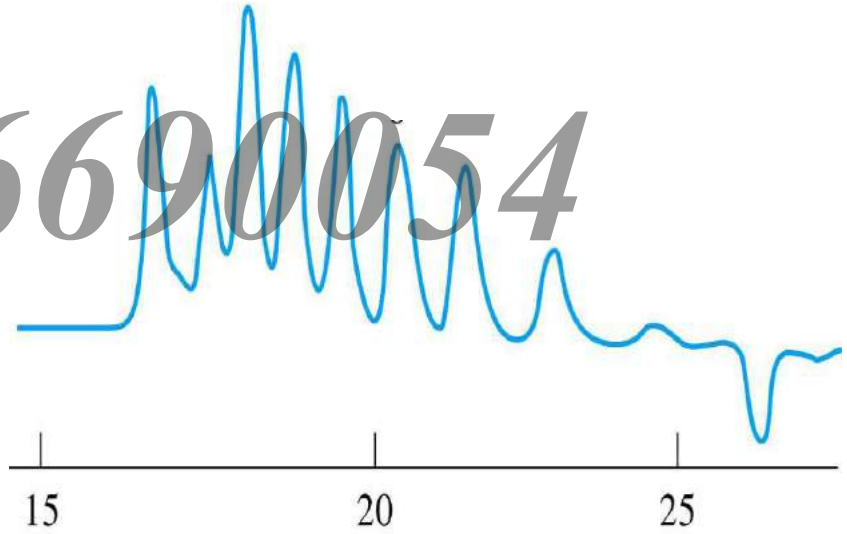
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Chromatography

Peak Resolution

- Shows that mobile phase has little interaction with the compounds.
- Analytes has greater affinity with stationary phase.
- May be elution speed is too low.
- Separated in a longer period of time.

Slow elution (~ 20 minutes) for all compounds in the mixture

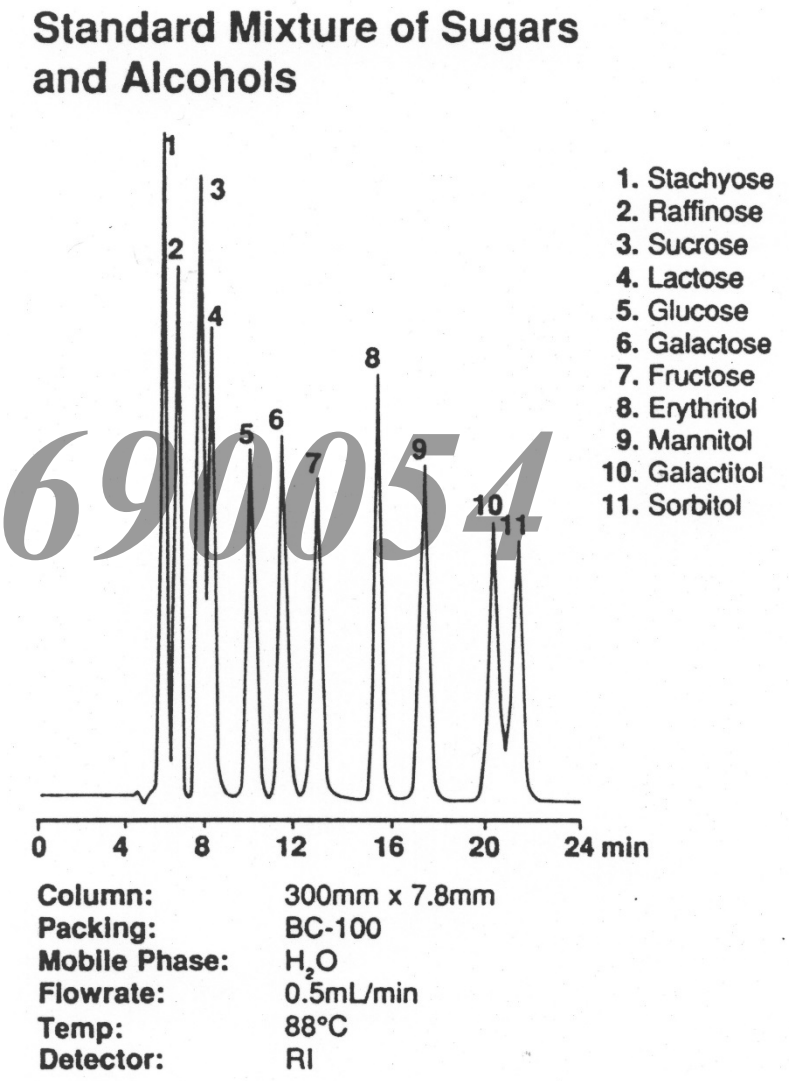


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Chromatography

Peak Resolution

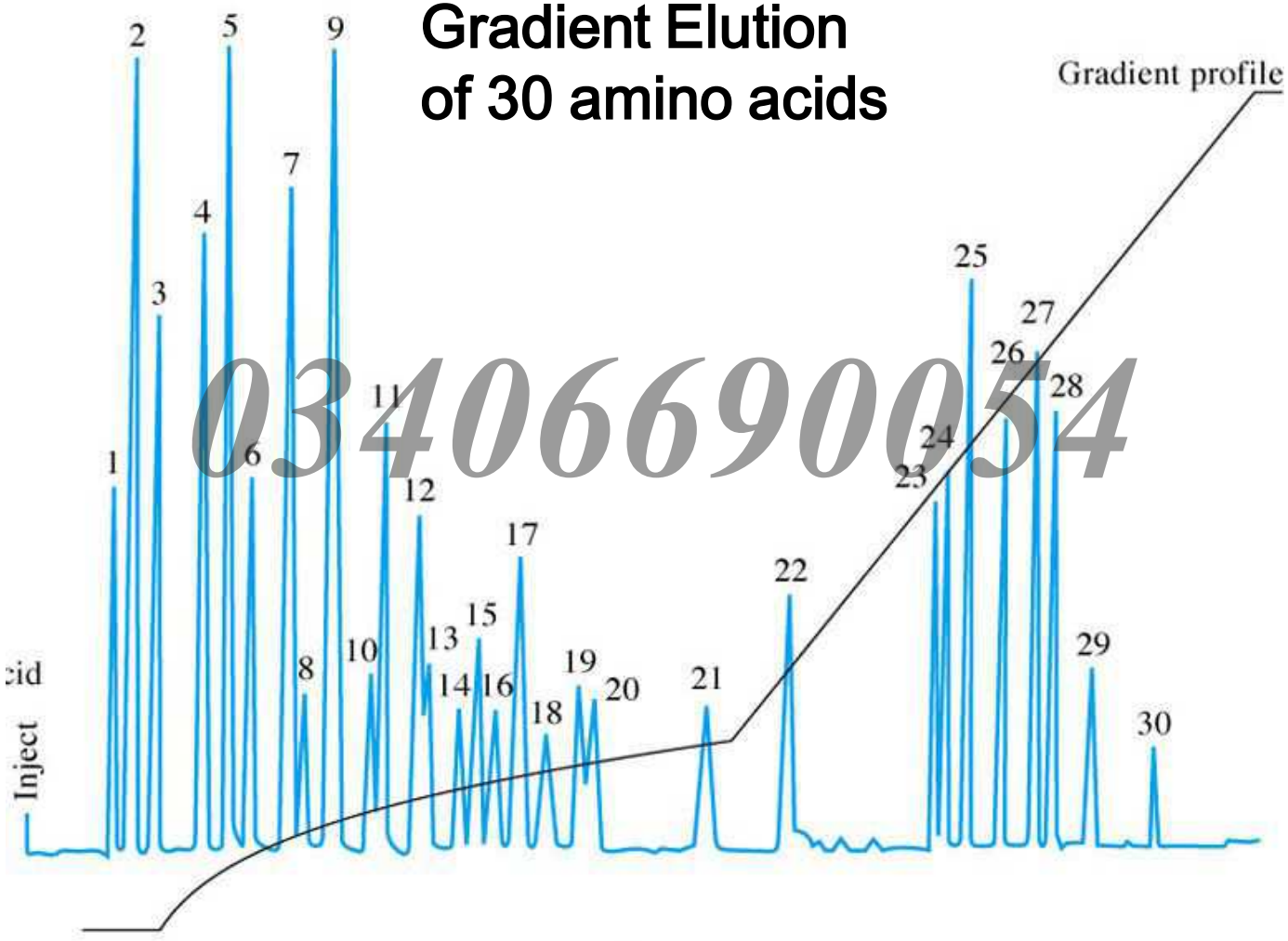
- For better peak resolution we have to select an appropriate column.
- Optimization of condition for the pH, flow rate, column oven temp., % composition.
- Some time a gradient elution is prefered.



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Chromatography

Gradient Elution of 30 amino acids



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Chromatography

Method development on HPLC:

- Followed previous guidelines we have to develop an analytical method.
- That method needs validation afterward.
- Only after validation it can be considered as an analytical method.

END

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Instrumentation
Chromatography-XI

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Chromatography

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Gas Chromatography

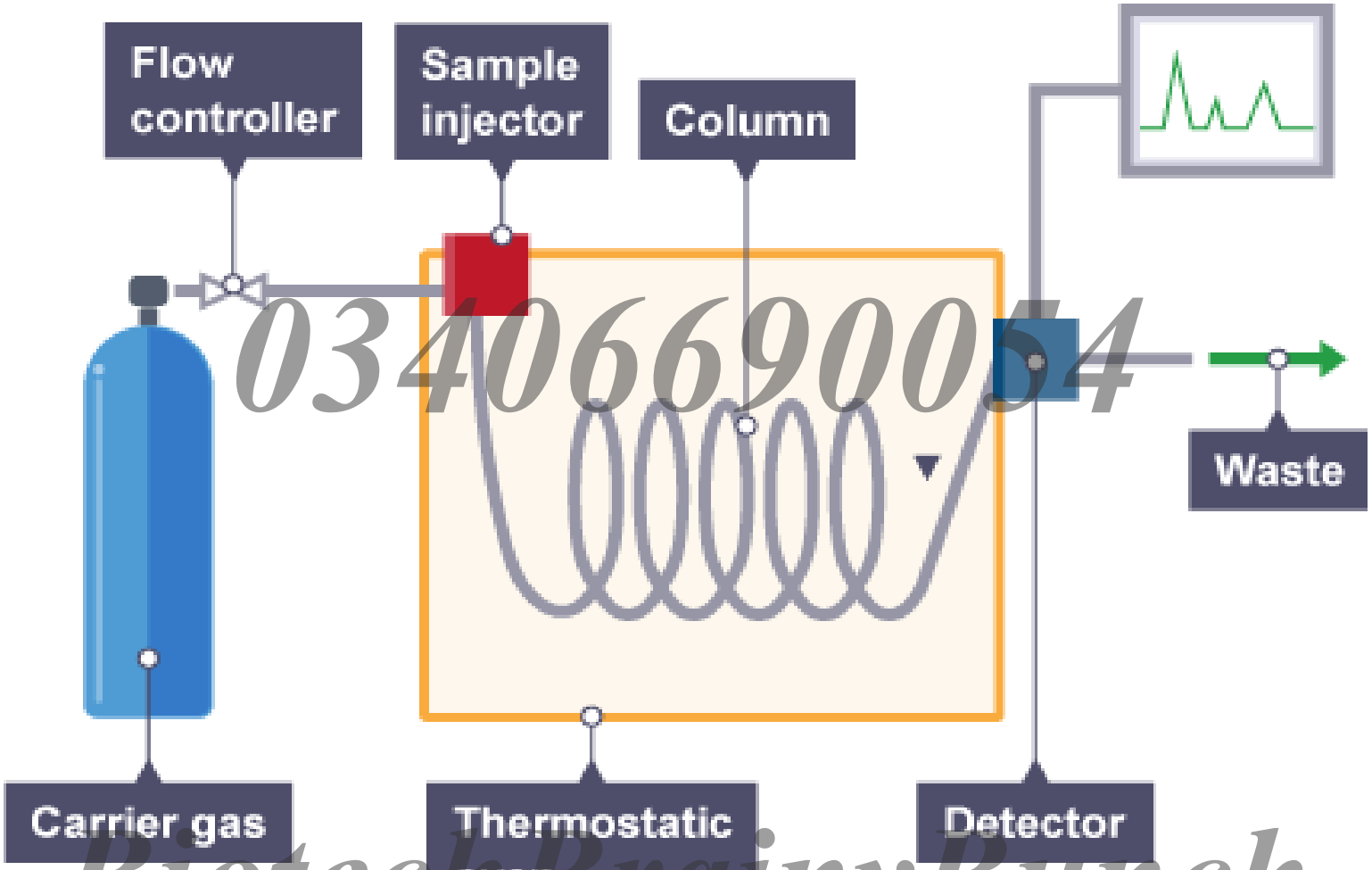
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- Partition of molecules between gas (mobile phase) and liquid or solid (stationary phase).

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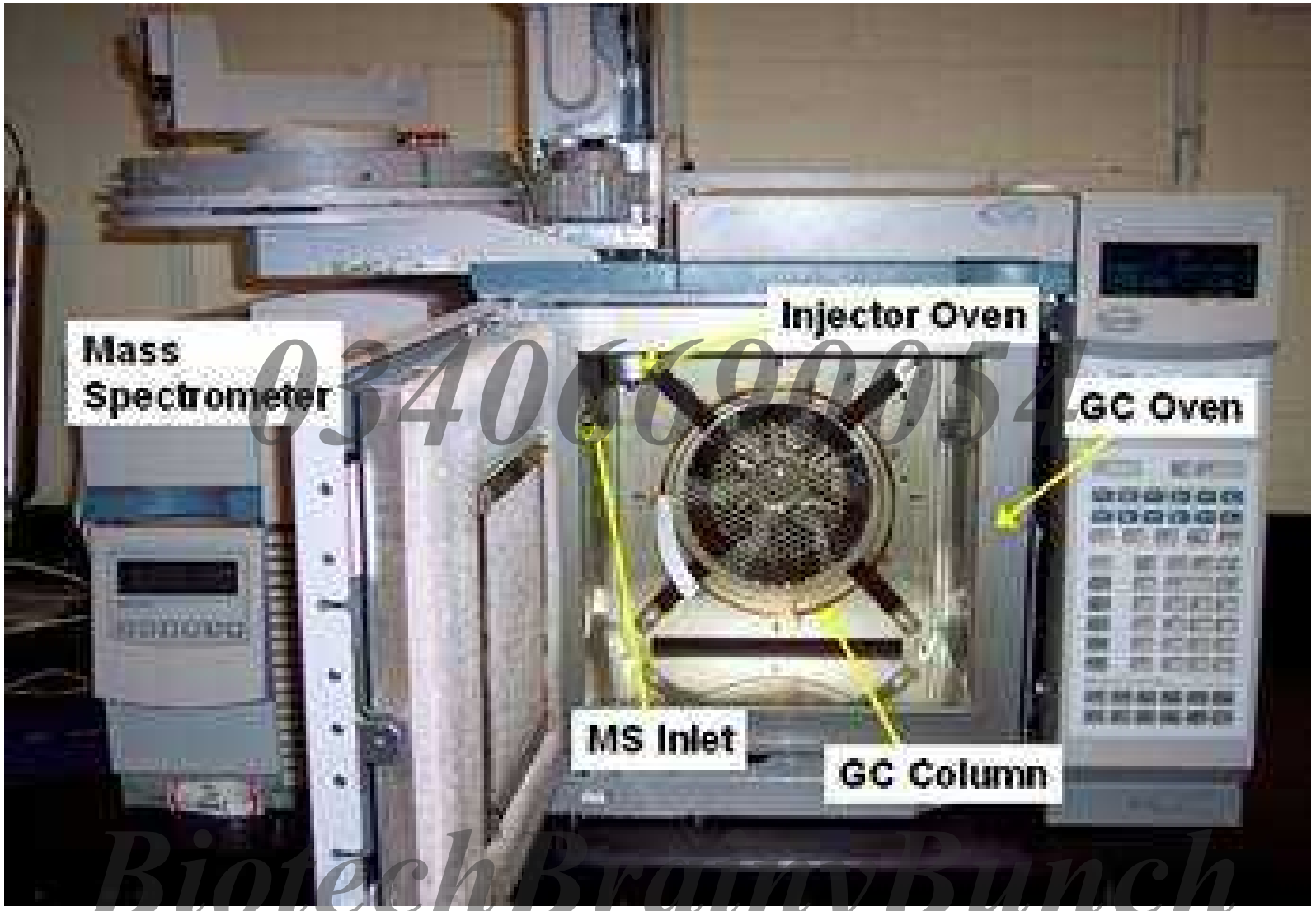
Chromatography

Schematic diagram of Gas Chromatograph



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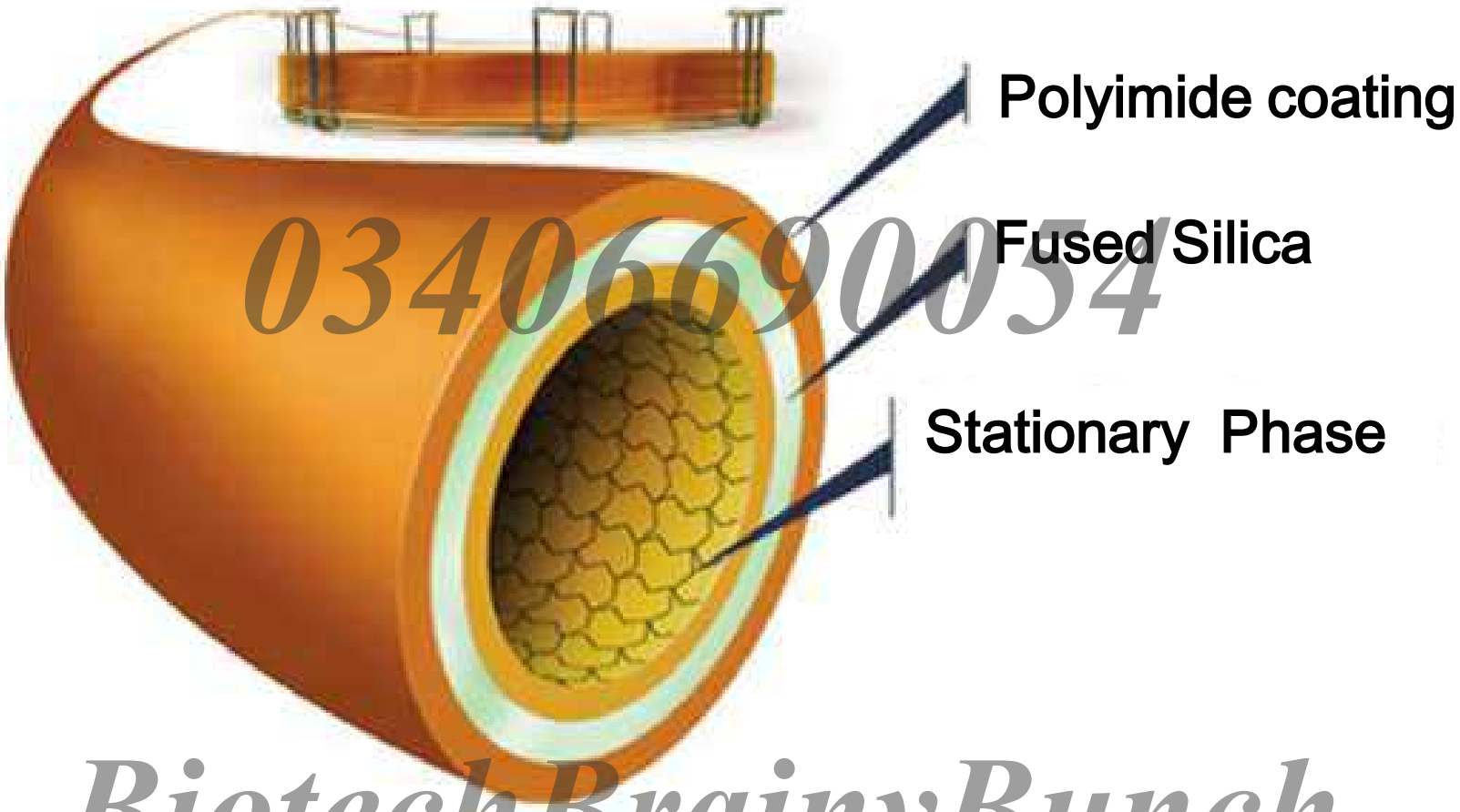
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GC Column



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Chromatography



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Chromatography

Common Detectors used with GC

- Flame Ionization Detector
- Thermal Conductivity Detector
- Electron Capture Detector
- Mass Spectrometer Detector

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Chromatography

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END

Applications of GC:

- Perfume industry
- Oil industry
- Aflatoxins and pesticide residue analysis
- Essential oils analysis

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Instrumentation
Solvent Extraction-I

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Solvent Extraction

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Solvent Extraction

- Liquid-liquid extraction is a useful method to separate components (compounds) of a mixture.

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Solvent Extraction

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Filtration

- To separate solid insoluble particles from solution.

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- For example separate sand from water,
- or purify salt from mixture of sand and salt or other impurities.

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Solvent Extraction





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Solvent Extraction

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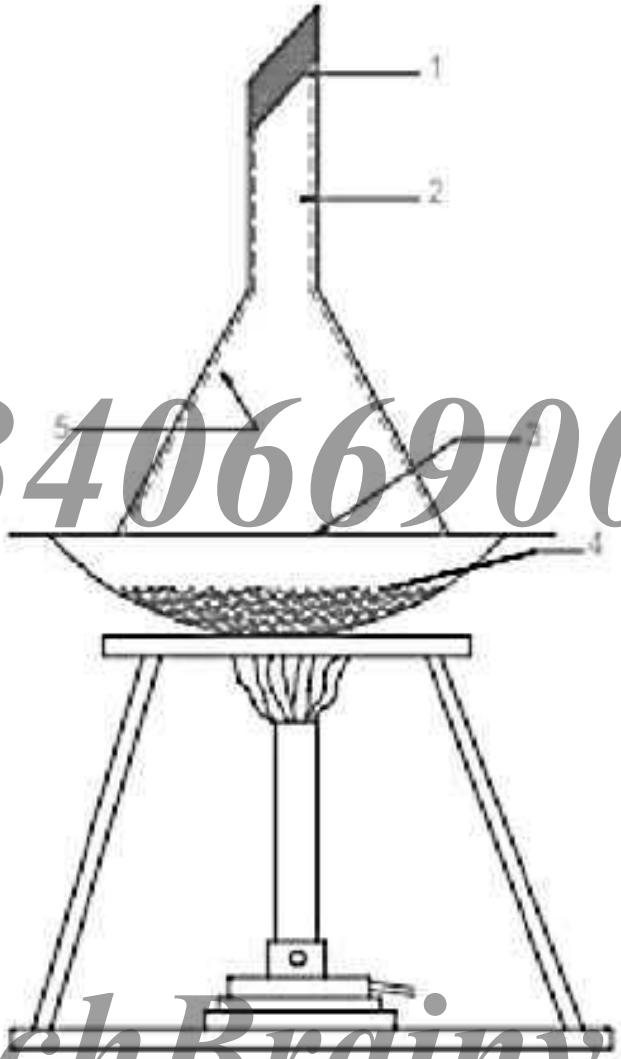
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Sublimation

- To separate one solid from other if it vaporize by heating.
- For example separate salt from benzoic acid or naphthalene.

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Solvent Extraction



1. Cotton
2. Inverted funnel
3. Perforated asbestos sheet
4. Mixture
5. Sublimate

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Solvent Extraction

Solvent extraction

- To separate two materials (solid or liquid) from immiscible liquids. One of the two liquids is water and other is any organic solvent.

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Solvent Extraction

- Suppose we have a mixture of sugar in vegetable oil and want to separate the sugar from the oil. (the sugar particles are too tiny to filter and you suspect that the sugar is partially dissolved in the oil).
- Sugar is much more soluble in water than in vegetable oil, and, water is immiscible with oil.
- The water phase is in the bottom layer because water is denser than oil.
- By shaking the layers (phases) well, you increase the contact area between the two phases. The sugar will move to the phase in which it is most soluble: the water layer.
- Remove the bottom water layer, evaporate water to obtain the sugar.

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Solvent Extraction

The success of this method depends upon the difference in solubility of a compound in various solvents. For a given compound, solubility differences between solvents is quantified as the "distribution coefficient"

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Instrumentation
Solvent Extraction-II

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Solvent Extraction

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Batch Extraction Method

- There are two possible ways to follow extraction procedure.
- One is to use high amount of solvent at one time.
- The second one is to use small amount of solvent in more than one extraction steps.

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Solvent Extraction

Partition Coefficient K_p (Distribution Coefficient K_d)

- **When a compound is shaken in a separatory funnel with two immiscible solvents, the compound will distribute itself between the two solvents.**
- **At a certain temperature, the ratio of concentrations of a solute in each solvent is always constant. And this ratio is called the distribution coefficient.**

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$$K = \frac{\text{concentration in solvent}_2}{\text{concentration in solvent}_1}$$

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Solvent Extraction

Distribution coefficient;

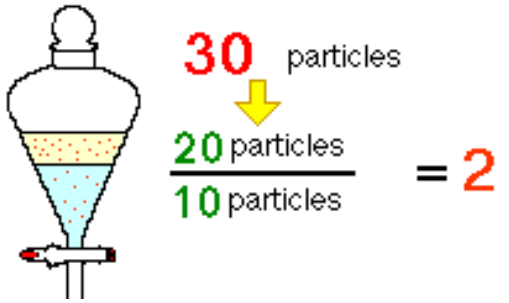
- **Where solvent$_1$ and solvent$_2$ are immiscible liquids.**
- **By convention we consider the organic solvent as Solvent 1 and water as solvent 2.**

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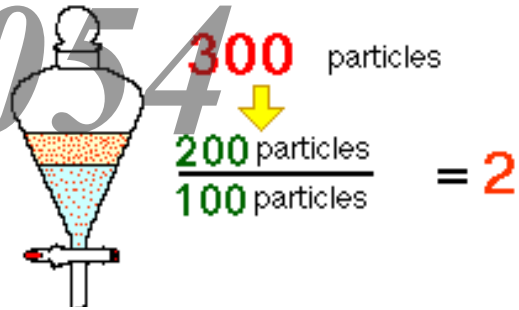
Solvent Extraction

Suppose a compound has a distribution coefficient K=2 between solvent$_1$ and solvent$_2$.

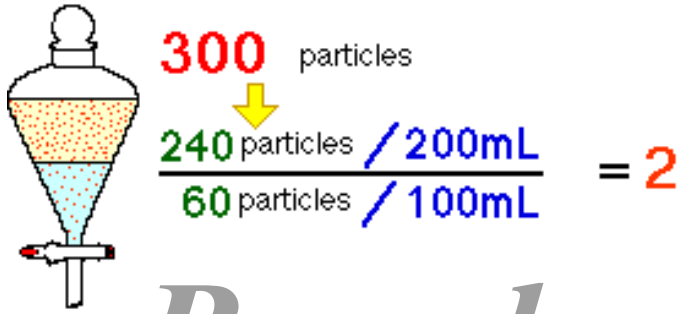
If there are 30 particles of comp. distributed between equal volumes of solvent$_1$ and solvent$_2$.



If there are 300 particles of comp., the same distribution ratio is observed in solvents 1 and 2



When we double the volume of solvent$_2$ (i.e., 200 mL of solvent$_2$ and 100 mL of solvent$_1$), the 300 particles of compound distribute As shown;



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Solvent Extraction

What happens if we extract twice with 100 mL of solvent$_2$?

If we use 100 mL of solvent$_2$ at one time then 200 particles extracted. Now we add a second 100 mL volume of fresh solvent$_2$. According to the distribution coefficient K=2, so we can extract 67 more particles from the remaining solution.

So, total 267 particles extracted as compared to 240 particles if we use 200mL of solvent$_2$ for one time extraction

Suppose **x** particles are extracted out of **100** dissolved in solvent1

$$K = \frac{x \;/100\text{mL solv2}}{(100-x)/100\text{mL solv1}} = 2$$

$$x = 2(100-x)\frac{100\text{mL solv1}}{100\text{mL solv2}}$$

$$x = 200 - 2x$$

$$3x = 200$$

$$x = 66.6\ldots \cong 67$$

$\therefore$ **67** particles are extracted with the second portion of solvent2

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Solvent Extraction

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Batch wise extraction:

- So, it is concluded that batch wise extraction is more convenient and efficient way of extraction.
- Statistically maximum three extractions are feasible.

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Instrumentation
Solvent Extraction-III

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Solvent Extraction

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Change in chemistry of solute

- Like dissolve like.
- Organic compounds dissolve in organic solvents.
- Inorganic salts dissolve in water.
- However, certain organic compounds can change their solubility from organic solvent to water.

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Solvent Extraction

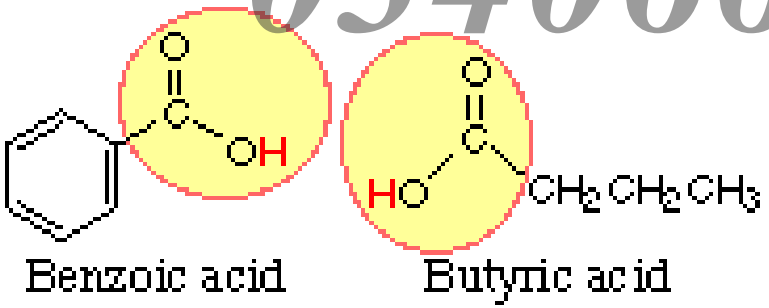
What kind of Organic compounds can change their solubility?

- **Organic acids include carboxylic acids (strong organic acids) and phenols (weak organic acids).**

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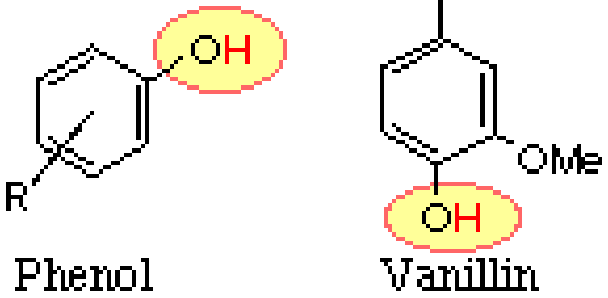
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Examples of **Carboxylic acids**



Benzoic acid Butyric acid

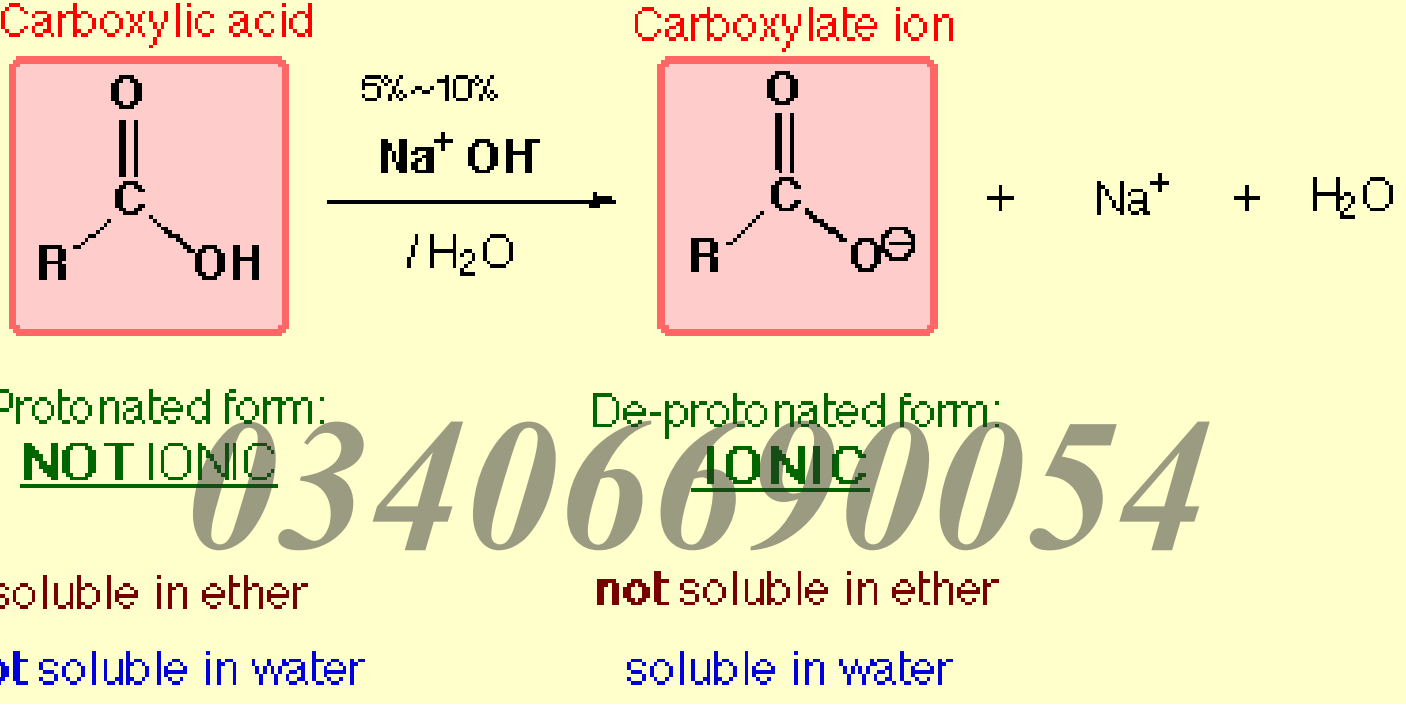
Examples of **Phenols**

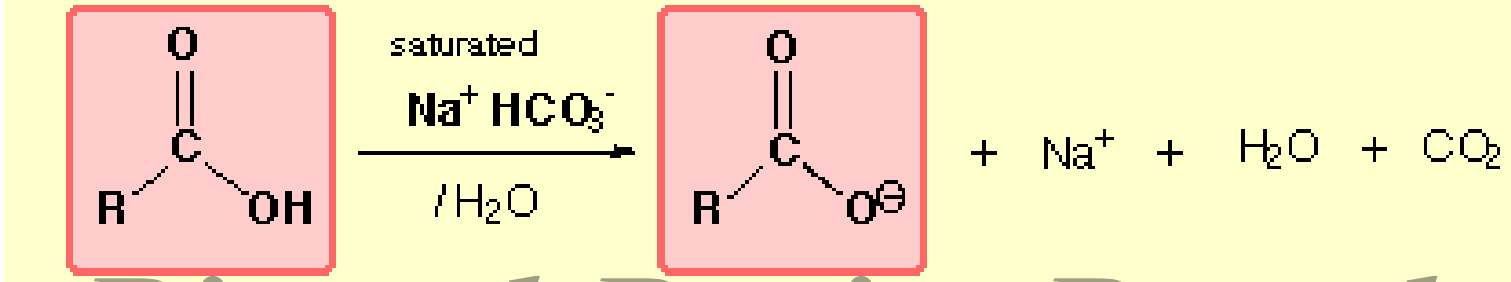


Phenol Vanillin

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Solvent Extraction





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Solvent Extraction

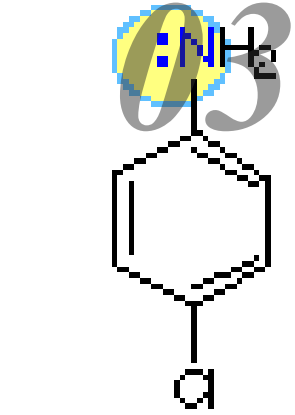
What kind of Organic compounds can change their solubility?

- **Organic bases includes amines**

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Examples of **Amines**

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p-Chloroaniline



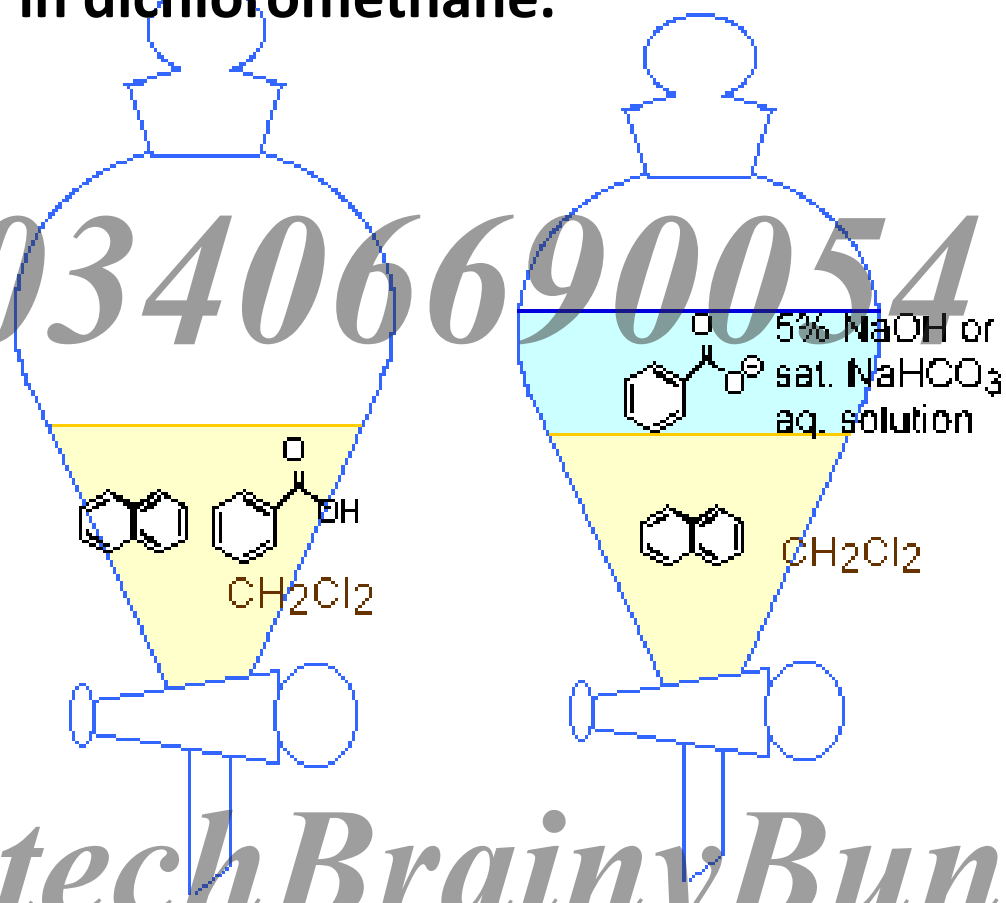
Tributylamine

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Solvent Extraction

Let's try a sample problem.

- **Here is a mixture of naphthalene and benzoic acid, dissolved in dichloromethane.**



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Solvent Extraction

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END

Chemistry of Solute:

- The phenols are weak acids and can react easily with strong bases e.g. NaOH.
- But they can't react with weak bases like sodium carbonate.

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Instrumentation
Solvent Extraction-IV

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Solvent Extraction

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Change in chemistry of solute

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Let's try another sample problem.

- We have a mixture of benzoic acid and p-methoxyphenol, dissolved in dichloromethane.

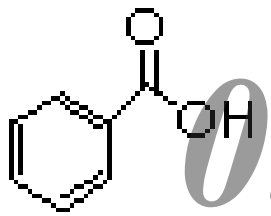
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Solvent Extraction

What should we do?

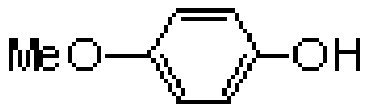
- Add 5% NaOH…

Benzoic acid

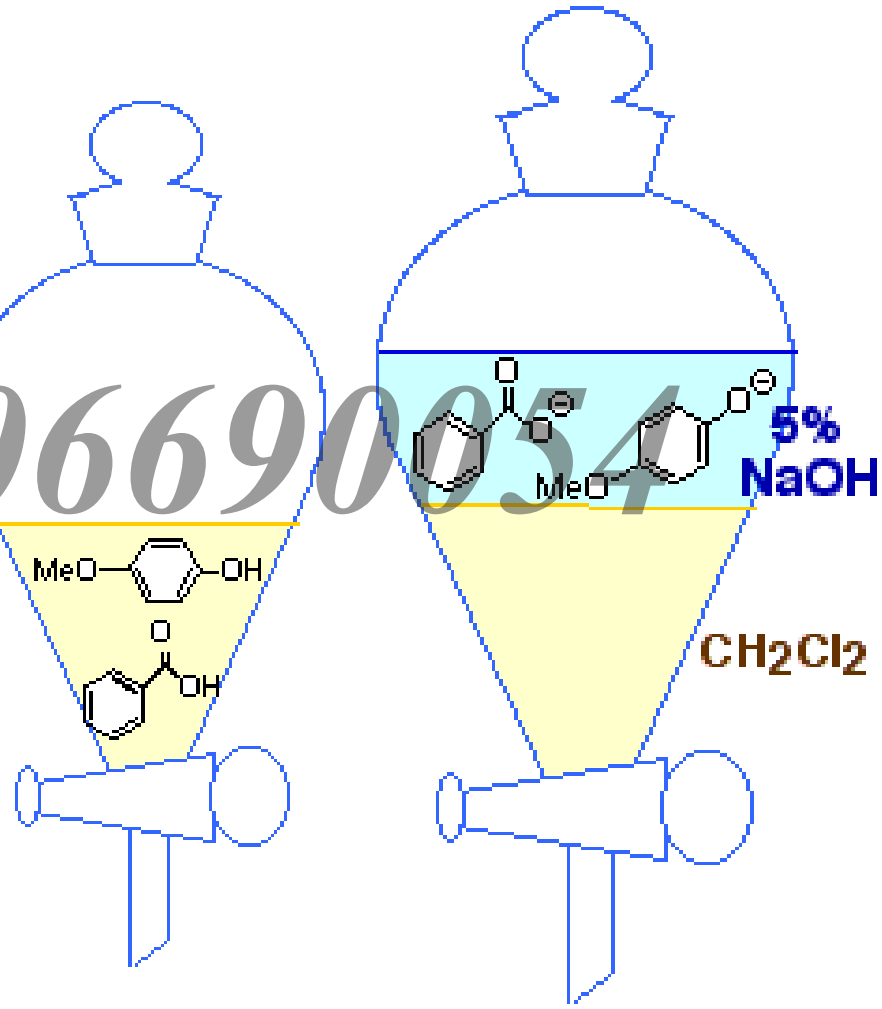


pKa = 4.2

p-Methoxyphenol



pKa = ~10



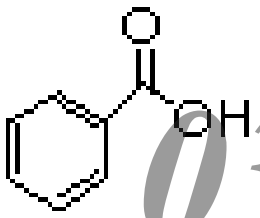
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Solvent Extraction

What should we do then?

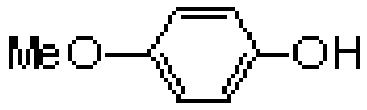
- Add saturated $NaHCO_3$…

Benzoic acid

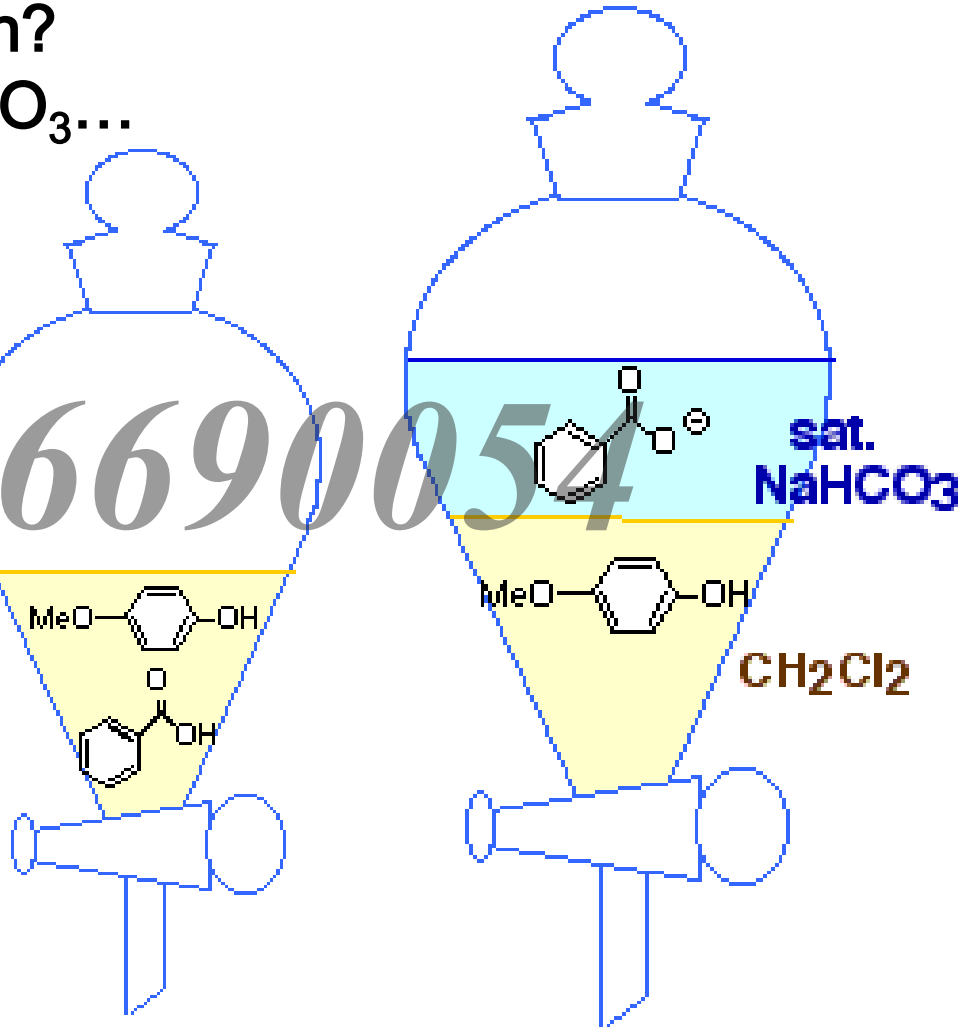


pKa = 4.2

p-Methoxyphenol



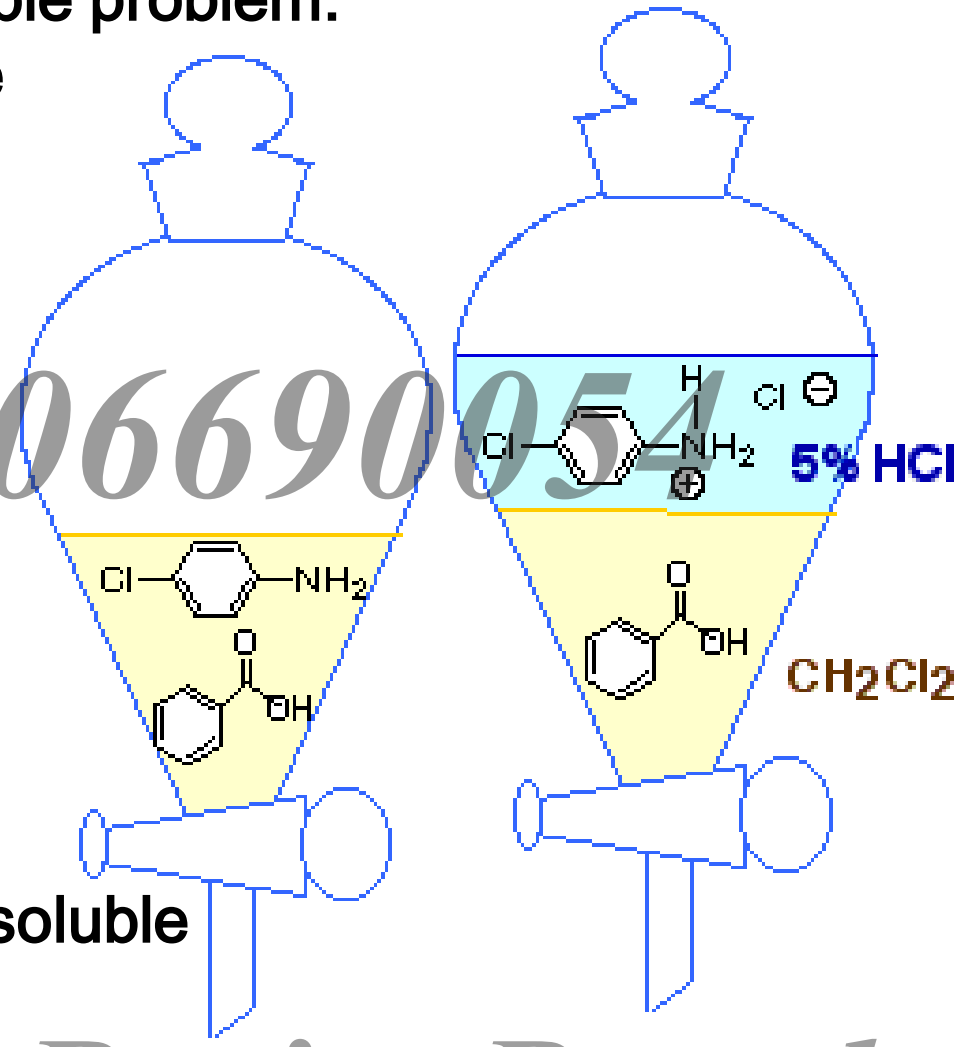
pKa = ~10



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Solvent Extraction

Let's try a third sample problem.
If we have a mixture of benzoic acid and p-chloroaniline, dissolved in dichloromethane.



We can do the either way, add 5% HCl to make amine water soluble or add 5% NaOH to make acid water soluble

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Solvent Extraction

Chemistry of Solute:

- By changing organic acids and bases into their salts we can revert the polarity from organic to aqueous soluble.
- We can separate alkaloids from plant extract or organic acids from food in this way.

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Instrumentation
Solvent Extraction-V

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Solvent Extraction

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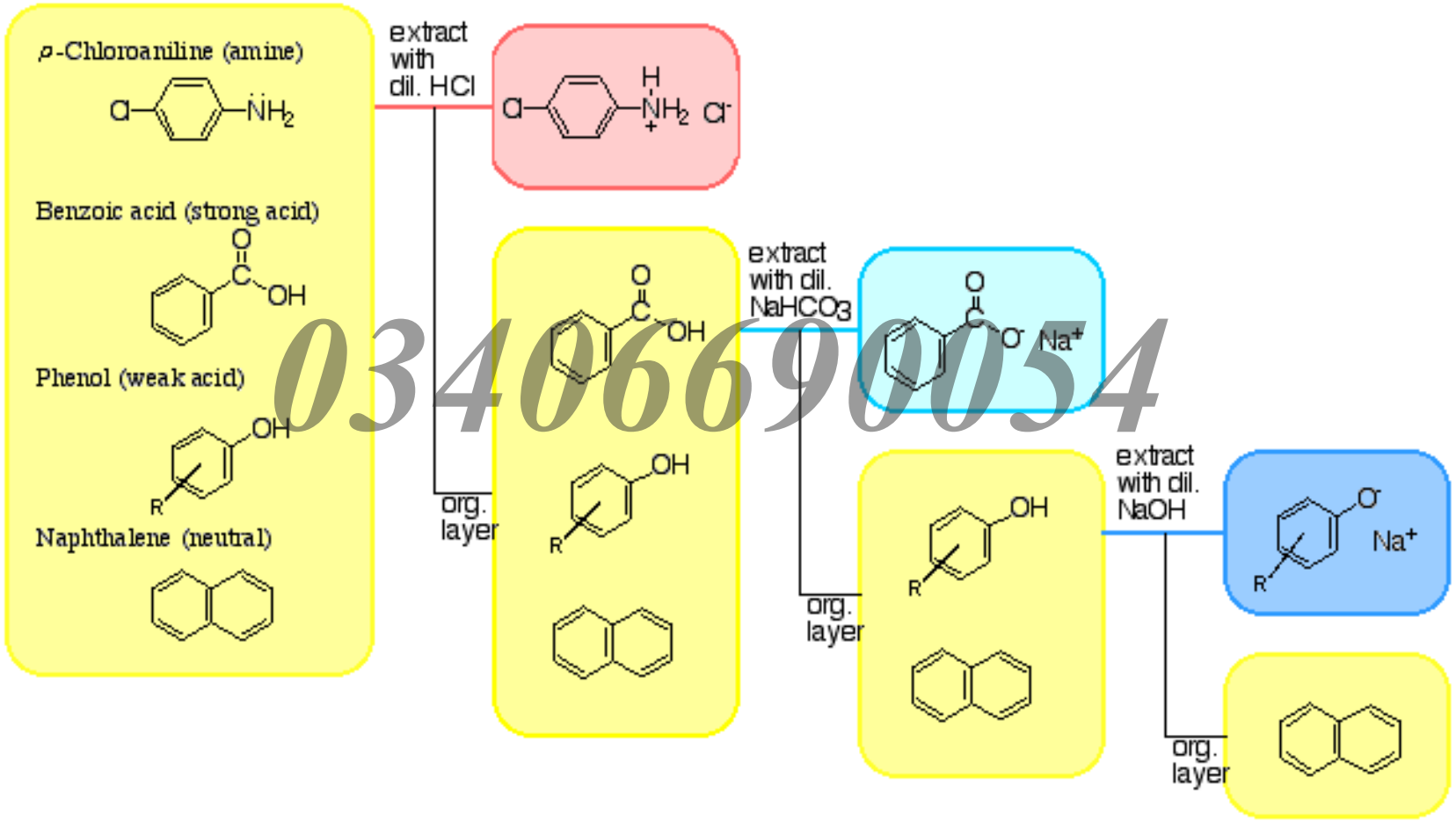
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Extraction Scheme

- We can Separate a complex mixture by designing an extraction scheme.

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Solvent Extraction



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Solvent Extraction

An example of Caffeine Extraction from Tea

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Solvent Extraction

An example of Caffeine Extraction from Tea

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Solvent Extraction

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Applications:

- For extraction of caffeine from tea leaves.
- For extraction of nicotine from tobacco.

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Instrumentation
Solvent Extraction-VI

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Solvent Extraction

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Recovery of organic compound

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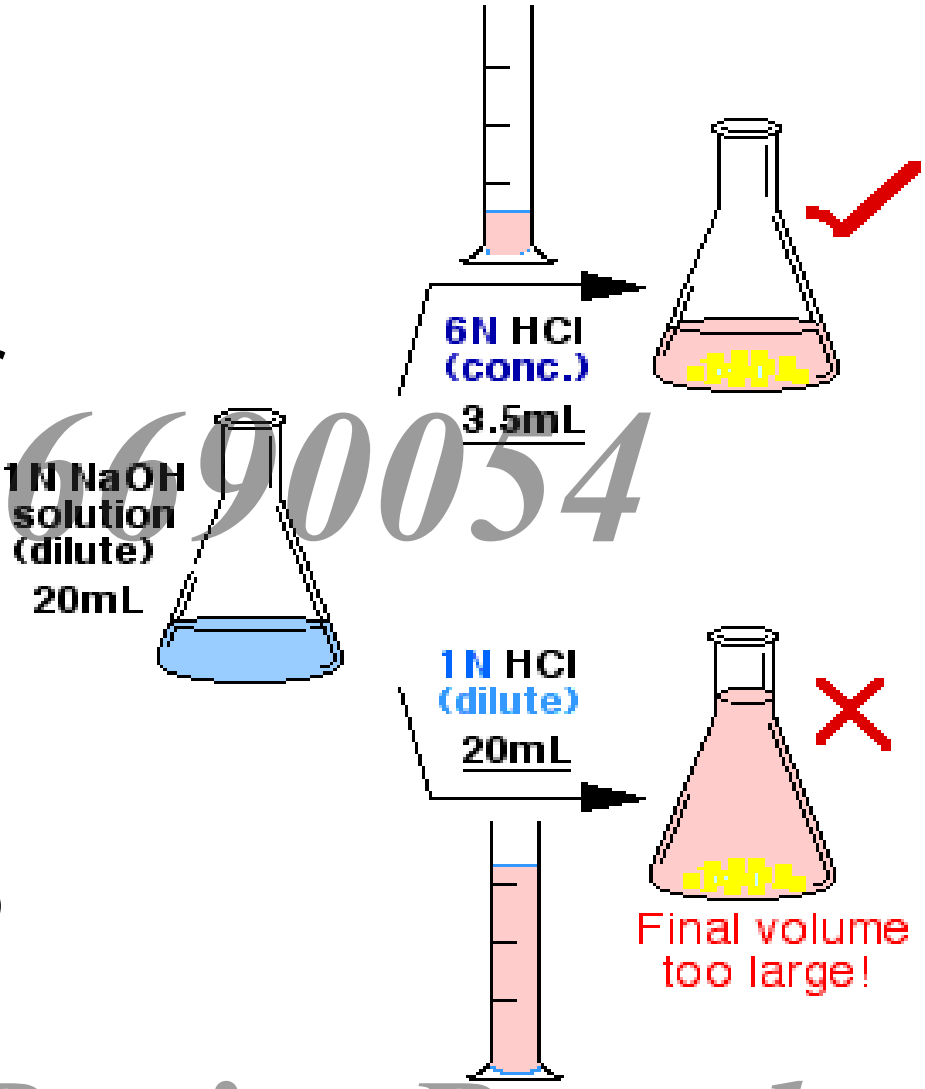
- We can recover an organic acid or base from their salt by reverting the pH of solution.

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Solvent Extraction

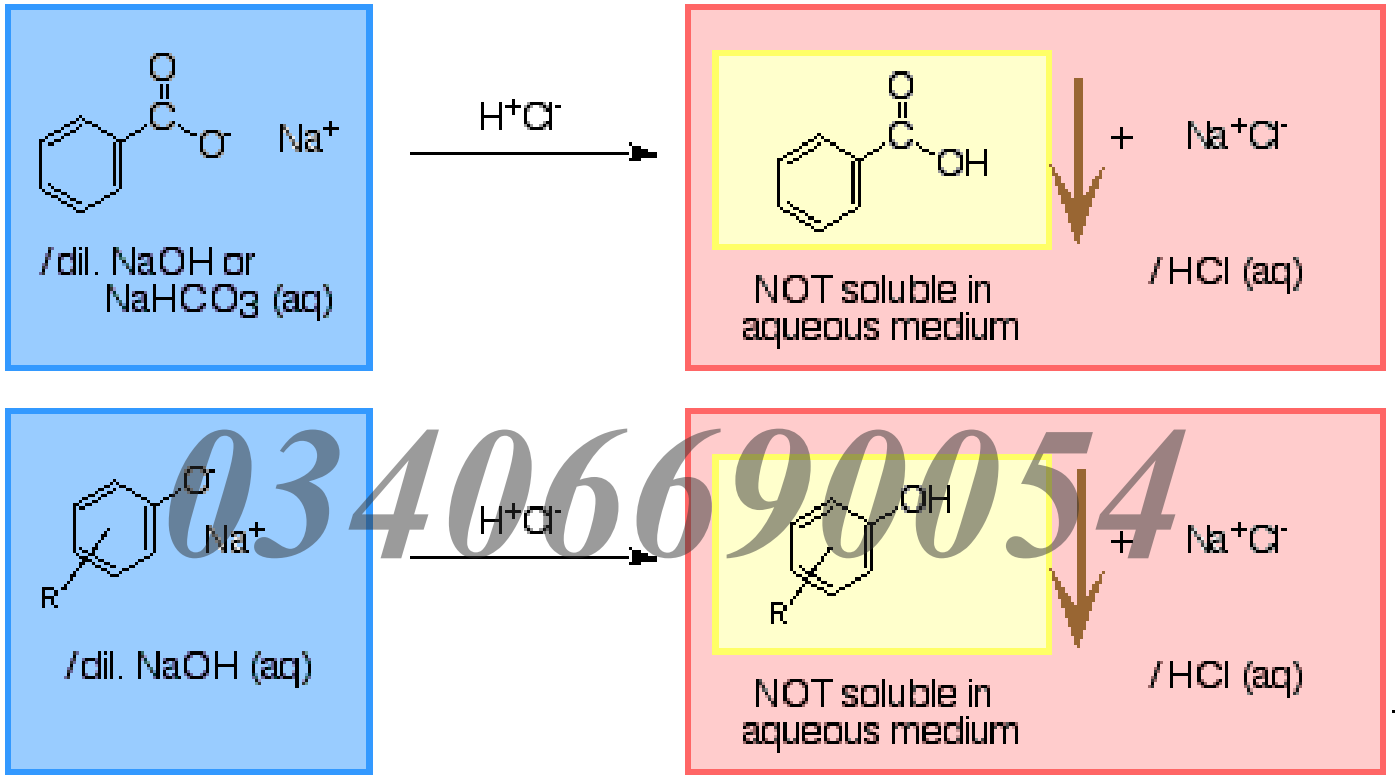
Recovery of acids and bases from salts

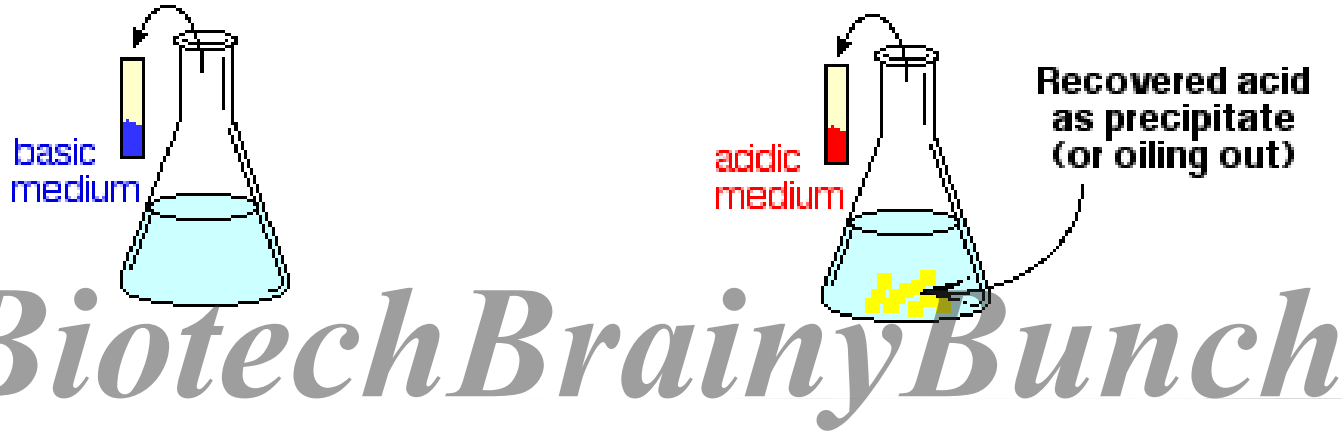
- Acids can be recovered from their salts by treating with HCl
- Decrease pH to acidic range~2
- For salts of bases we have use solution of a base to increase pH~10



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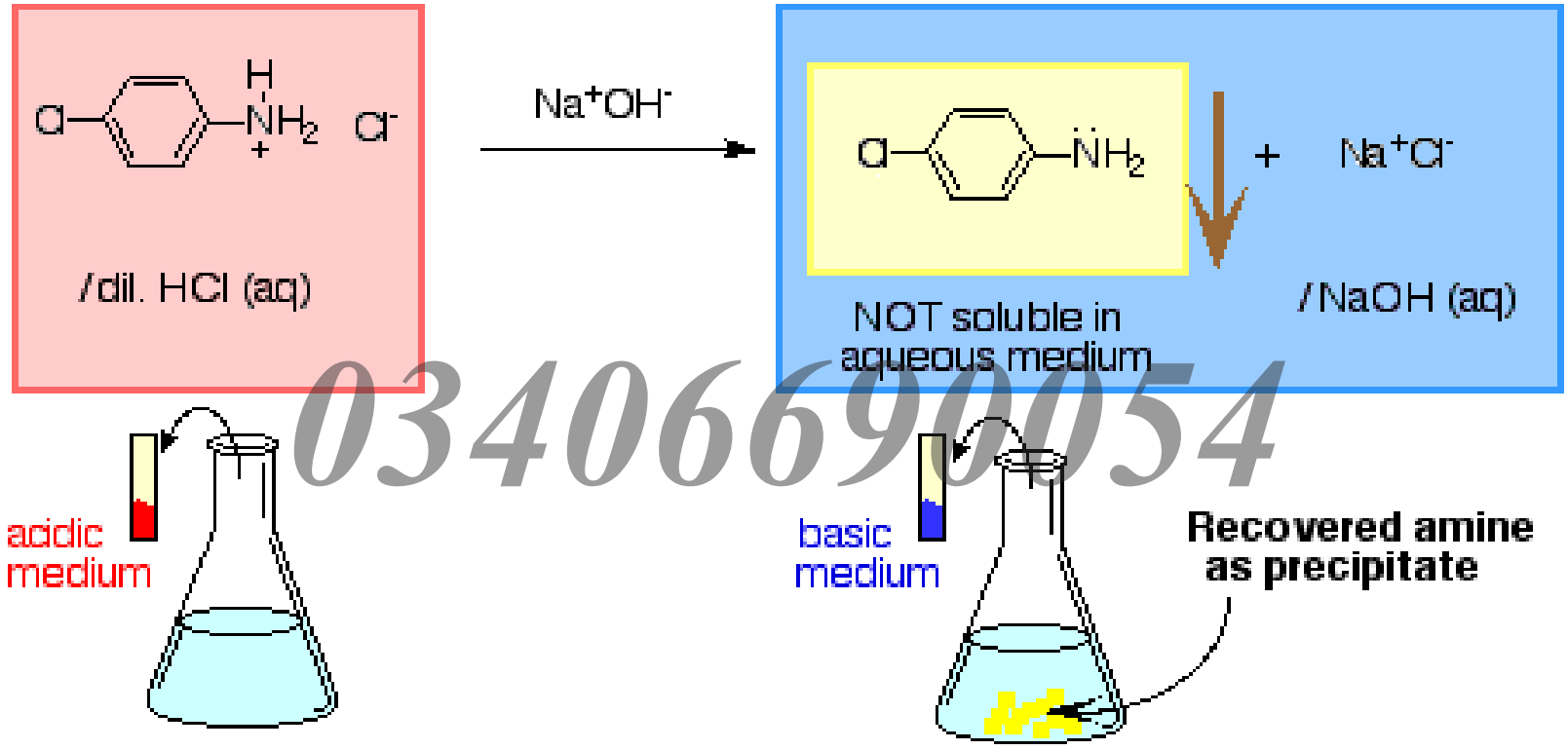
Solvent Extraction





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Solvent Extraction





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Solvent Extraction

Applications:

- Extraction of drugs from formulation to conduct assay.
- Extraction of benzoic acid (preservative) from food samples.
- Extraction of food colours.
- Extraction of oil and grease from waste water.

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Instrumentation

Solid Phase Extraction-I

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Solid Phase Extraction

- Solid-phase extraction (SPE) is a sample preparation process by which compounds that are dissolved or suspended in a liquid mixture are separated from other compounds in the mixture according to their physical and chemical properties.

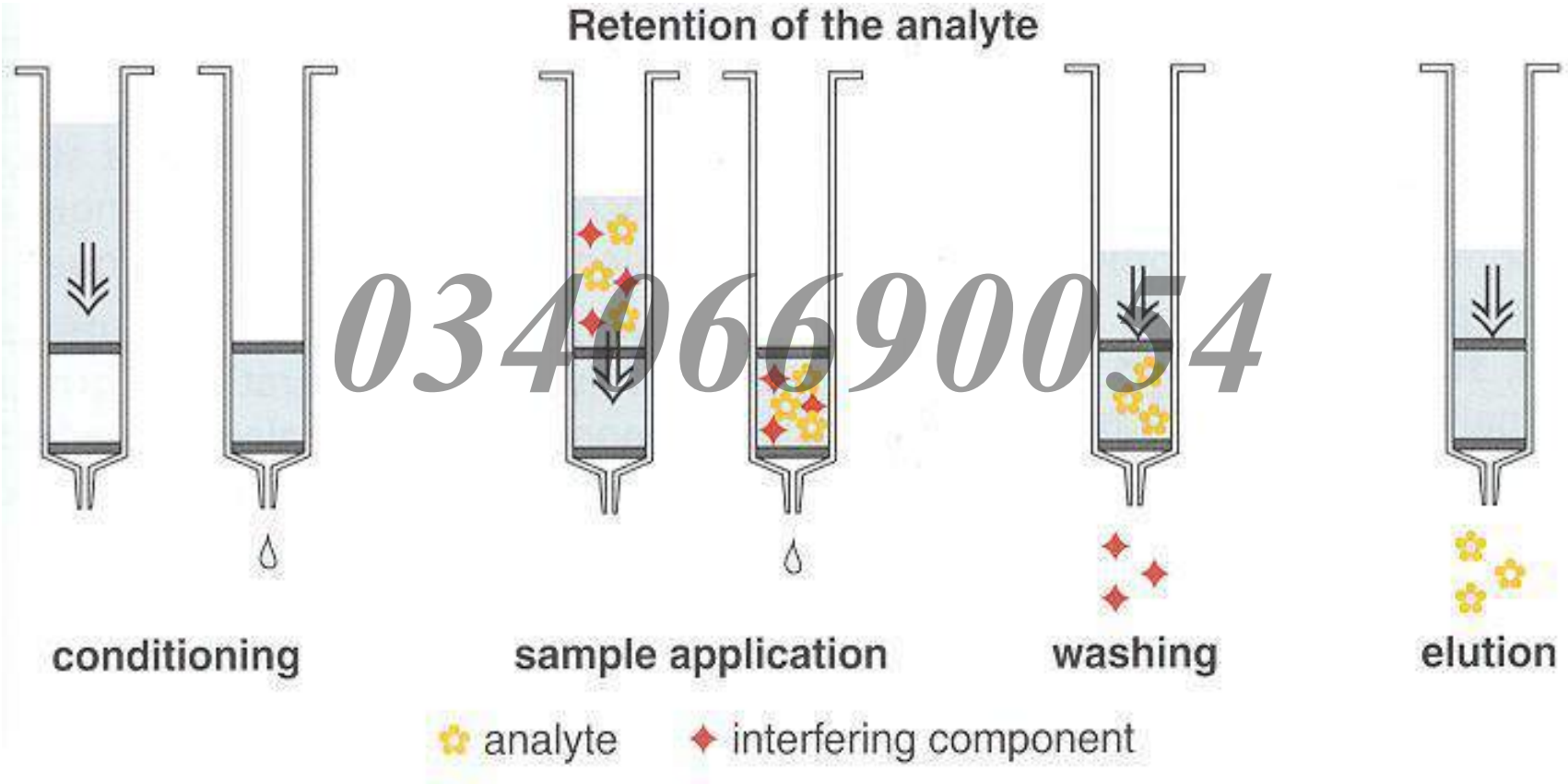
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Solid Phase Extraction



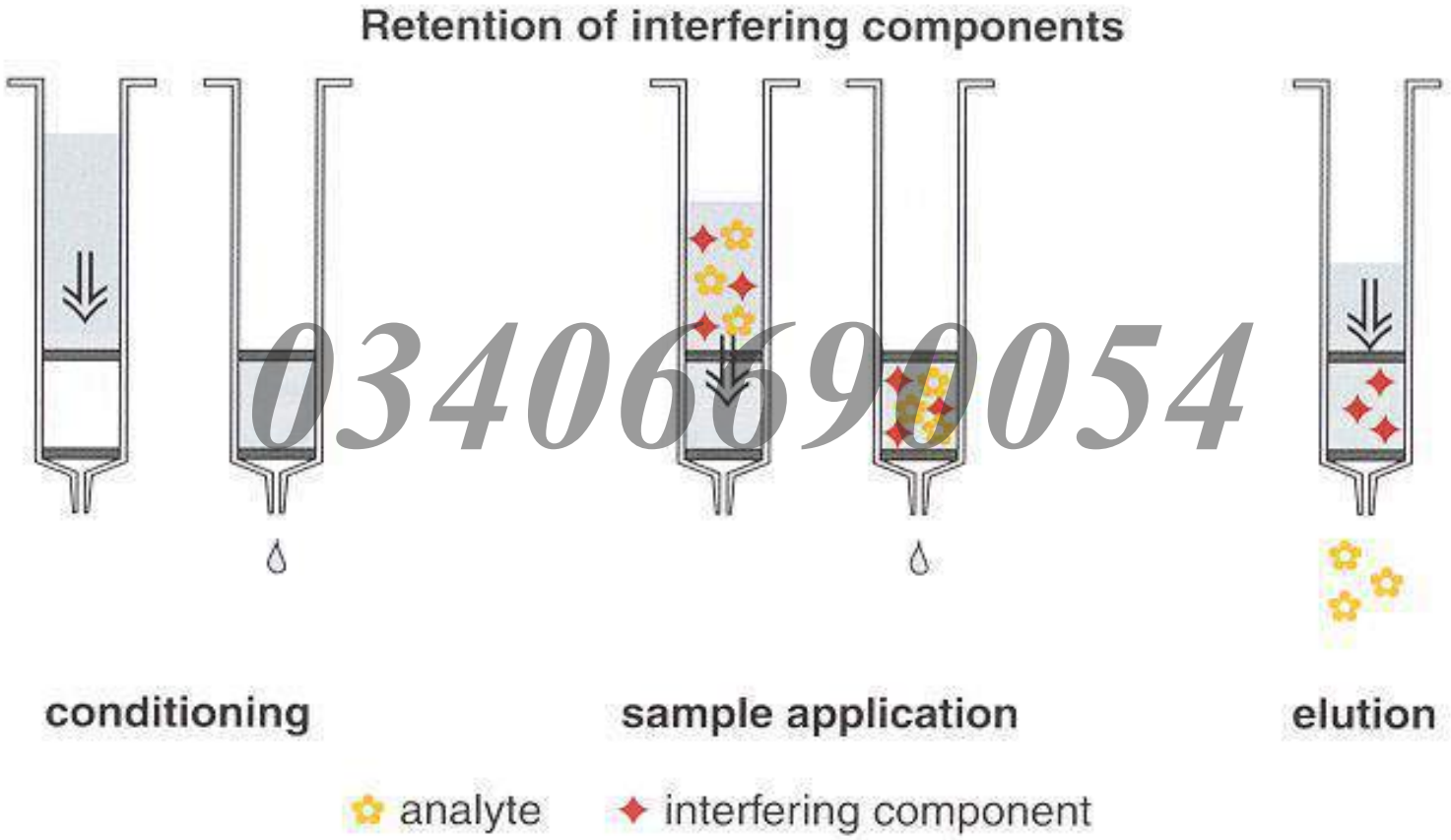
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Solid Phase Extraction



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Solid Phase Extraction



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Solid Phase Extraction

- Sorbent: The solid phase use for extraction
- Matrix: Our sample
- Analyte: A compound or type of compounds
- Washing: The solvent used for washing out the interferences
- Elution solvent: The solvent use to elute out the analyte

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Instrumentation

Solid Phase Extraction-II

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Solid Phase Extraction

Types of Sorbent:

- Polar (Normal Phase)
- Non-Polar (Reverse Phase)
- Ion Exchange
- Copolymeric
- Size exclusion
- Chiral

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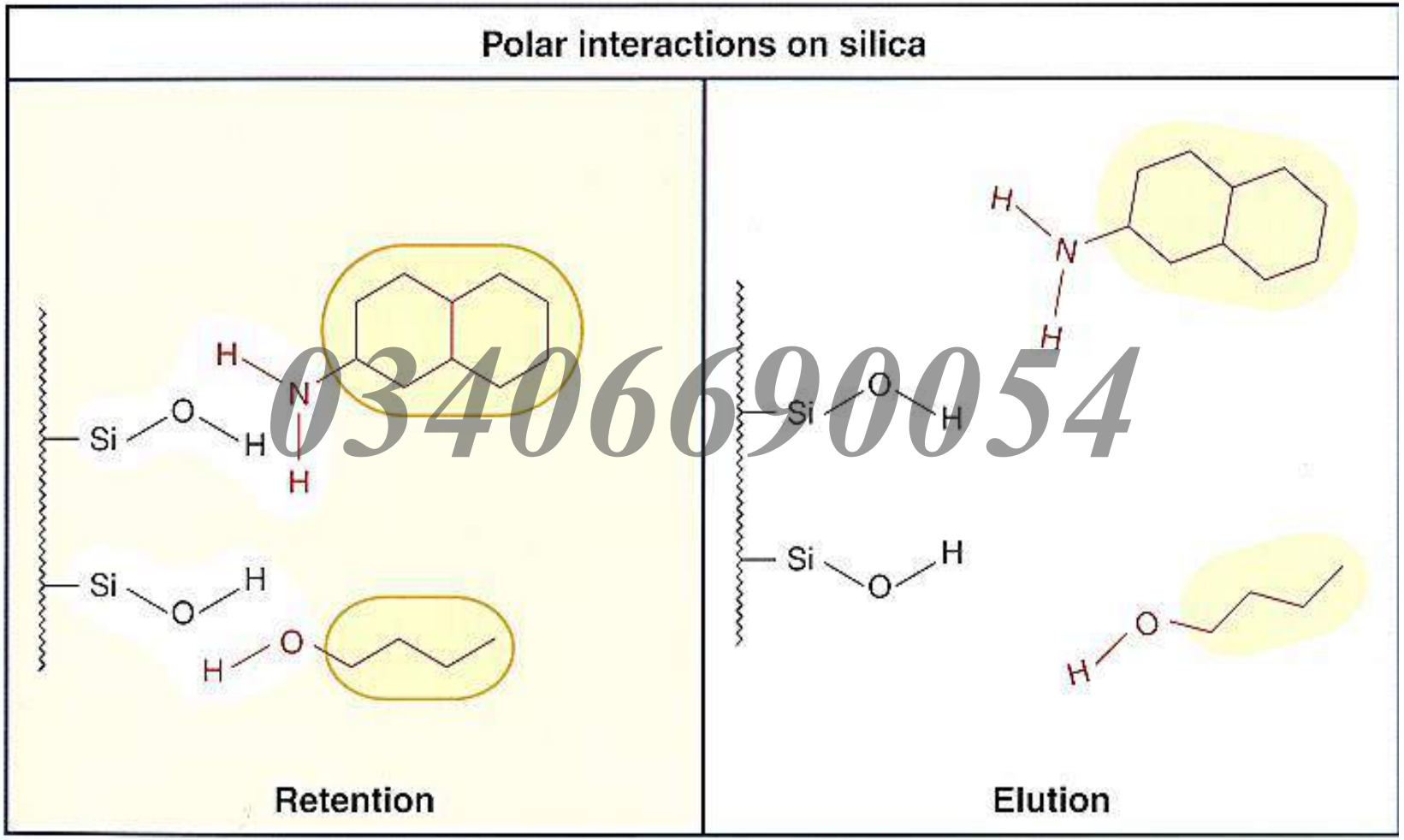
Solid Phase Extraction

POLAR INTERACTION

- Also called hydrophilic or normal phase
- Involves hydrogen bonding, pi-pi and dipole/ dipole interactions
- Sorbents - silica, diol, diethylamino, cyanopropyl
- Applications - lipids, oil additives, carbohydrates, phenols, oil soluble vitamins
- Analytes - amines, hydroxyls, carbonyls, aromatic rings, heteroatoms (O, S, N, P)
- Matrix - non-polar, organic
- Elution solvents - medium to high polarity

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Solid Phase Extraction



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Solid Phase Extraction

Applications:

- Vegetable oil, ointment, cereals, crude oil products, soil and sludge may use for pretreatment; dissolution of sample in non-polar solvent and removal of polar interfering compounds are needed before application.

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Instrumentation

Solid Phase Extraction-III

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Solid Phase Extraction

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Non-Polar Sorbent:

- It is also called reverse phased.
- Where polar silica's ends are replaced with non-polar carbon chain.
- Same as in HPLC.

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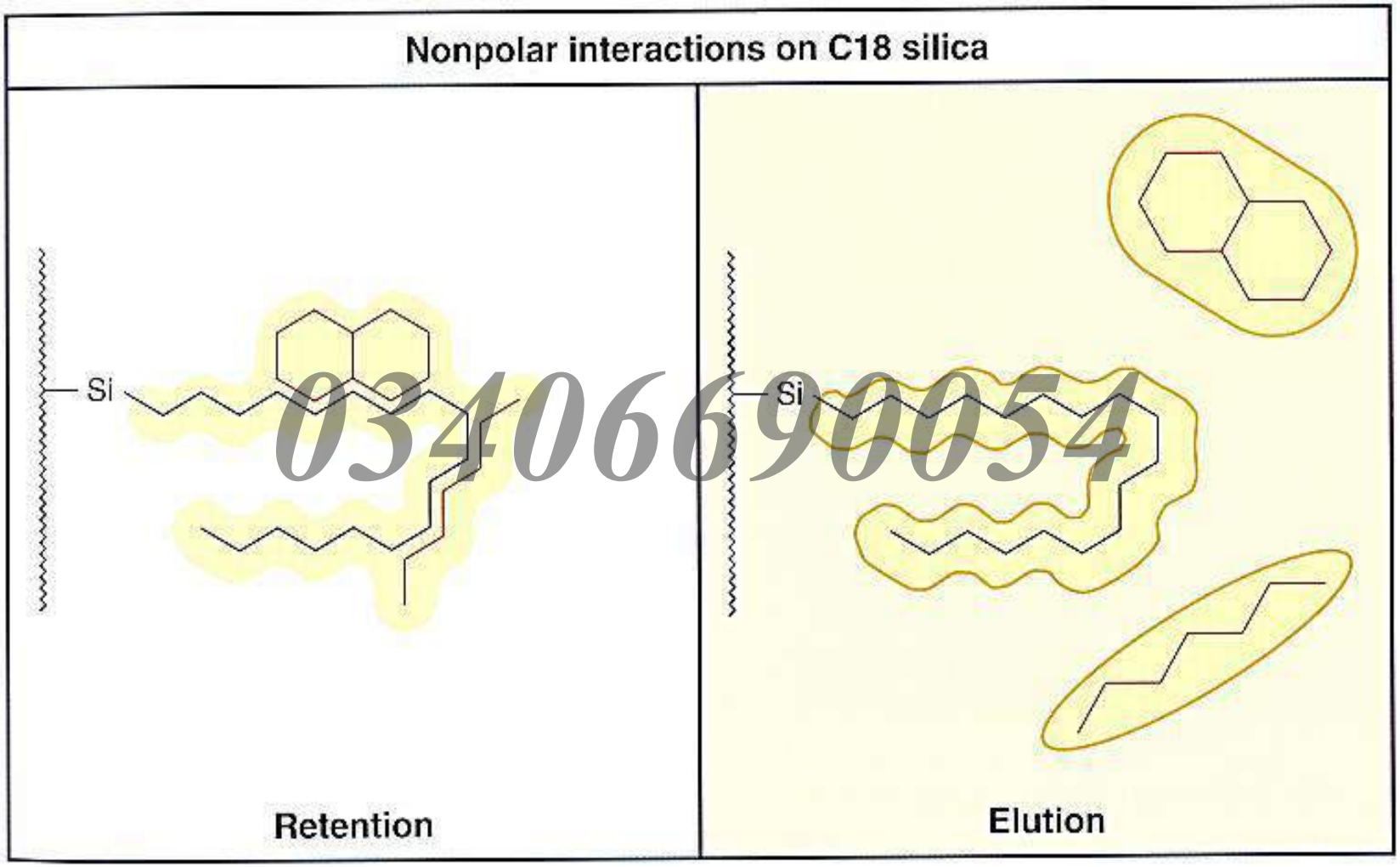
Solid Phase Extraction

Non-POLAR INTERACTIONS

- Also called hydrophobic or reverse phase
- Involves van der Waals / dispersion forces
- Sorbents - C2, C3, C4, iC4, tC4, C5, C6, C7, C8, C10, C12, C18, C20, C30 phenyl and cyclohexyl
- Applications - drugs of abuse, pesticides
- Analytes - protonated / neutral state, aromatics & alkyl chains
- Matrix - biologicals, water, aqueous buffers
- Elution solvents - typically non-polar to moderately polar

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Solid Phase Extraction



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Solid Phase Extraction

Applications:

- **Sample containing vitamins, steroids, pesticides, aflatoxins, antibiotics and other class of drugs may be isolated from hospital waste, body fluids or irrigation water using this technique.**

END

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Instrumentation

Solid Phase Extraction-IV

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Solid Phase Extraction

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Ion exchange sorbent:

- Cation exchange resins are negatively charged and adsorb positively charged cations from matrix.
- Anion exchange resins are positively charged and adsorb negatively charged conjugate bases from matrix.

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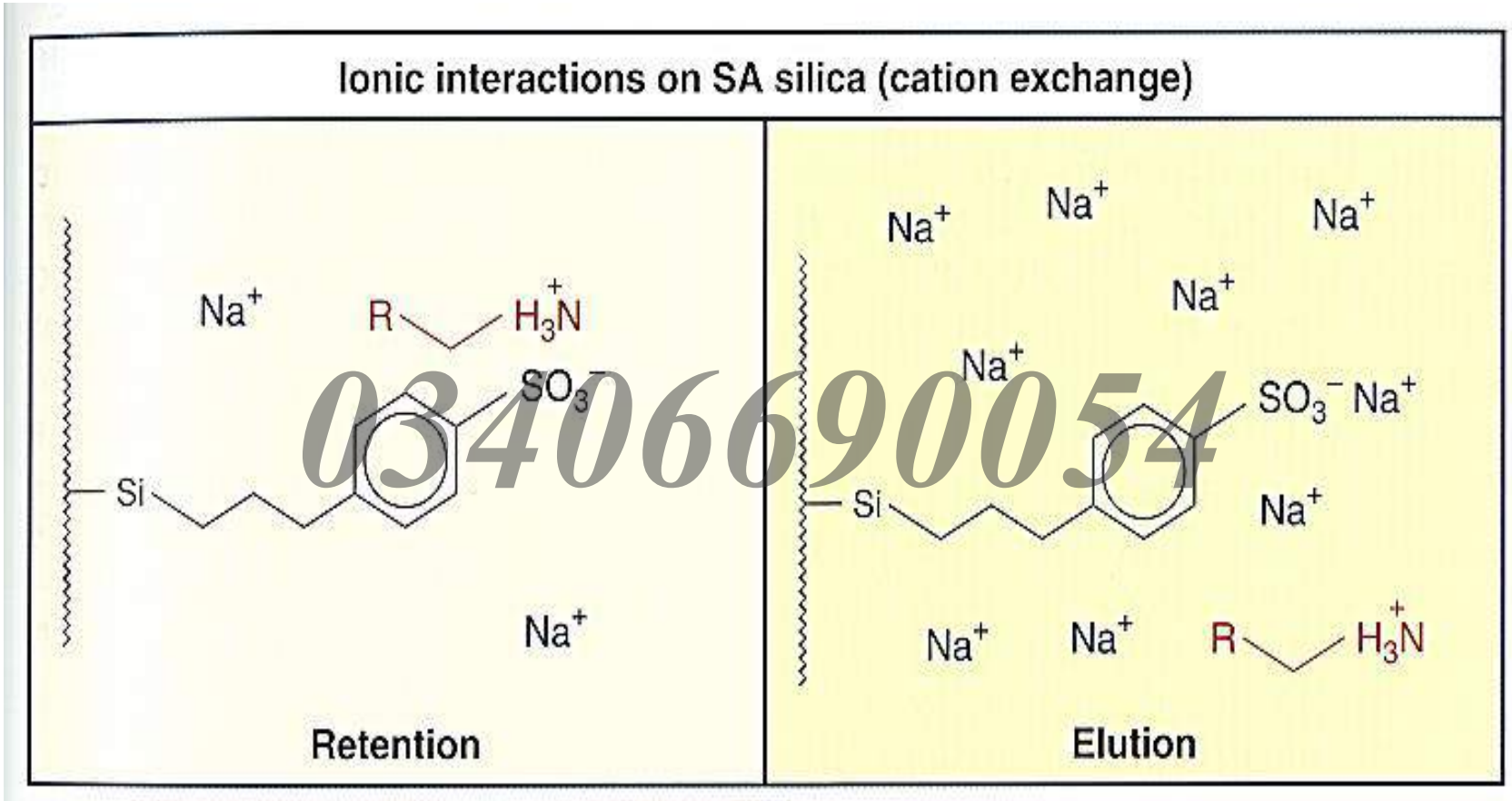
Solid Phase Extraction

CATION EXCHANGE SORBENTS

- Cation exchange sorbents are negatively charged
- Basic analytes manipulated to carry positive charge
- Opposites attract forming strong bonds
- Sorbents
  - Benzenesulfonic acid (strong)
  - Propylsulfonic acid (strong)
  - Carboxylic acid (weak)
- Applications: include basic drugs, catecholamines, pharmaceuticals, herbicides
- Analytes: Amines, Pyrimidines (cations)
- Matrix: aqueous
- Elution: Basic elution solvents to neutralize analyte

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Solid Phase Extraction



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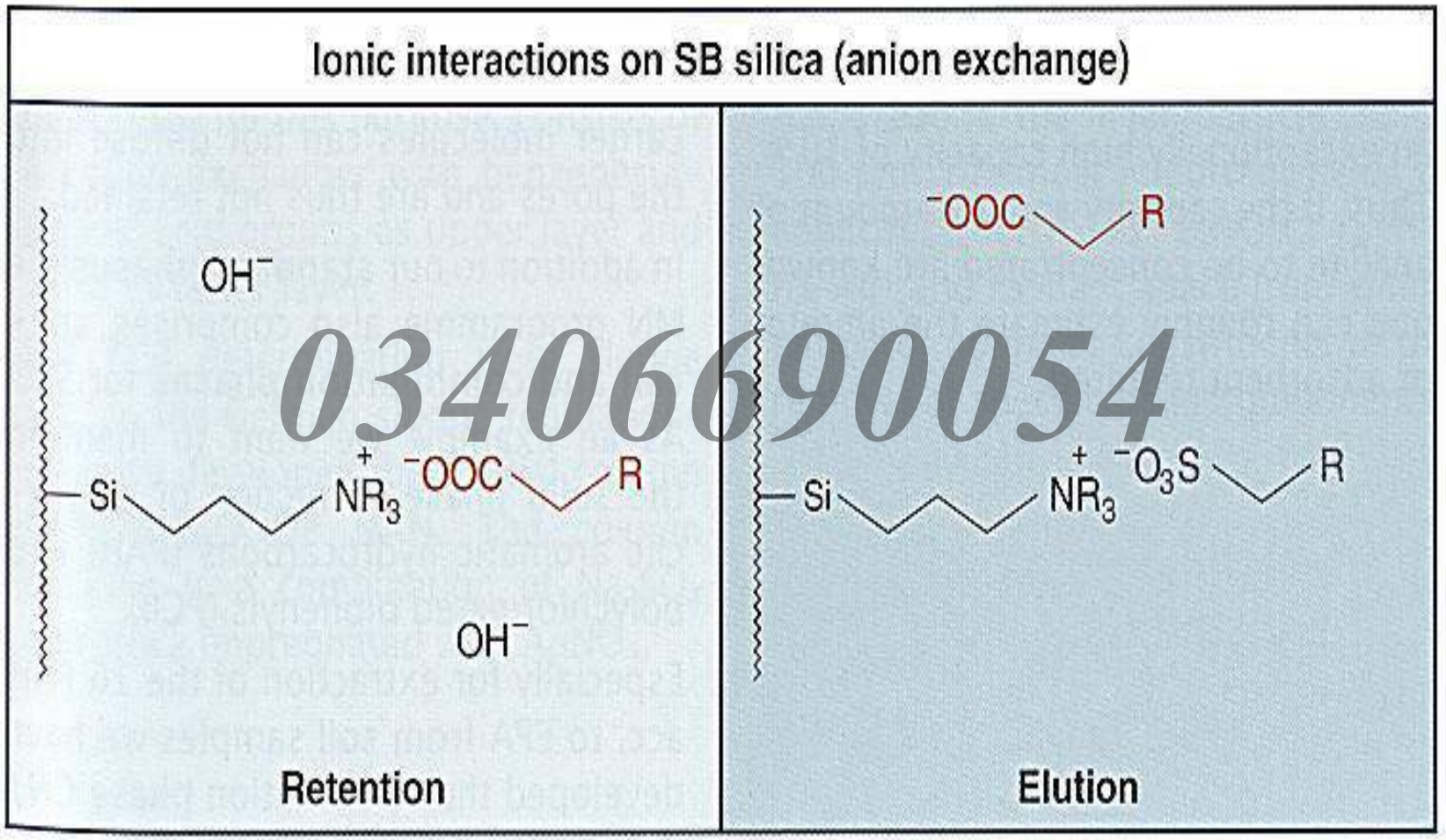
Solid Phase Extraction

ANION EXCHANGE SORBENTS

- Anion exchange sorbents positively charged
- Acidic analytes manipulated to carry negative charge
- Opposites attract forming strong bonds
- Sorbents
  - 1°, 2° amine
  - Aminopropyl (weak)
  - Quaternary amine (strong)
  - Diethylamino (weak)
- Applications: include phosphates, acidic drugs, organic acids, fatty acids, vitamins
- Analytes: Phosphates, Carboxylic acids, Sulfonic acids
- Matrix - aqueous
- Acidic elution solvents to neutralize analyte

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Solid Phase Extraction



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Solid Phase Extraction

Applications:

- **Sample containing vitamins, caffeine, saccharine, amino acids, organic acids from effluent waste, textile and leather products or food sample can be extracted using this technique.**

END

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Instrumentation

Solid Phase Extraction-V

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Solid Phase Extraction

Copolymeric sorbent:

- Hydrophobic & ionic retention mechanisms
- Reverse phase sorbent with cation or anion exchange
- Acidic, basic & neutral analyte applications
- Matrix - aqueous
- Elution solvents mixture of organics with acid or base

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Solid Phase Extraction

SIZE EXCLUSION

- Porous silica or polymers, where separation is based on differences between the volumes of the molecules, corresponding to steric exclusion

CHIRAL CHROMATOGRAPHY

- Special chemically modified stationary phases, for example cellulose or amylose derivatives, proteins or peptides, cyclodextrins etc., for the separation of enantiomers

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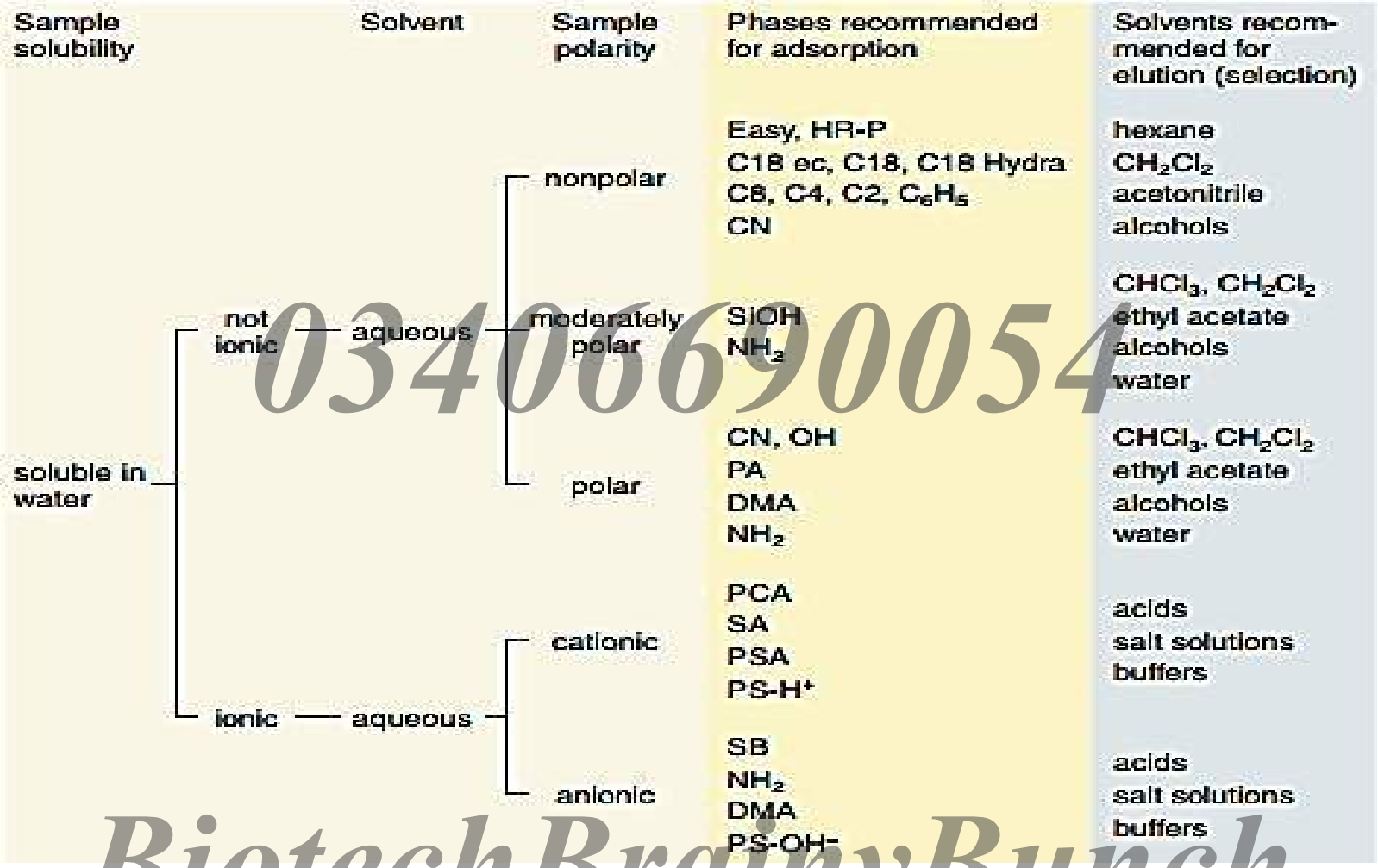
- Most separations are based upon partition mechanisms utilizing chemically modified silica as the stationary phase and polar solvents as the mobile phase. The surface of the support, for example the silanol groups of silica, is reacted with various silane reagents to produce covalently bound silyl derivatives covering a varying number of active sites on the surface of the support. The nature of the bonded phase is an important parameter for determining the separation properties of the chromatographic system.

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Solid Phase Extraction

Selection guide for SPE phases and solvents



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Solid Phase Extraction

Selection guide for SPE phases and solvents

Sample solubility	Solvent	Sample polarity	Phases recommended for adsorption	Solvents recommended for elution (selection)
soluble in organic solvents	aqueous	nonpolar	Easy, HR-P C18 ec, C18, C18 Hydra C8, C4, C2, C_6H_5 CN PS-RP	hexane CH_2Cl_2 acetonitrile alcohols
	organic	moderately polar	SiOH NH_2	$CHCl_3$, CH_2Cl_2 ethyl acetate alcohols
	organic	polar	CN, OH PA DMA NH_2	$CHCl_3$, CH_2Cl_2 ethyl acetate alcohols

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Solid Phase Extraction

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Method Selection Guidelines:

- **Based on above sketch we can easily frame the methodology of extraction for different kind of analytes.**

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Instrumentation
UV Spectroscopy-I

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UV Spectroscopy

Introduction

- Electromagnetic radiation
- Wavelength and energy
- Electronic transitions in organic molecules
- Conjugated systems and colours
- Chromophores and auxochrome
- Beer Lambert Law; explanation and application

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UV Spectroscopy

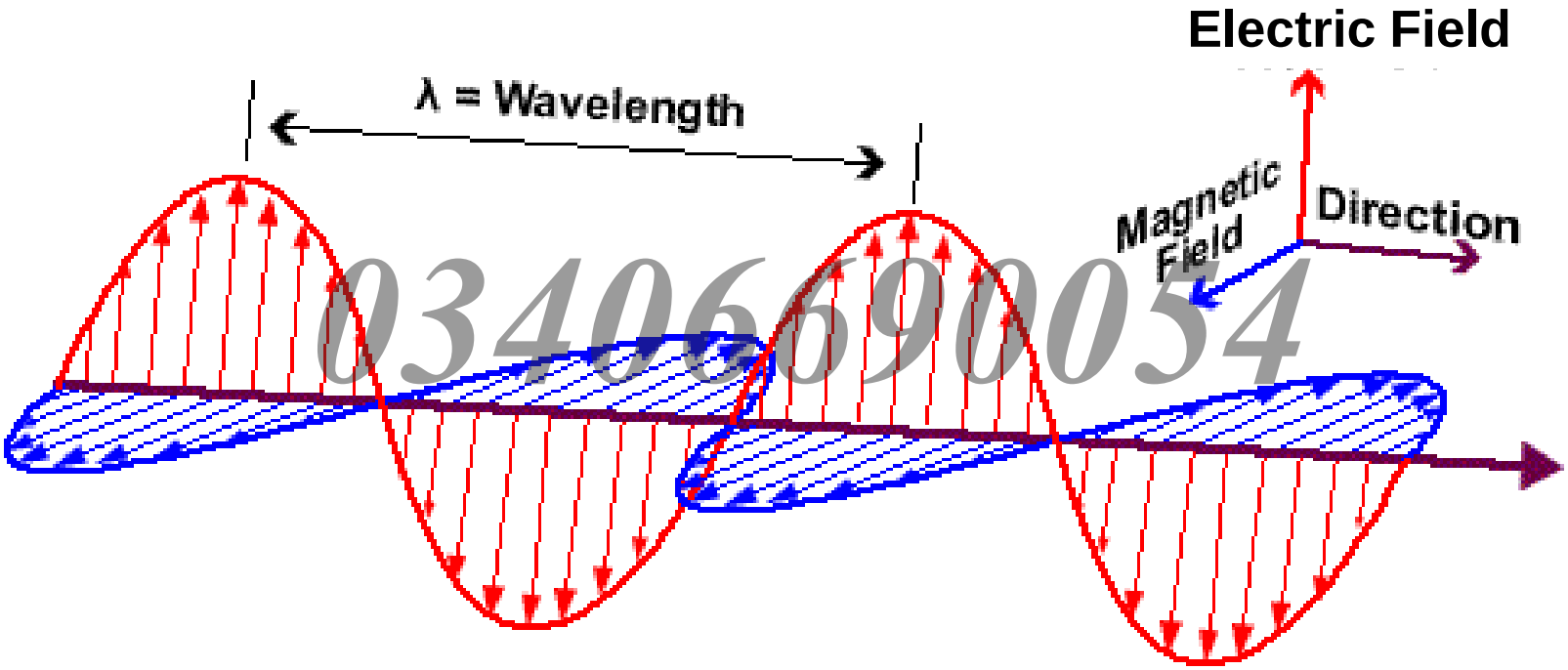
Introduction

- Effect of substituents on UV Spectrum
- Schematic diagram of UV Spectrophotometer
- Woodward Fieser Rule (theoretical calculation of wavelength maxima) for conjugated diene and α,β-Unsaturated diene systems.

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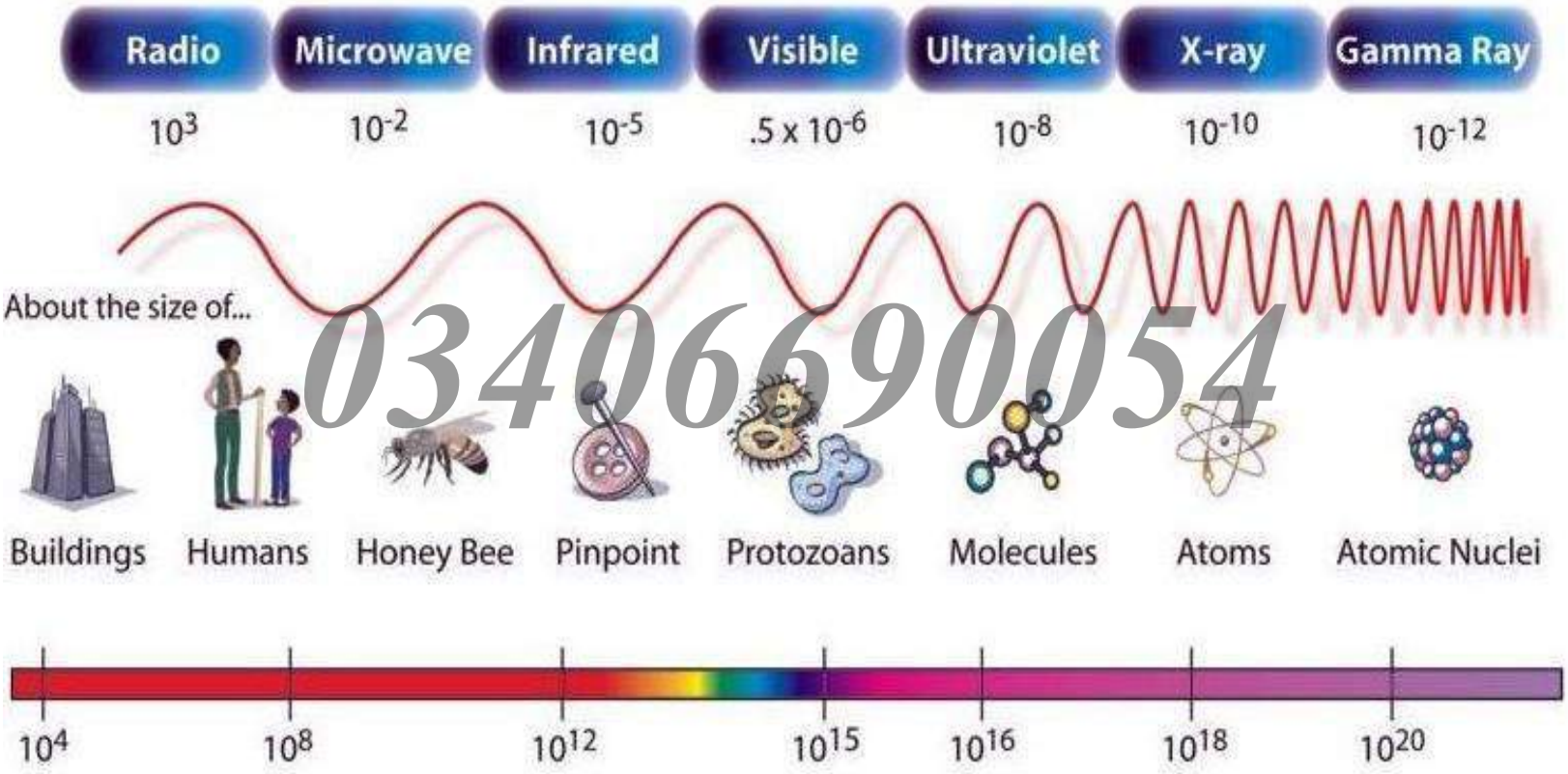
UV Spectroscopy

Electromagnetic radiation



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UV Spectroscopy



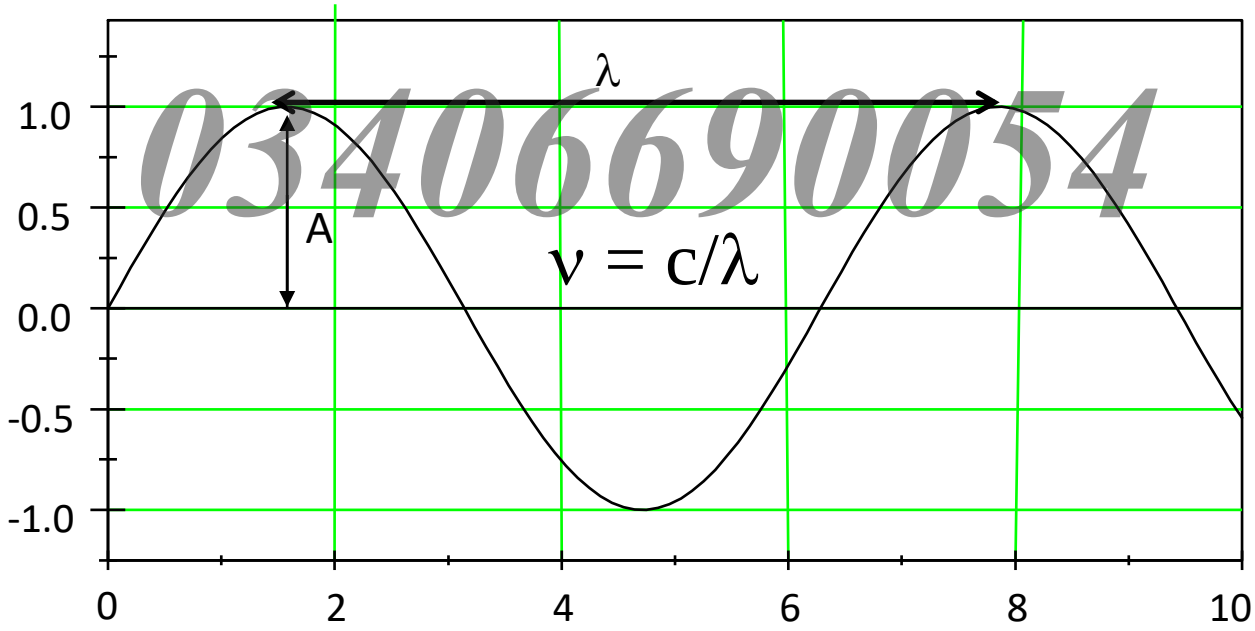
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UV Spectroscopy

λ = wavelength (nm / cycle, cm / cycle)

ν = frequency (cycle / second)

c = speed of light ($3.0 * 10^{10}$ cm / sec)



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UV Spectroscopy

- Amplitude (A) of the sinusoidal wave is defined as the length of the electrical vector at the maximum in the wave.
- wavelength (λ) is the linear distance between successive maxima or minima of a wave.
- Frequency (ν) refers to the number of oscillations of the field per second.

END

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Instrumentation
UV Spectroscopy-II

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UV Spectroscopy

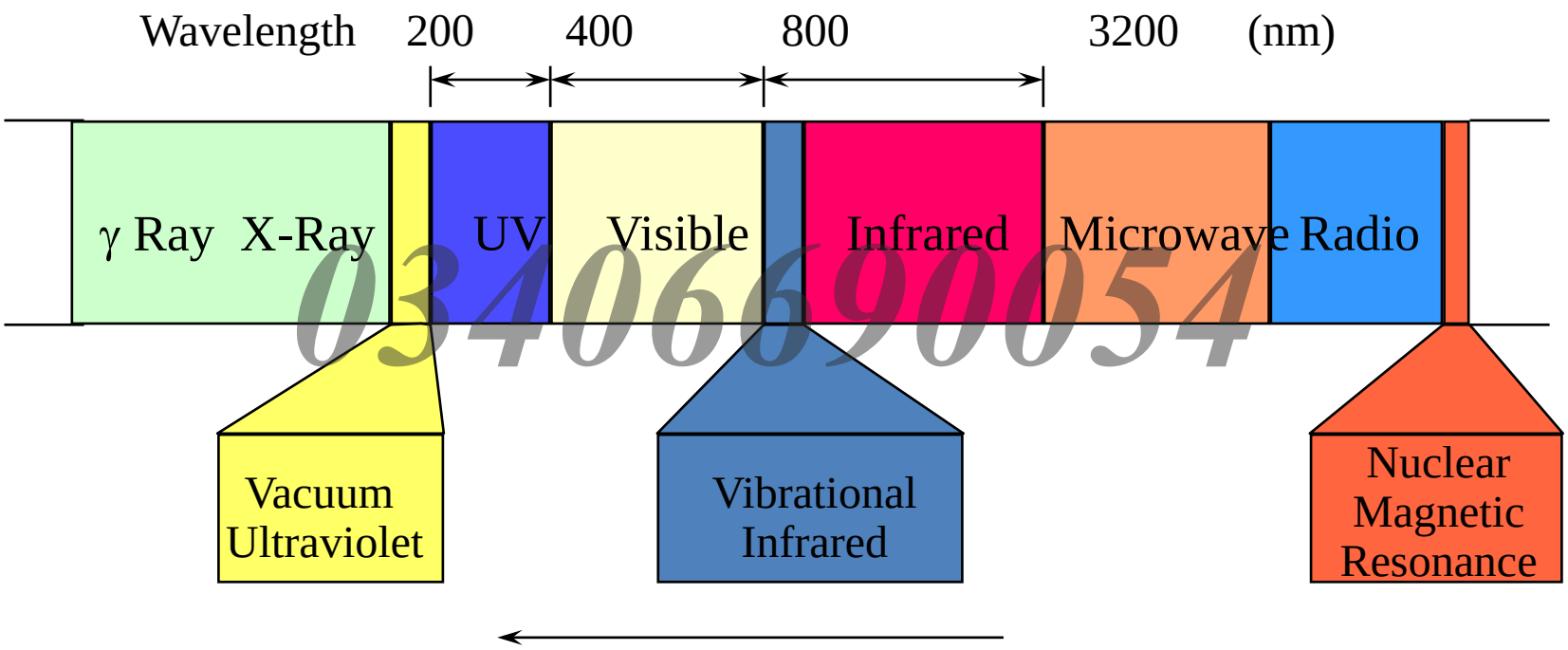
Wavelength and Energy relation

- $E = h\nu = hc/\lambda$
- Where E = energy
- h = Plank's constant
- Energy and frequency are directly proportional
- Wavelength and energy are inversely proportional

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UV Spectroscopy

Electromagnetic radiations



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UV Spectroscopy

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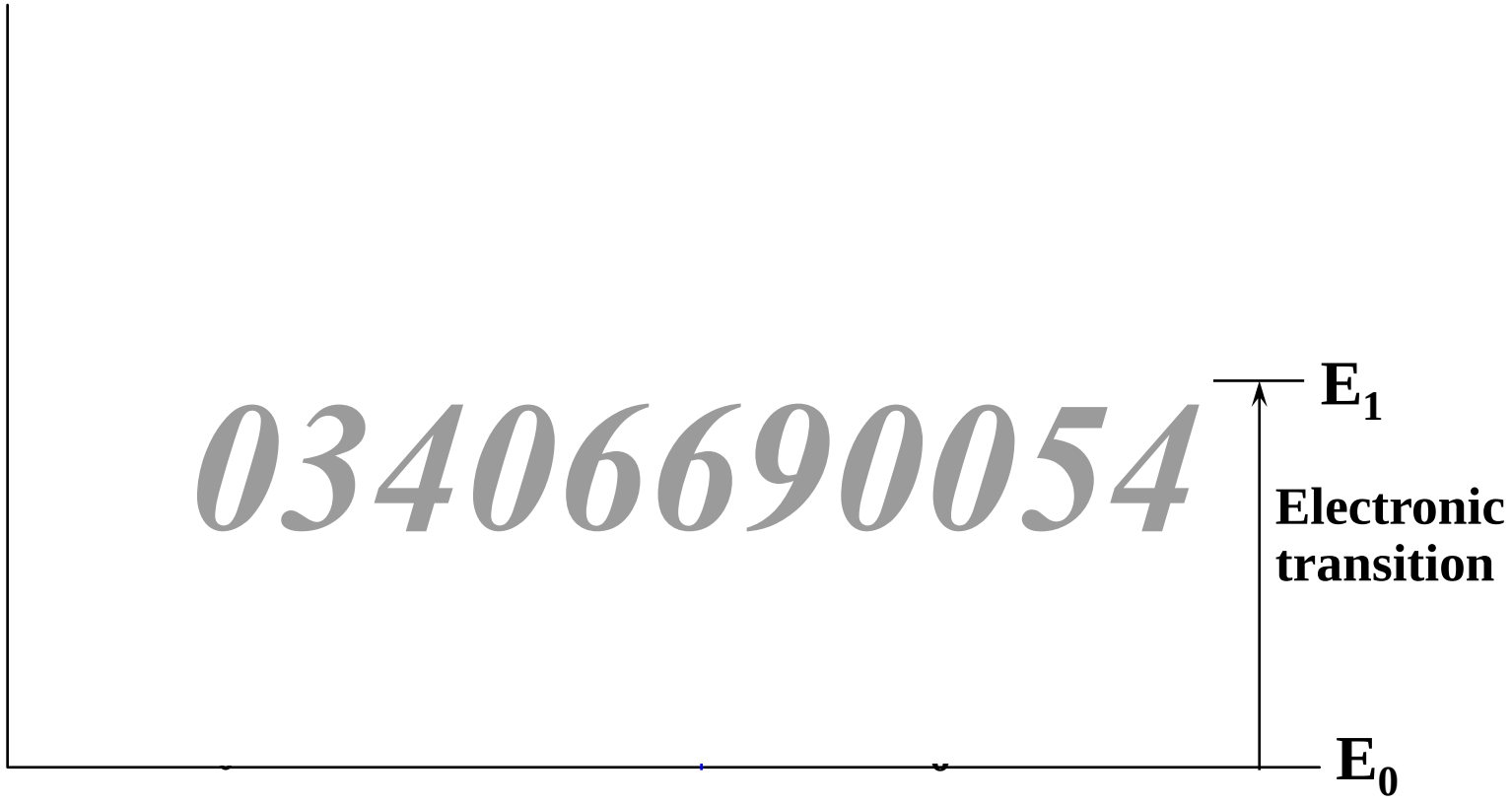
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Absorption of light by molecules or ions causes three types of energy changes:

- **Rotational (Low energy change): molecule rotates around a center of gravity**
- **Vibrational (Low energy change): average internuclear distance of two or more atoms in the molecule changes**
- **Electronic (High energy change): change in energy of the electrons or transitions**

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UV Spectroscopy



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E = hν in which h is Planck's constant (6.626 * 10^{-34} Joules and ν is the frequency of the radiation

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UV Spectroscopy

- UV radiation bring electronic transitions in molecules.
- Electromagnetic radiations of visible region bring electronic transition where we have conjugated system of colour bearing moieties.

END

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Instrumentation
UV Spectroscopy-III

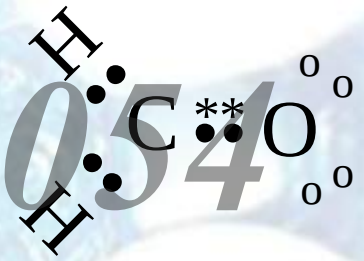
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UV Spectroscopy

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In organic compounds, three types of ground state electrons may be involved:

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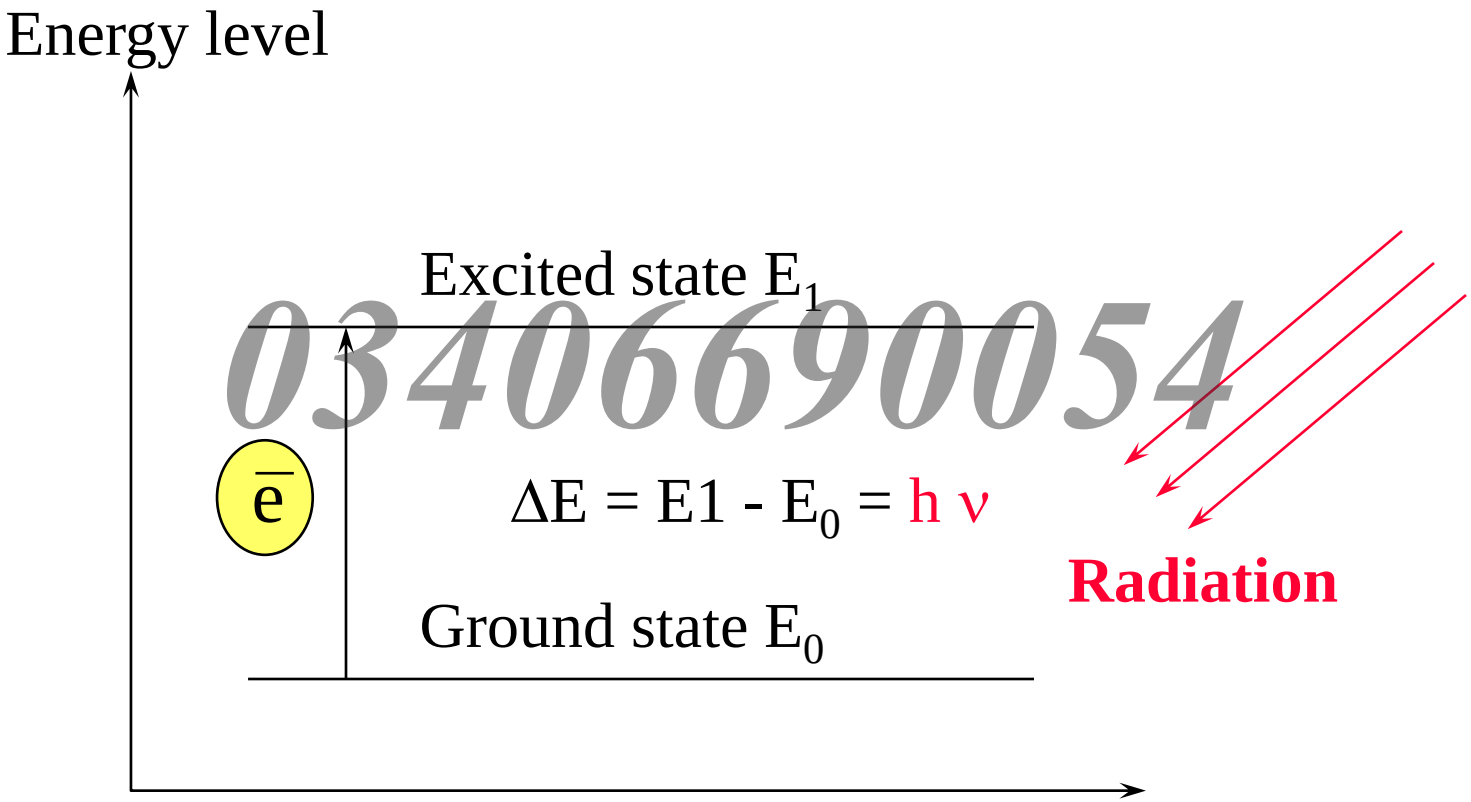


formaldehyde

- **σ (bonding) electrons**
- * **π (bonding) electrons**
- o **n (non-bonding) electrons**

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UV Spectroscopy

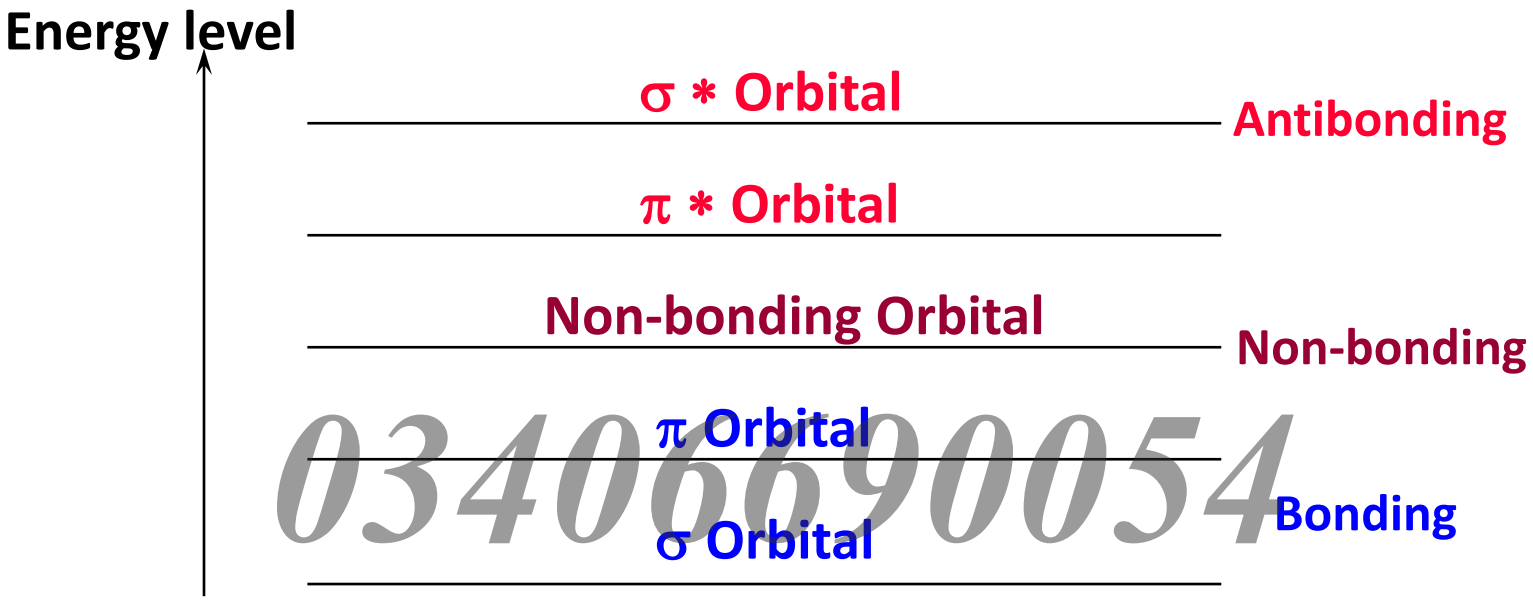


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UV Spectroscopy



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σ ⟶ σ* Transition : Compounds with single bonds

n ⟶ σ* Transition : O, N, S and halogen compounds

n ⟶ π* Transition : Carbonyl compound

π ⟶ π* Transition : Alkenes, carbonyl, azo compounds, alkynes

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UV Spectroscopy

Wavelength and transitions:

- **For sigma to sigma transition required energy can be achieved with electromagnetic radiation of < 100 nm.**
- **Pi to pi transition is possible with 171 nm (vacuum UV range)**
- **For an absorption in normal UV range the molecule must have conjugated system or non-bonding electron along with double bond.**

END

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Instrumentation
UV Spectroscopy-IV

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UV Spectroscopy

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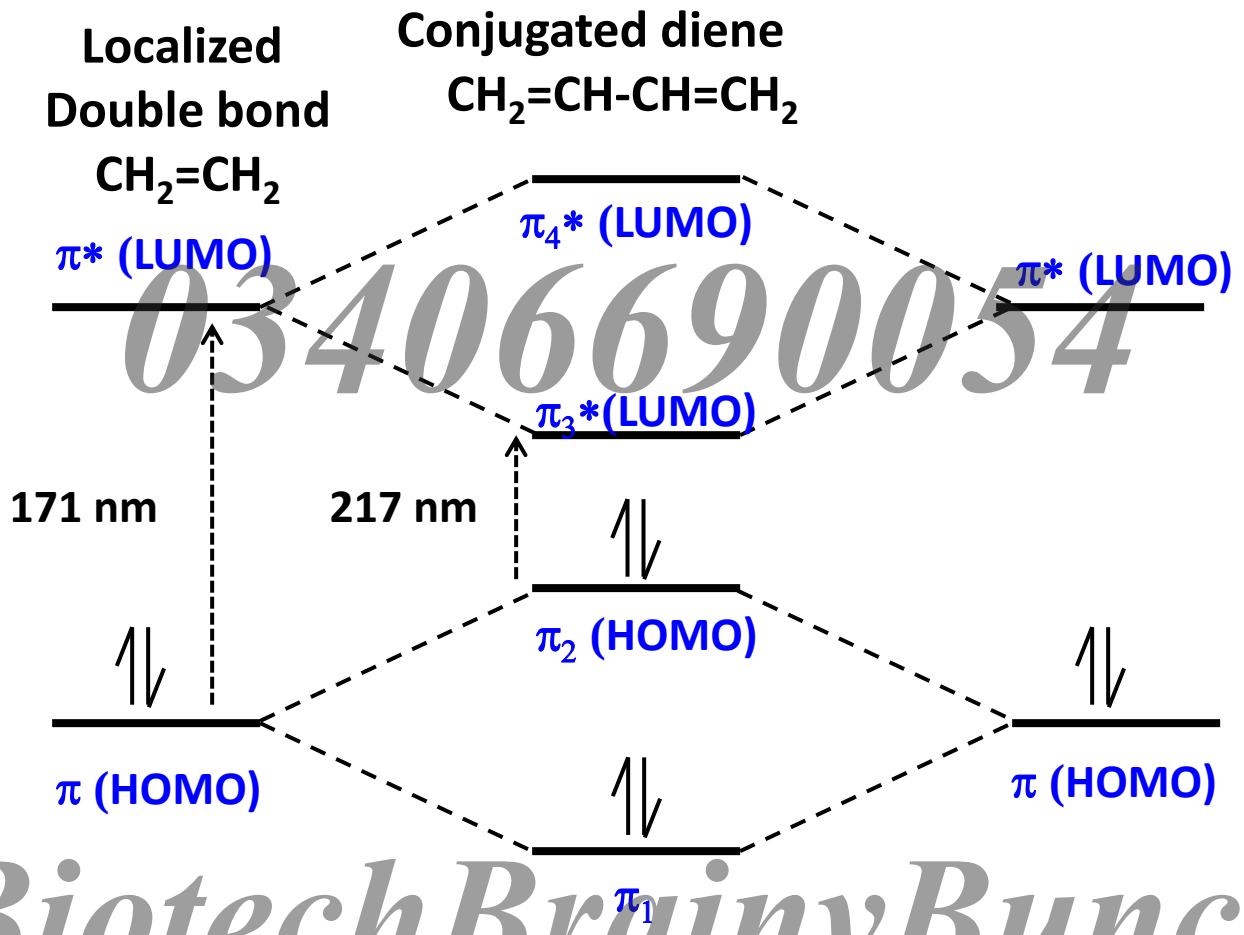
Why molecules display colours:

- Transition metals are coloured because of d-d transitions while organic molecules are coloured because of conjugation.
- Higher wavelength or lower energy.

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UV Spectroscopy

Transition in Conjugated diene system;



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UV Spectroscopy

Complementary Colours

Color Absorbed	Color observed	Absorbed radiation (nm)
Violet	Yellow-green	400-435
Blue	Yellow	435-480
Green-blue	Orange	480-490
Blue-green	Red	490-500
Green	Purple	500-560
Yellow-green	Violet	560-580
Yellow	Blue	580-595
Orange	Green-blue	595-605
Red	Blue-green	605-750

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UV Spectroscopy

Complementary colours:

- When white light falls upon a sample, either reflection or absorption takes place
- If only a portion of the light is absorbed, the color of the sample is determined by the reflected light (is called complementary colours).

END

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Instrumentation
UV Spectroscopy-V

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UV Spectroscopy

Chromophores

- Color-bearing functional group
- Exhibits a characteristic absorption spectrum in the UV / Vis region
- Chromophore conjugated with others, display more intense absorption band at a longer wavelength

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UV Spectroscopy

Chromophore	Example	λ_{max} (nm)	Type of transition
Alkene	$C_6H_{13}CH=CH_2$	177	$\pi \rightarrow \pi^*$
Alkyne	$C_5H_{11}C\equiv C\text{-}CH_3$	225	$\pi \rightarrow \pi^*$
Carbonyl	$CH_3\overset{\overset{O}{\|}}{C}CH_3$	186 280	$n \rightarrow \sigma^*$ $n \rightarrow \pi^*$
	$CH_3\overset{\overset{O}{\|}}{C}H$	180 293	$n \rightarrow \sigma^*$ $n \rightarrow \pi^*$
Carboxyl	$CH_3\overset{\overset{O}{\|}}{C}OH$	204	$n \rightarrow \pi^*$

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UV Spectroscopy

Chromophore	Example	Solvent	λ_{max} (nm)	Type of transition
Amido	CH_3CNH_2 (O double-bonded to C)	**Water**	214	n –> π*
Azo	$CH_3N{=}NCH_3$	**Ethanol**	339	n –> π*
Nitro	CH_3NO_2	**Isooctane**	280	n –> π*
Nitroso	C_4H_9NO	**Ethyl ether**	300 665	---- n –> π*
Nitrate	$C_2H_5ONO_2$	**Dioxane**	270	n –> π*

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UV Spectroscopy

Compound	Type	λ_{max} (nm)
$CH_3(CH_2)_2CH=CH_2$	**Olefin**	184
$CH_2=CH(CH_2)_2CH=CH_2$	**Diolefin (unconjugated)**	185
$H_2C=CHCH=CH_2$	**Diolefin (conjugated)**	217
$H_2C=CHCH=CHCH=CH_2$	**Triolefin (conjugated)**	250

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UV Spectroscopy

Auxochromes

- Functional group that does not absorb in the UV/Vis region
- It has the effect of shifting chromophore peaks to longer wavelength as well as increasing their intensities
- For example -OH, -OR, -NHR, -SH, -Cl, -Br, -I

END

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Instrumentation
UV Spectroscopy-VI

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UV Spectroscopy

Beer Lambert Law

$$\text{Absorbance A} = \log \frac{1}{T} = \varepsilon b c$$

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T : Transmittance

ε : Molar absorptivity

***b* : Path length**

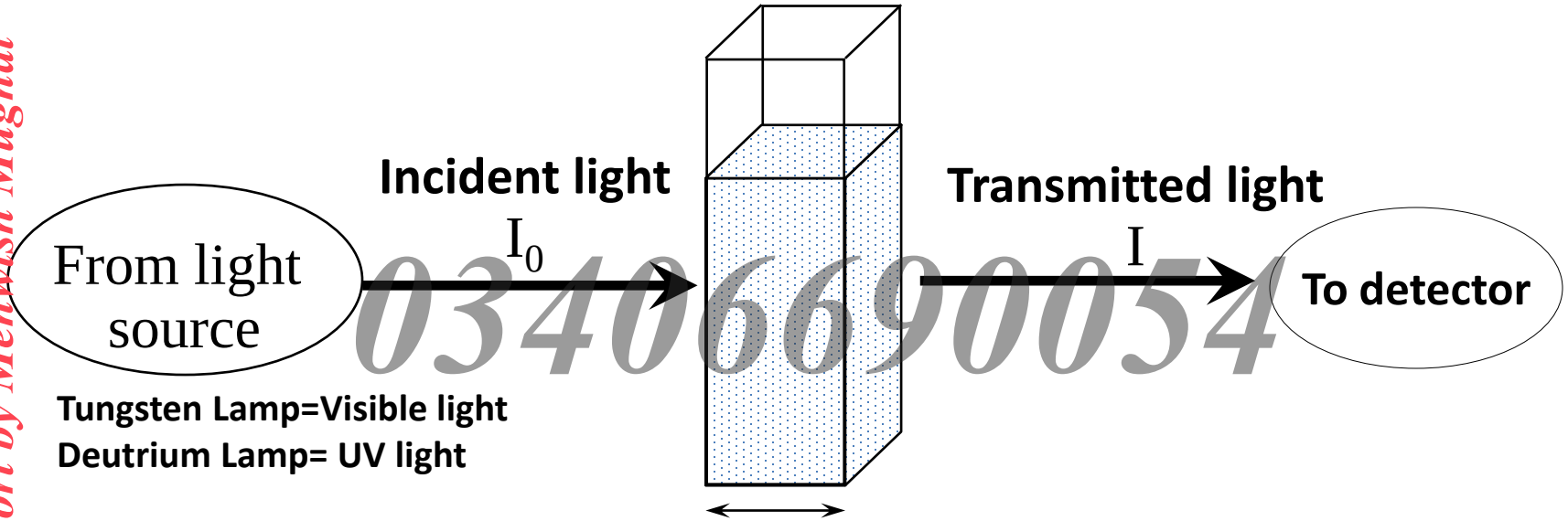
C : Analyte concentration

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UV Spectroscopy

Cuvette containing analyte



Tungsten Lamp=Visible light
Deutrium Lamp= UV light

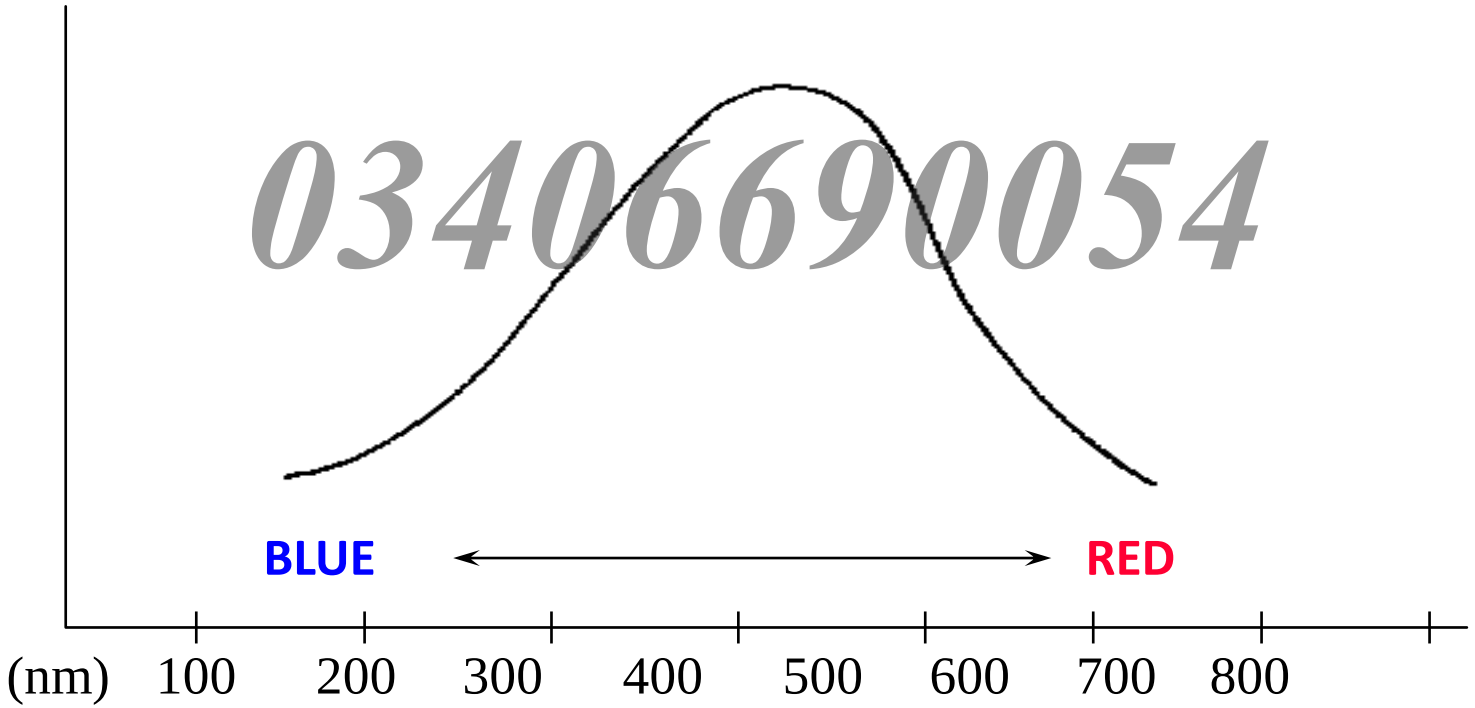
$$\text{Transmittance } T = \frac{I}{I_0}$$

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UV Spectroscopy

Effect of Constituents on UV Spectrum:

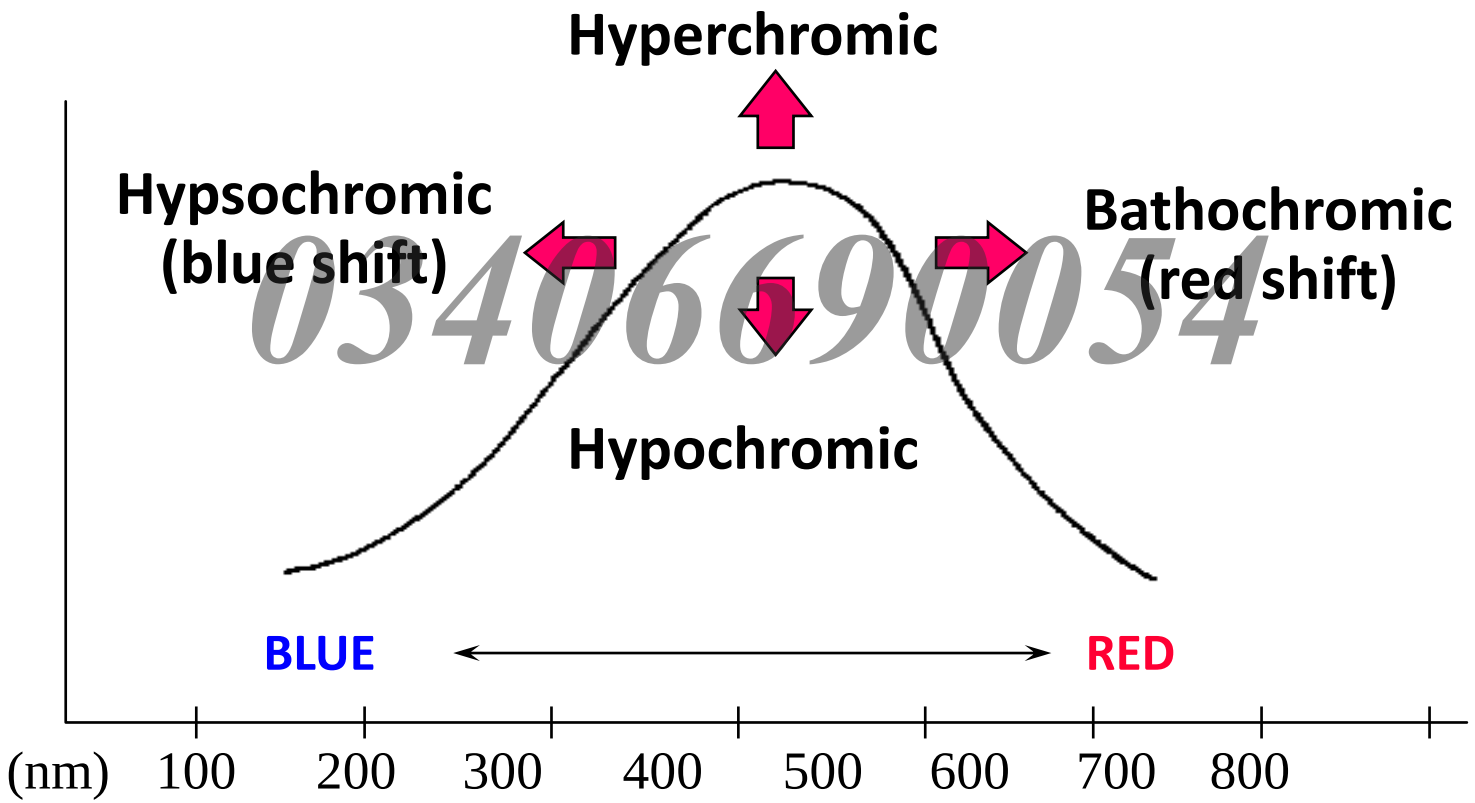
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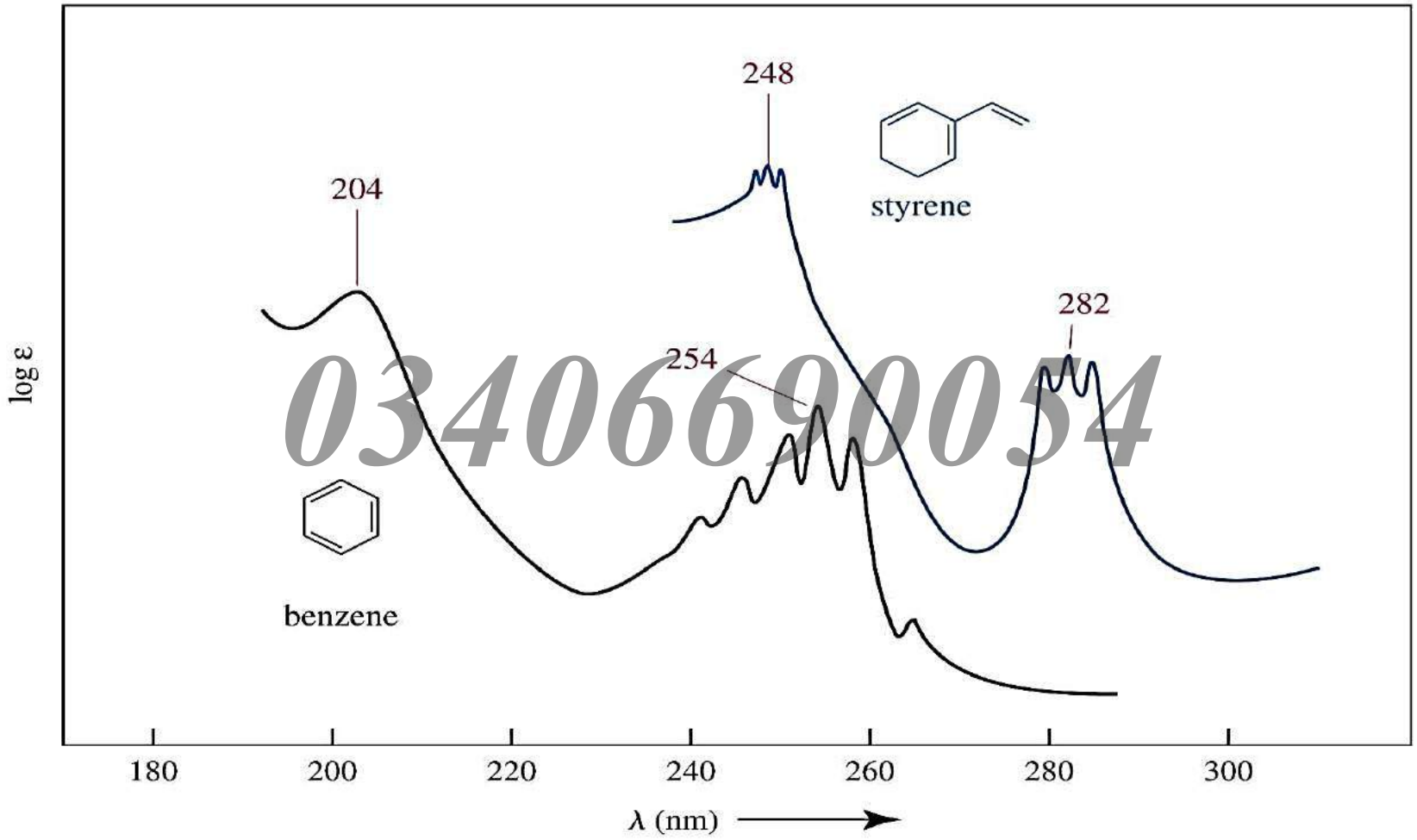
UV Spectroscopy

Effect of Constituents on UV Spectrum:



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UV Spectroscopy



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UV Spectroscopy

Dimethylpolyenes $CH_3 (CH = CH)_n CH_3$

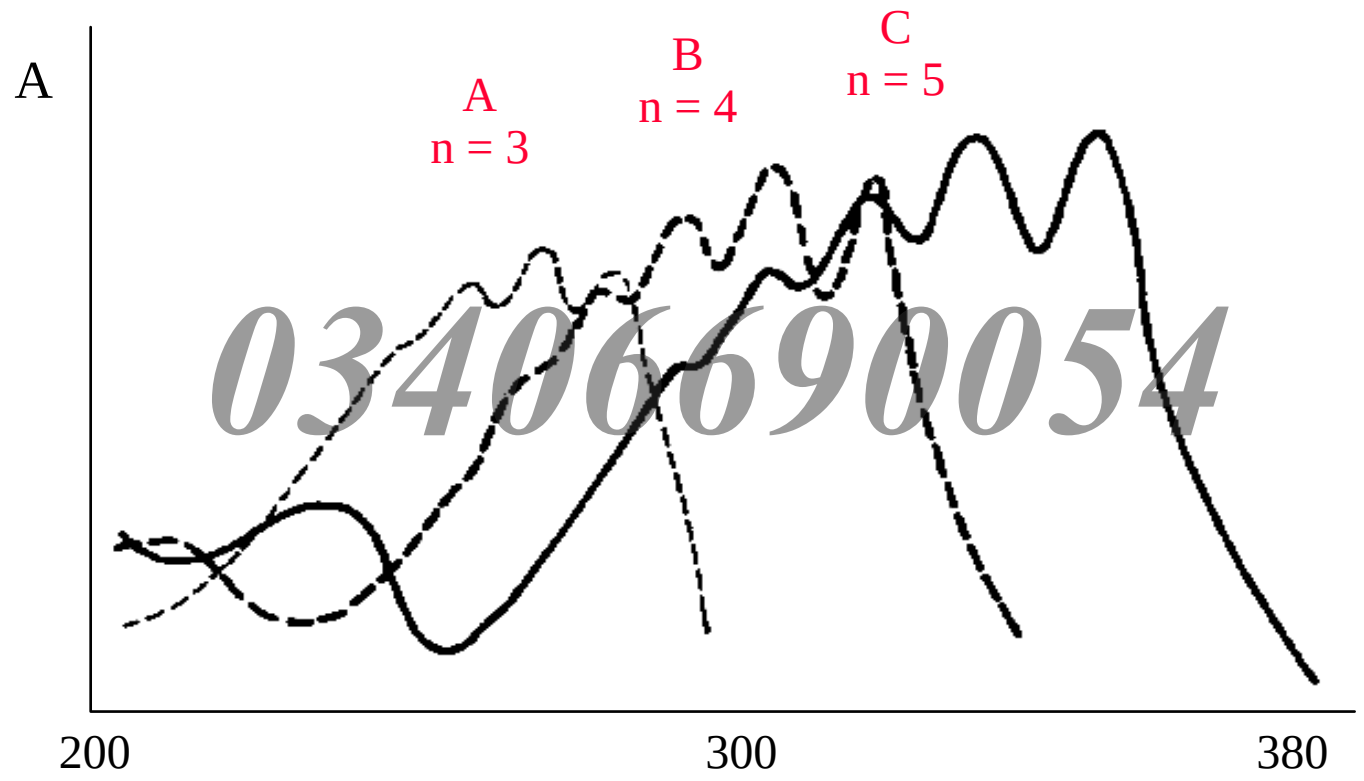


Diagram was taken from : Donald L. Pavia et. al., Introduction to Spectroscopy, Saunders Golden Sunburst Series, 1979, page 192

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UV Spectroscopy

- Substitution of simple alkyl group produces a small bathochromic effect (red shift) along with hyperchromic shift.
- Substitution of auxochromic groups (OH, NH_2 etc.) causes a pronounced red shift with large intensification (hyperchromic shift).
- Altering the availability of the non-bonding electrons (e.g. conversion of phenol to its anion) results in pronounced red shift and hyperchromic shift.
- Attachment of an unsaturated group such as ethylenic, acetylenic, nitrile, aldehydic, carboxylic and nitro to the benzene ring produces a red shift, often to a greater extent.

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UV Spectroscopy

- When substituents of electronic opposite type, such as hydroxyl and nitro groups, are situated *para* to each other, an unexpectedly large red shift is observed, due to extension of conjugation.

- As the number of condensed rings increases, the absorption is shifted progressively to longer wavelength.

END

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Instrumentation
UV Spectroscopy-VII

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UV Spectroscopy

Beer Lambert Law

$$\text{Absorbance A} = \log \frac{1}{T} = \varepsilon b C$$

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T : Transmittance

ε : Molar absorptivity

***b* : Path length**

C : Analyte concentration

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UV Spectroscopy

The two main characteristics of an absorption band are;

- Its position: Corresponds to wavelength of radiation whose energy is equal to that required for the electronic transition.

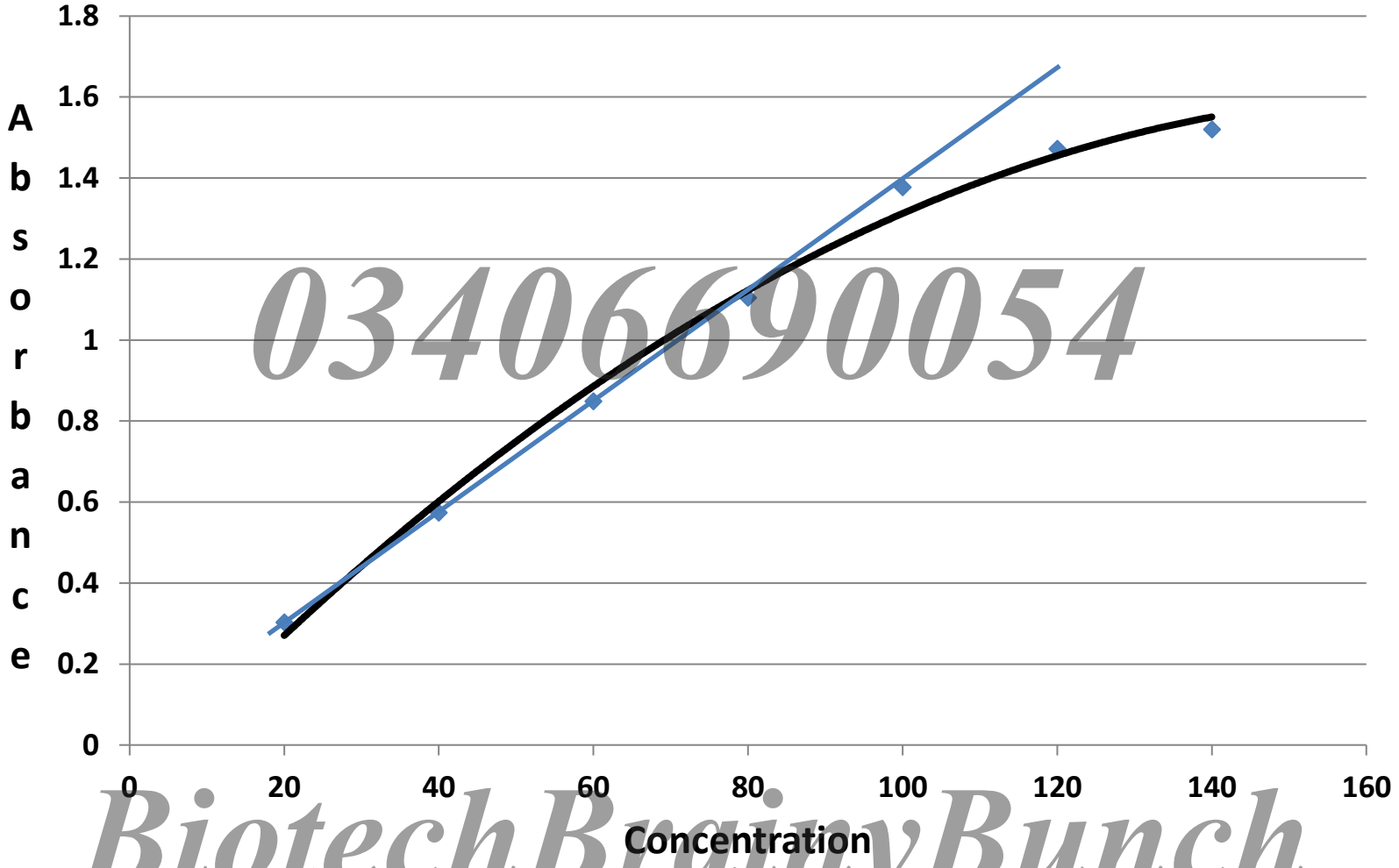
- Its intensities: Corresponds to the number of molecules which absorb the radiation of that wavelength, depends largely on probability of transition. It may be measured as Transmittance (T) or Absorbance (A).

Note: If a compound contains more then one chromophore usually corresponds to the sum of the expected spectra for all the chromophores. If the chromophore are separated by at least two single bonds they behave independently. If two chromophores in the same molecule are identical, a single absorption band is observed

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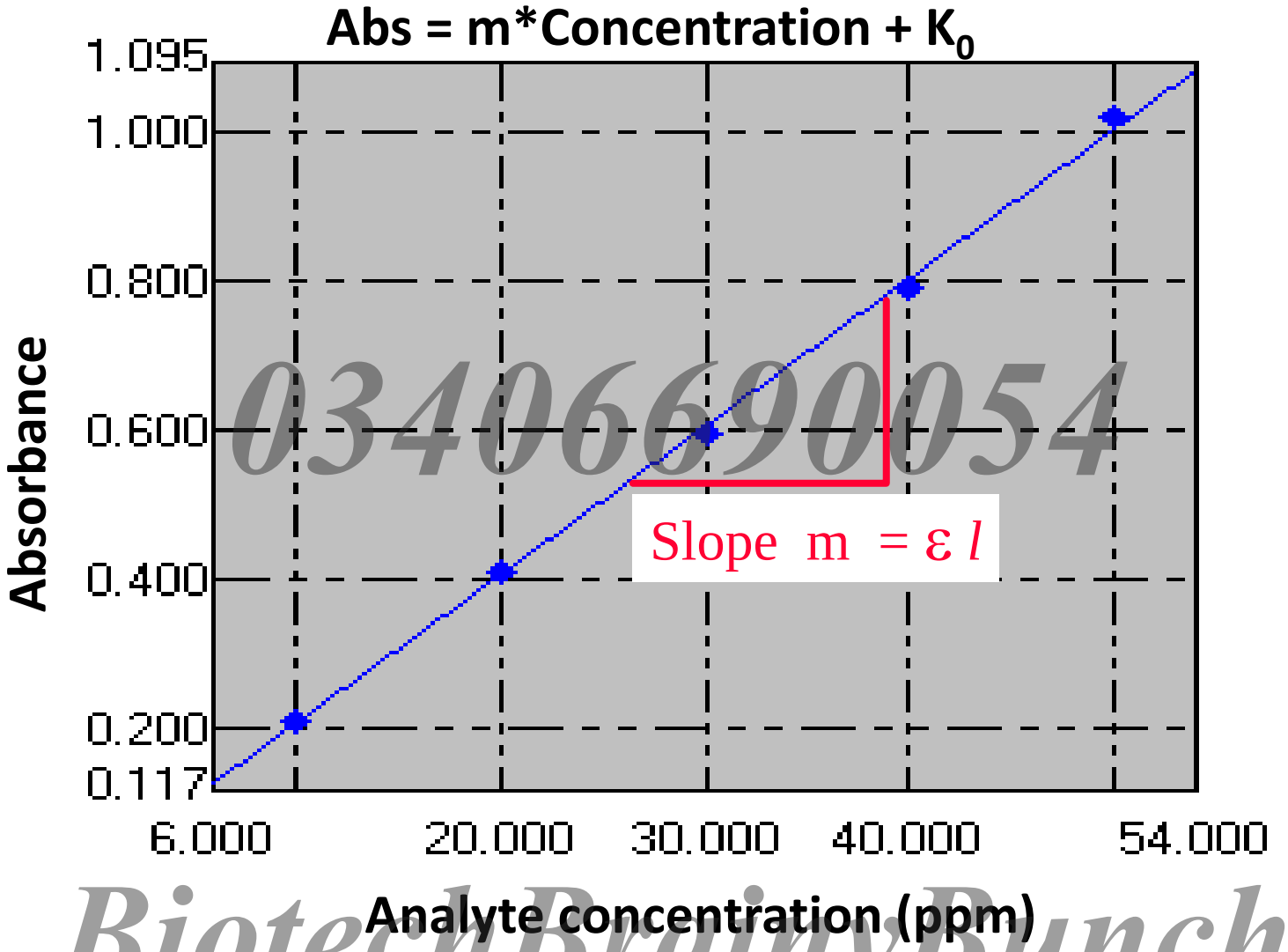
UV Spectroscopy

Calibration Curve at higher wavelength



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UV Spectroscopy



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UV Spectroscopy

- Analysis of an unknown is done by using a calibration curve of known concentration.
- Finding wavelength maximum (λ_{max}) is a qualitative estimation (spectrum mode).
- Analysis of unknown by using calibration curve is quantitative analysis (photometric mode).

END

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Instrumentation
UV Spectroscopy-VIII

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UV Spectroscopy

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UV Spectrophotometer

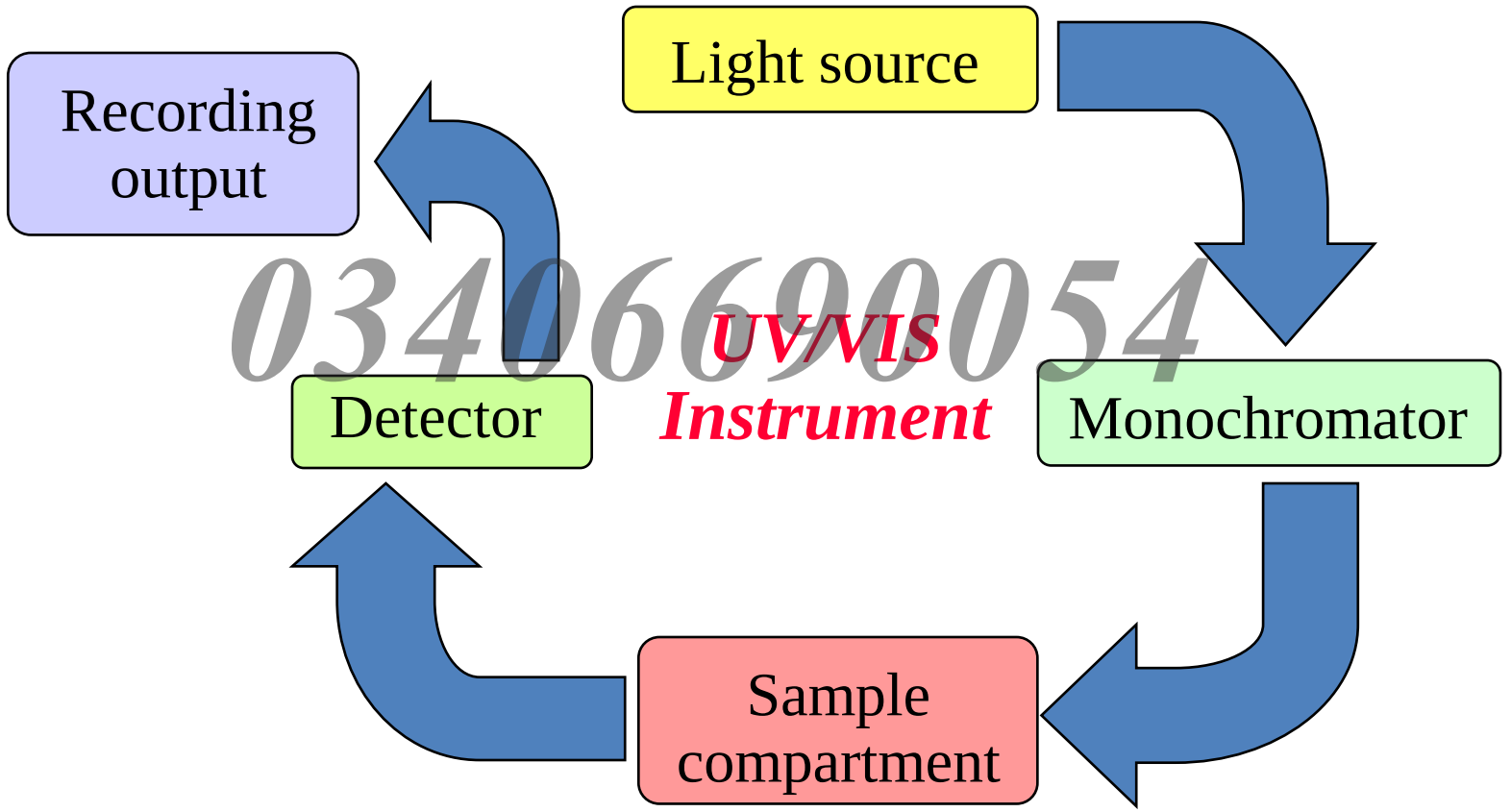
- Light source
- Monochromator
- Sample compartment
- Detector
- Read out device

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UV Spectroscopy

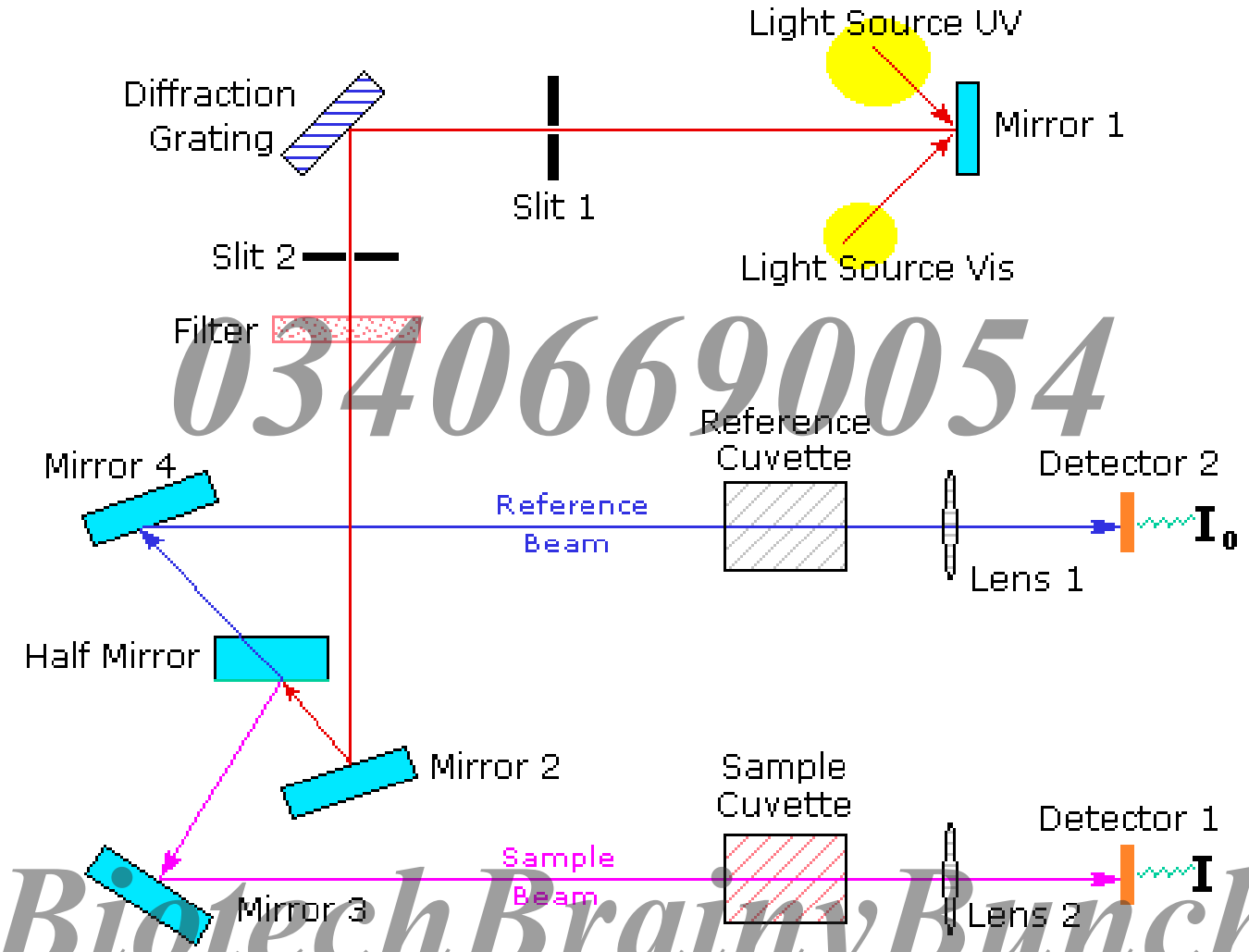
Schematic diagram of UV Spectrophotometer



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UV Spectroscopy

Double beam UV Spectrophotometer



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UV Spectroscopy

Tungsten and Deuterium Lamp





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UV Spectroscopy

- Tungsten lamp emit light of visible region (350-800 nm) while Deuterium lamp emit light in ultraviolet region (190-350 nm).
- UV spectrophotometer normally cover the range from 190 to 1100 nm.

END

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Instrumentation
UV Spectroscopy-IX

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UV Spectroscopy

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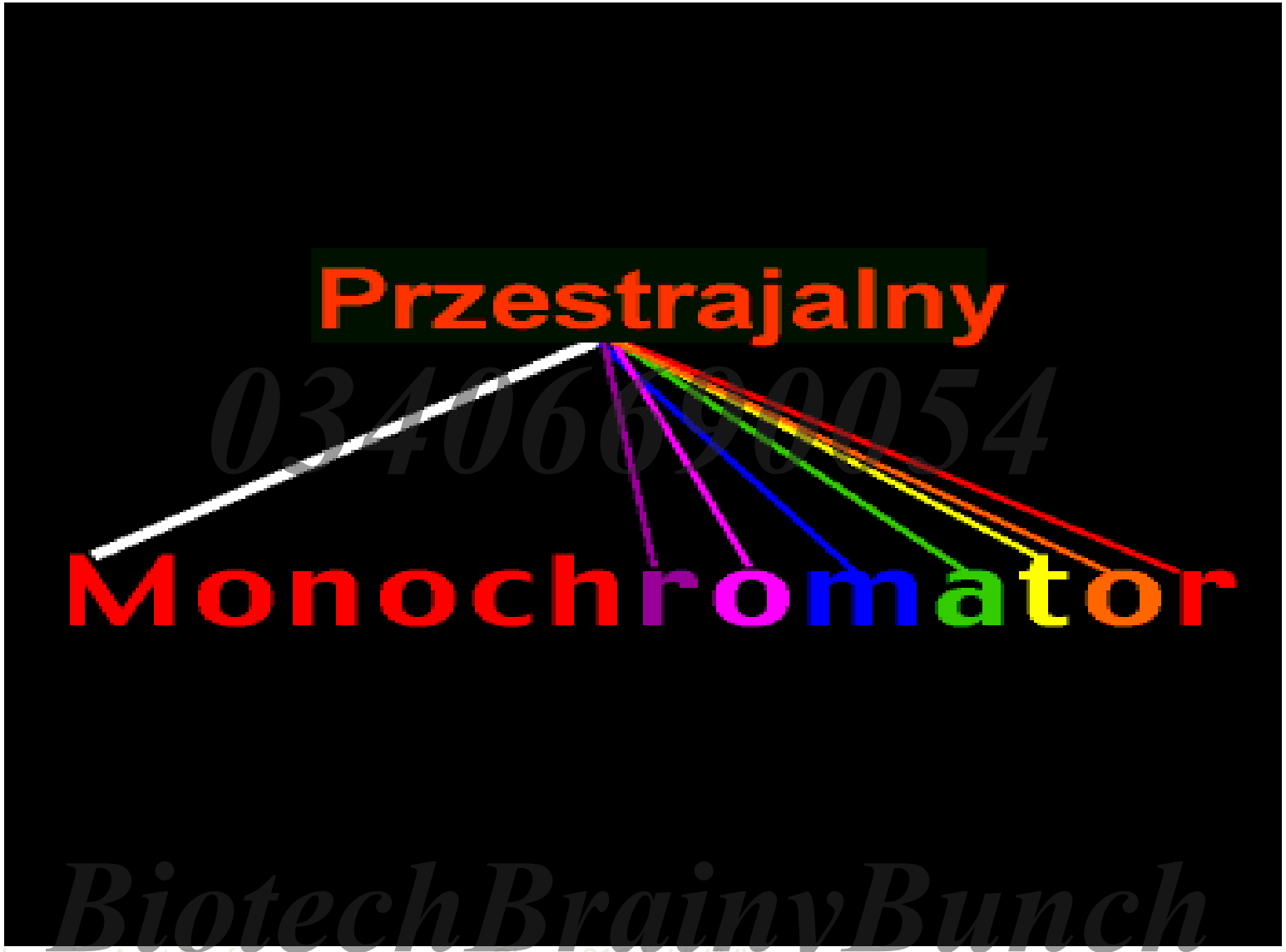
Monochromators used in UV Spectrophotometer

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- **Diffraction gratings**
- **Prism**
- **Filters**

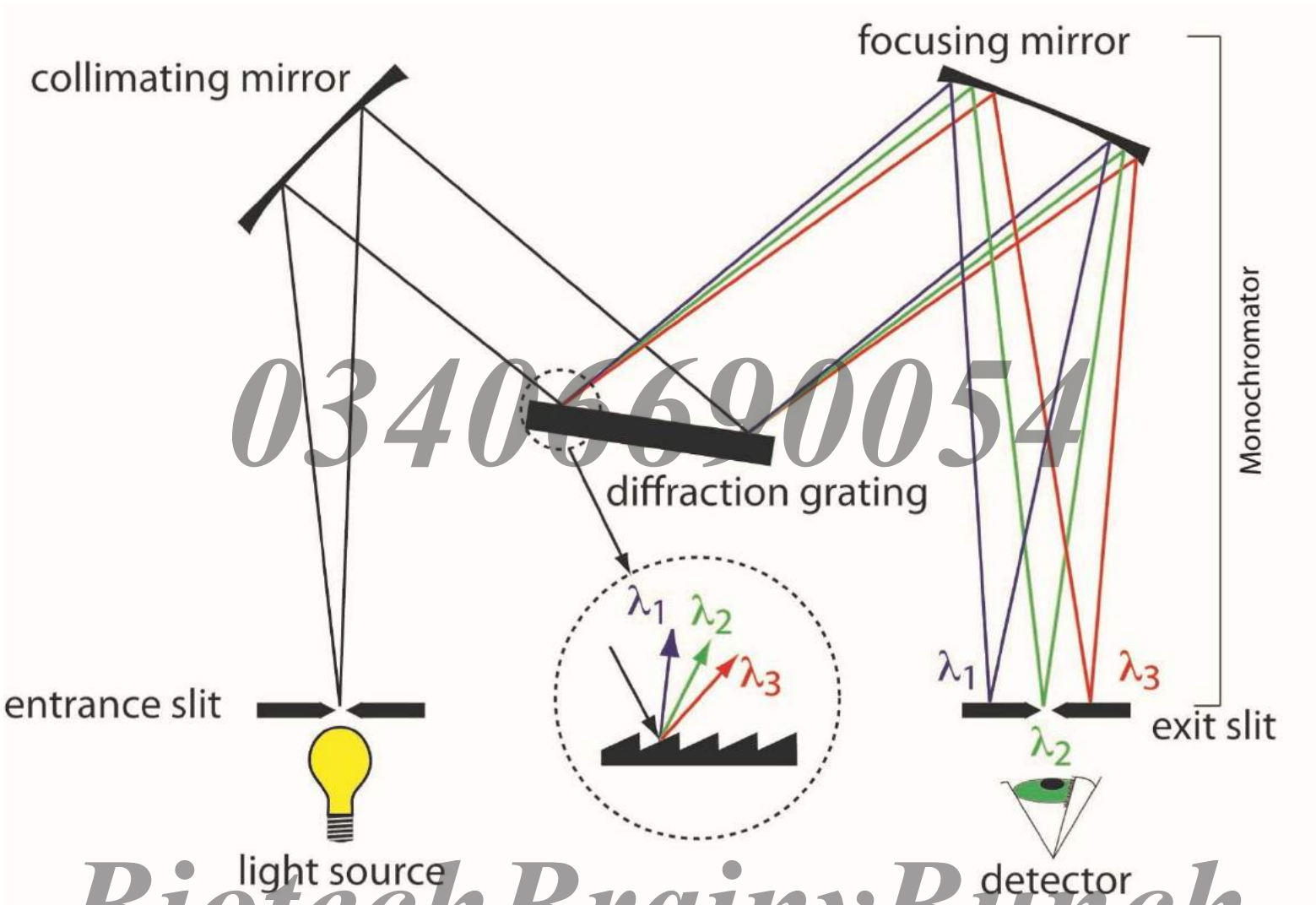
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UV Spectroscopy



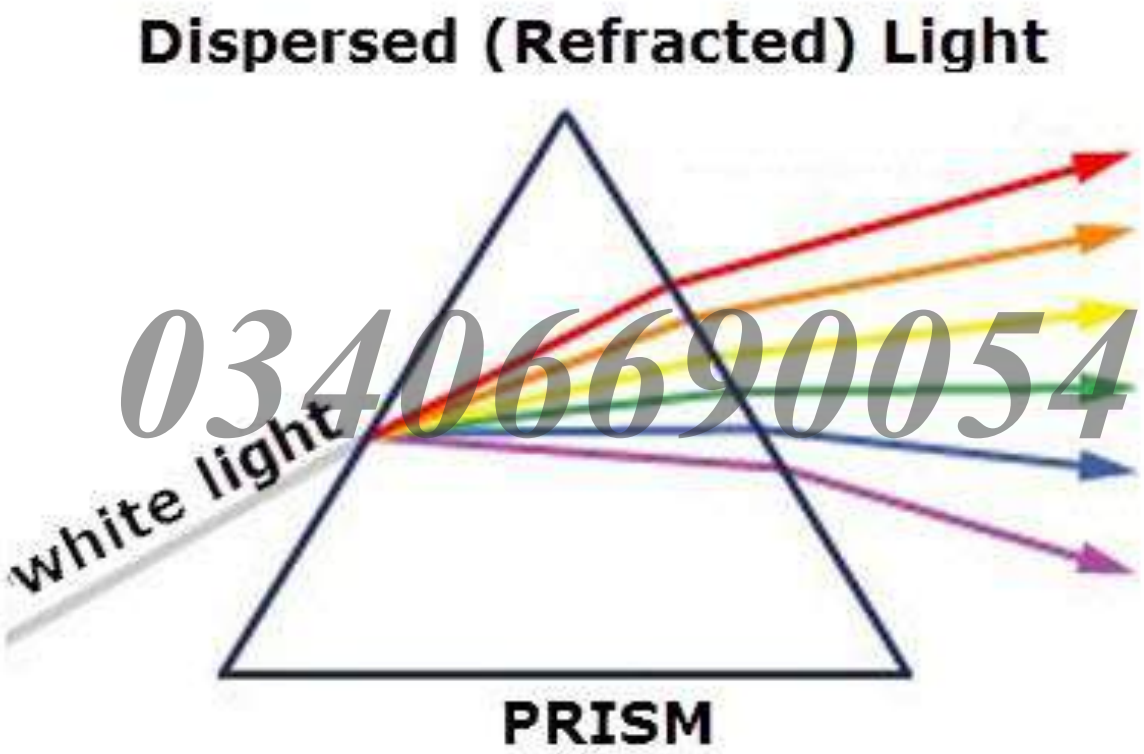
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UV Spectroscopy



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UV Spectroscopy

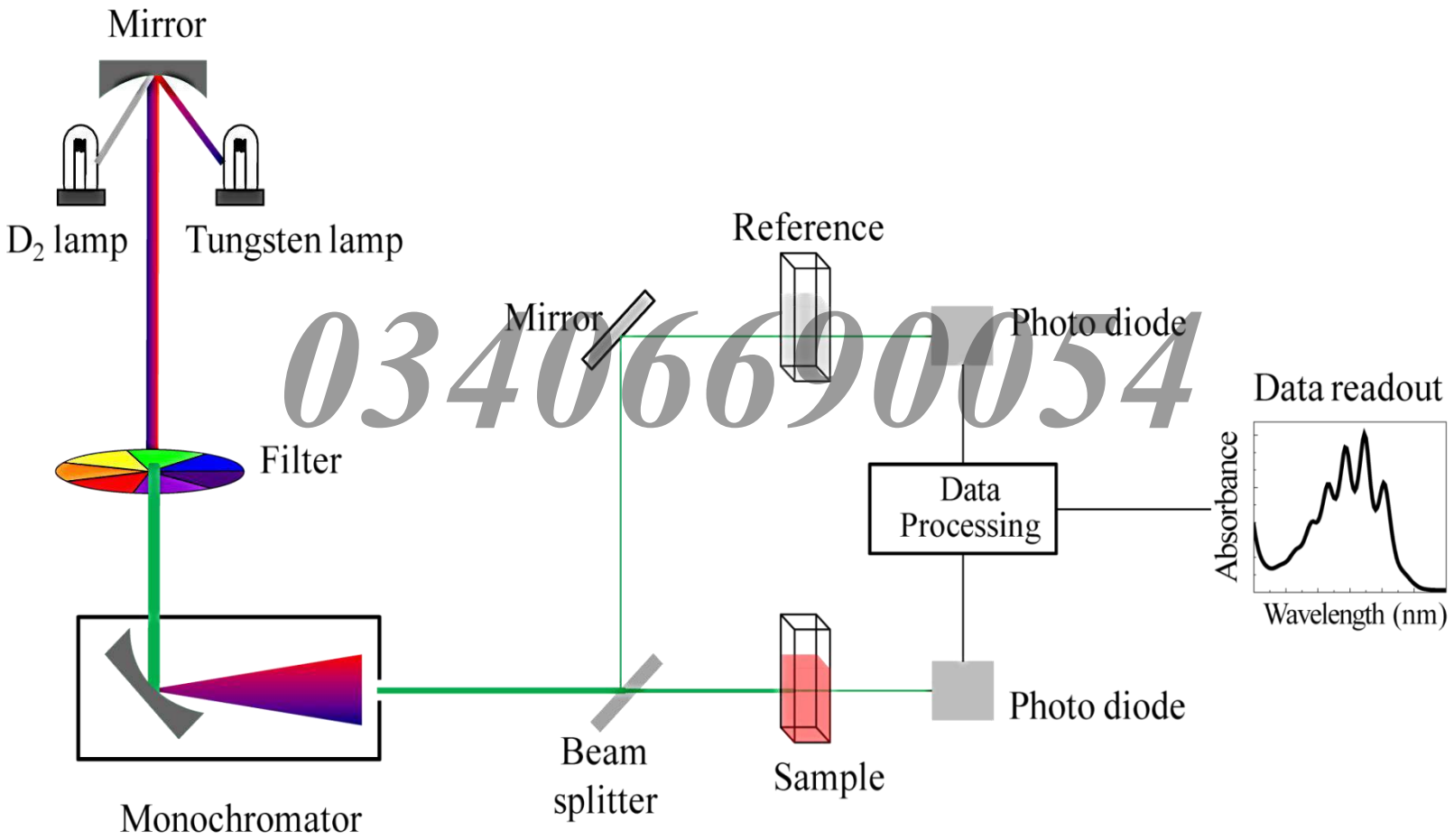


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UV Spectroscopy



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UV Spectroscopy

- In UV Spectrophotmeter the monochromators are installed before sample compartment.
- UV/Vis Spectrophotometers Used in two ways;
  1. Determination of wavelength maximum (where we have to scan full range spectrum).
  2. To determine absorbance values at a particular wavelength.

END

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Instrumentation
UV Spectroscopy-X

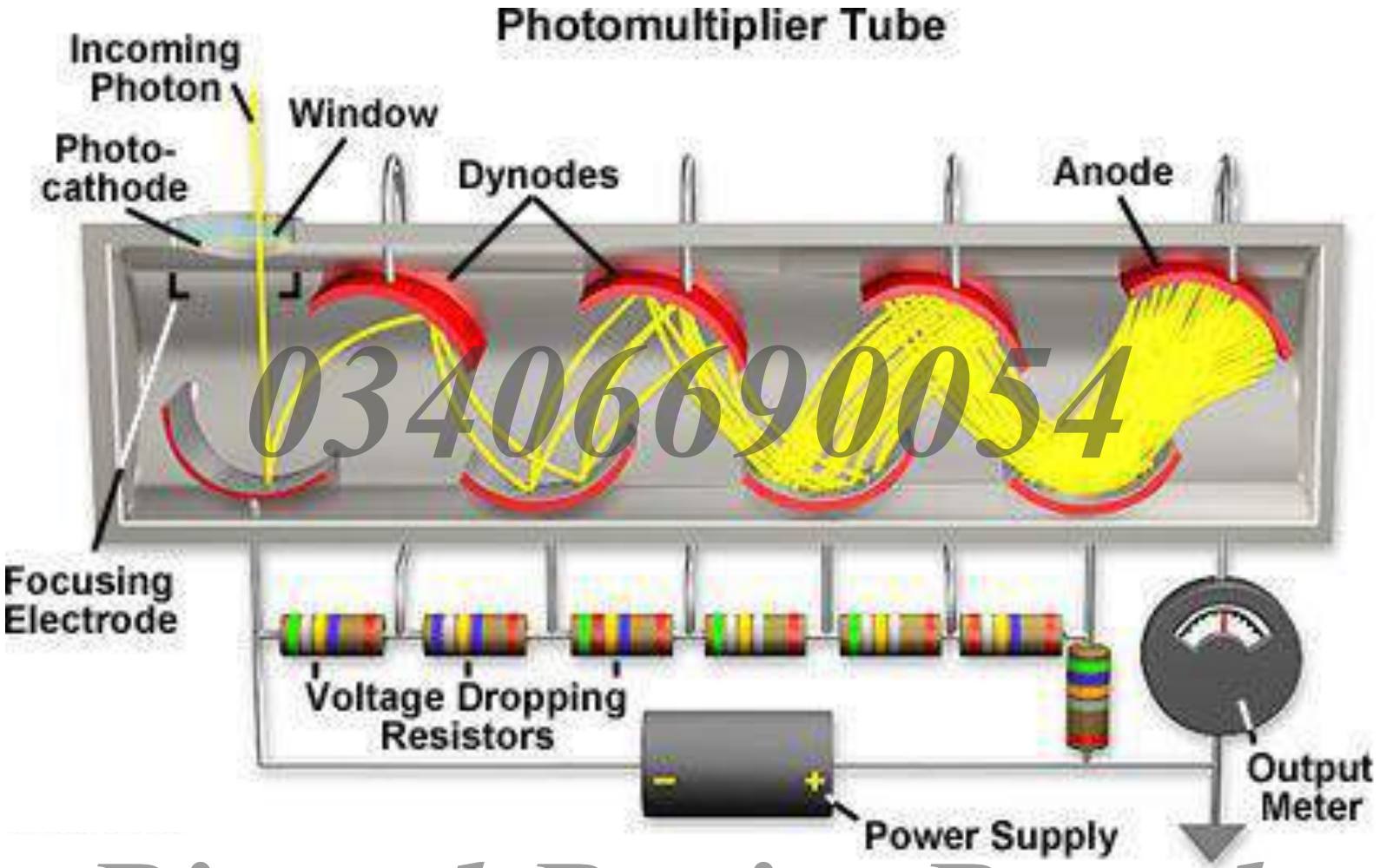
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UV Spectroscopy

Photomultiplier tubes (photomultipliers o r PMTs), members of the class of vacuum tubes, and more specifically vacuum phototubes, are extremely sensitive detectors of light in the ultraviolet, visible, and near-infrared ranges of the electromagnetic spectrum.

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UV Spectroscopy



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UV Spectroscopy

Woodward-Fieser Rule for Conjugated Dienes

- For theoretical calculation of wavelength maxima for a synthesized compound.
- Used as QC tool.
- Calculated values may be checked and correlated with experimental values.

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UV Spectroscopy

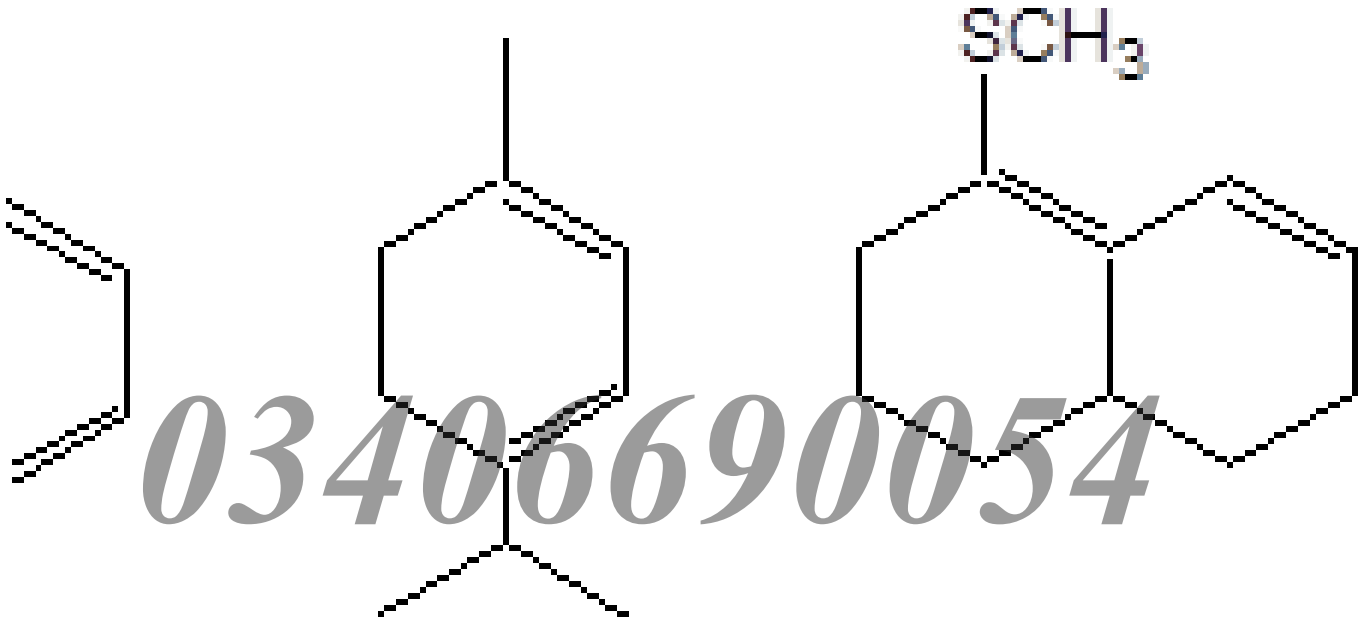
Woodward-Fieser Rule for Conjugated Dienes (in ethanol)

- Base Value for
  - Acyclic diene---------- 217
  - Heteroannular diene---------- 214
  - Homoannular diene---------- 253
- Increment for each
  - Double bond extending conjugation---------- 30
  - Exocyclic double bond---------- 5
  - Alkyl group or ring residue---------- 5
  - Cl or Br---------- 5
  - OAc---------- 0
  - OR---------- 6
  - SR---------- 30
  - NR_2---------- 60

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UV Spectroscopy

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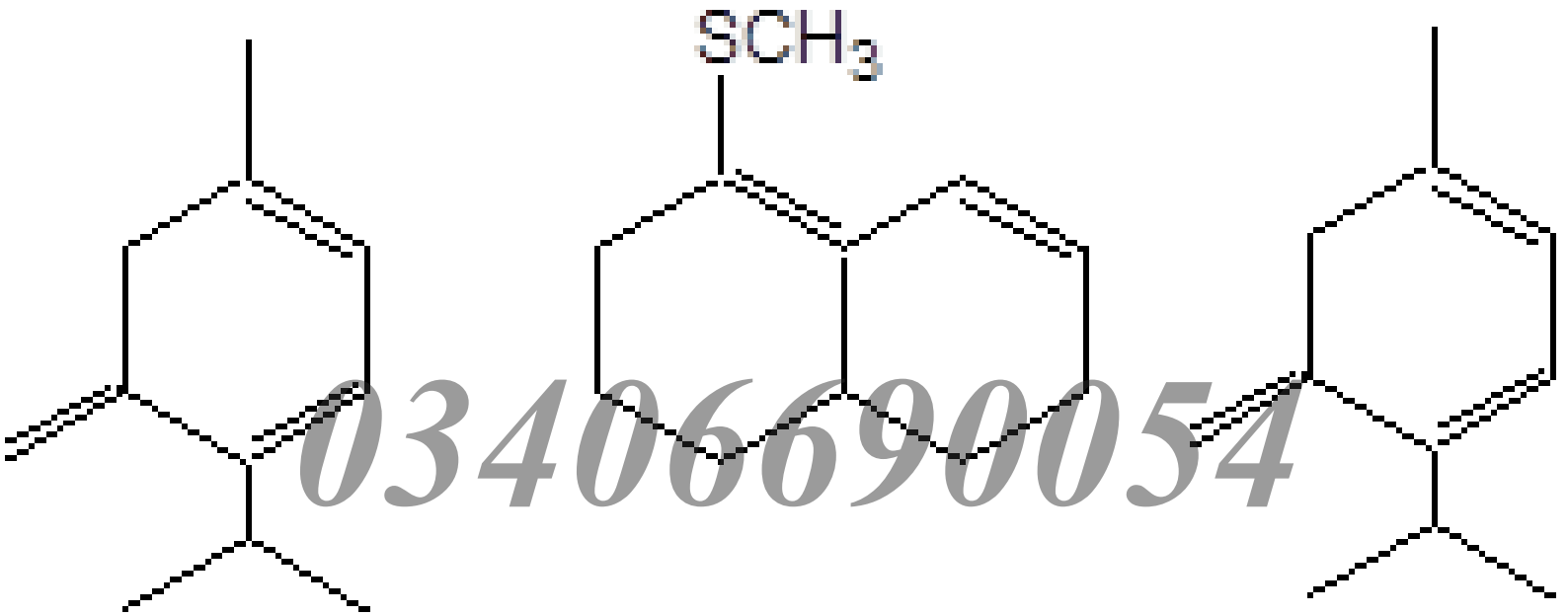
Acyclic diene

Homoannular diene

Heteroannular diene

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UV Spectroscopy



Exocyclic Double Bond

Extending Conjugation

Alkyl residue

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UV Spectroscopy

- We can assign values for base, exocyclic double bonds, extending conjugation and alkyl / ring residues.
- Sum of these values will consider as theoretical value of wavelength maximum.

END

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Instrumentation
UV Spectroscopy-XI

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UV Spectroscopy

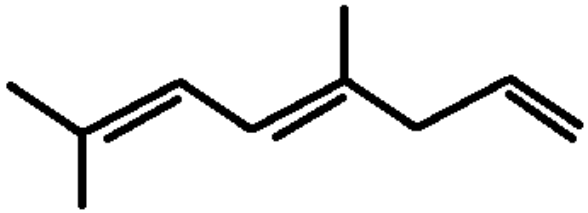
Woodward-Fieser Rule for Conjugated Dienes

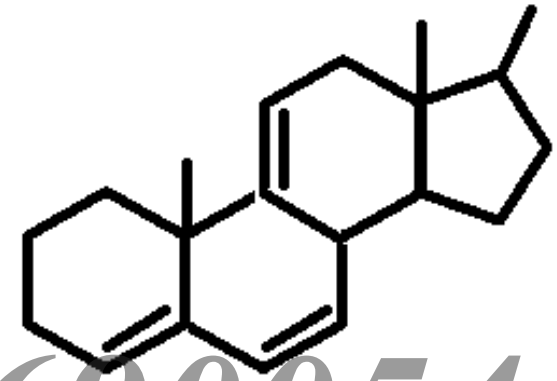
- Now we calculate some reported diene systems to make some understanding.
- And compare them with reported observed values.

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UV Spectroscopy

Effort by Mehwish Mughal





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Base Value: 214
DB Ext:
ExoCyc DB:
Alk Res:
Residue:

λ_{max} Predicted:
Observed: 236 nm

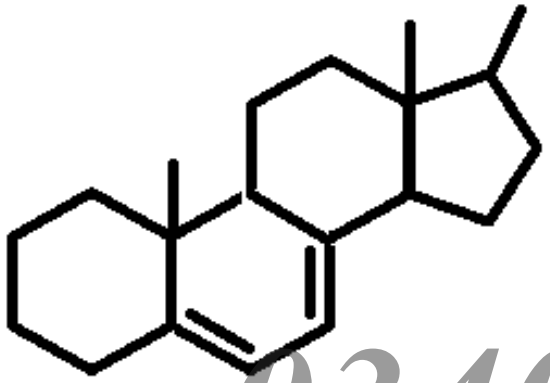
Base Value: 214
DB Ext:
ExoCyc DB:
Alk Res:
Residue:

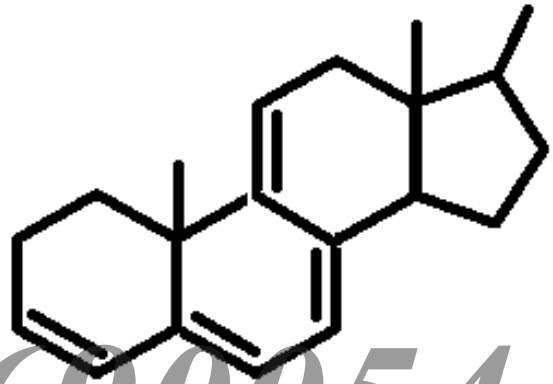
λ_{max} Predicted:
Observed: 235 nm

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UV Spectroscopy

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Base Value:	253
DB Ext:	
ExoCyc DB:	
Alk Res:	
Residue:	
λ_{max} Predicted:	
Observed:	282 nm

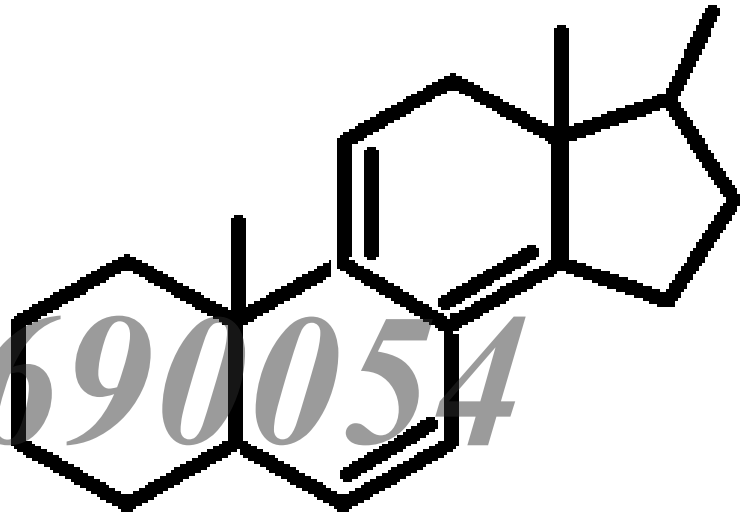
Base Value:	253
DB Ext:	
ExoCyc DB:	
Alk Res:	
Residue:	
λ_{max} Predicted:	
Observed:	354 nm

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UV Spectroscopy

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Base Value: 253
DB Ext:
ExoCyc DB:
Alk Res:
Residue:



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λ_{max} Predicted:
Observed: 291 nm

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UV Spectroscopy

- In this way these predicted values may be used to confirm the synthesized compounds.
- We may finalize structure on the basis of different spectroscopic techniques like NMR and MS but have to report UV λ_{max} values for confirmation.

END

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Instrumentation
UV Spectroscopy-XII

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UV Spectroscopy

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Woodward-Fieser Rule for α,β-Unsaturated Carbonyl Compounds

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- For theoretical calculation of wavelength maxima of conjugated carbonyl systems.

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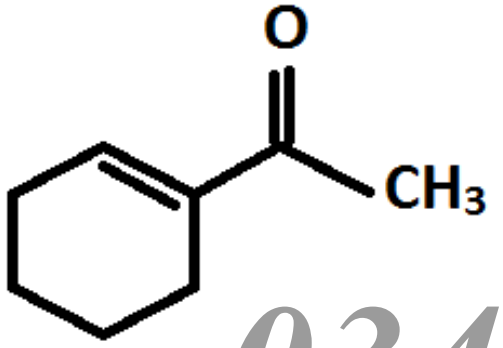
UV Spectroscopy

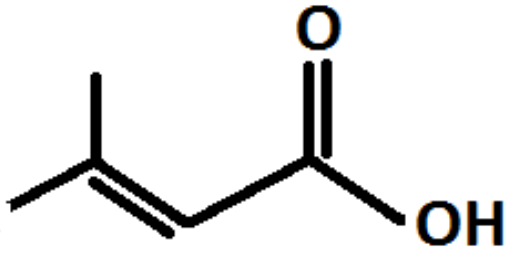
Woodward-Fieser Rule for α,β-Unsaturated Carbonyl Compounds (in Ethanol)

- Base Value for
  - Acyclic or six membered cyclic ketones ---------- 215
  - Five membered cyclic ketones ---------- 202
  - Aldehydes ---------- 207
  - Carboxylic acids and esters ---------- 195
- Increment for each
  - Double bond extending conjugation ---------- 30
  - Homodienic component ---------- 39
  - Exocyclic double bond ---------- 5
  - Alkyl group or ring residue, α ---------- 10
    β ---------- 12
    γ (or higher) ---------- 18

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UV Spectroscopy



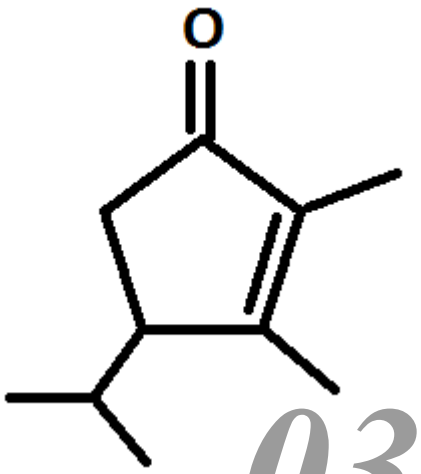


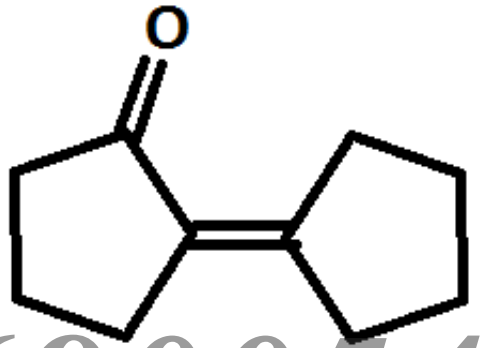
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Base Value:	**215**	**Base Value:**	**195**
DB Ext:		**DB Ext:**	
ExoCyc DB:		**ExoCyc DB:**	
Alk Res, α:		**Alk Res, α:**	
β:		**β:**	
	———		———
λ_{max} Predicted:		**λ_{max} Predicted:**	
Observed:	**234 nm**	**Observed:**	**216 nm**

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UV Spectroscopy





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Base Value:	**202**
DB Ext:	
ExoCyc DB:	
Alk Res, α:	
β:	
λ_{max} Predicted:	
Observed:	**235 nm**

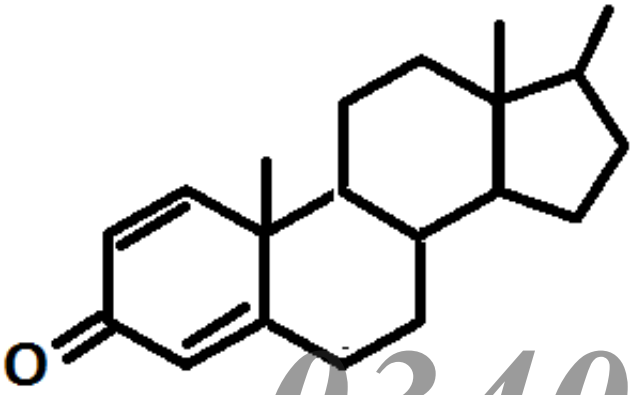
Base Value:	**215**
DB Ext:	
ExoCyc DB:	
Alk Res, α:	
β:	
λ_{max} Predicted:	
Observed:	**258 nm**

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UV Spectroscopy

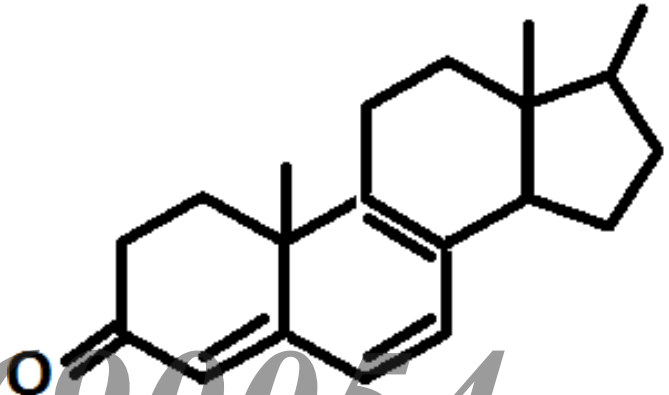
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Base Value: 215
DB Ext:
ExoCyc DB:
Alk Res, α:
β:

λmax Predicted:
Observed: 245 nm



Base Value: 215
DB Ext:
ExoCyc DB:
Homodiene:
Alk Res, α:
β:

λmax Predicted:
Observed: 388 nm

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UV Spectroscopy

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- All this data is copied from Textbook of Organic Chemistry by Dr. M. Younas.
- You may also found in any other organic chemistry book available in market or Introduction to spectroscopy by “Pavia”.

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END

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Instrumentation
UV Spectroscopy-XIII

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UV Spectroscopy

Applications:

1. **Structure Determination**
2. **Analytical applications**
3. **Kinetic measurements**
4. **Ionization constants**
5. **Stereochemical study**

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UV Spectroscopy

Structure Determination

- **Nature of Conjugation**
  - As discussed in Woodword-Fieser Rules
- **Confirm availability of chromophores**
  - Spectrum plot before and after reaction may lead to this conclusion
- **Structure Verification**
  - Structure suggested based on chemical testing / NMR / mass fragments may be verified

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UV Spectroscopy

Analytical Applications

- **Food – color and consistency**
  - **In quality control testing of foods**
- **Beverage – impurity and decomposition**
  - **Stability of product with aging or help in determining expiry dates of products**
- **Leaf analysis – chlorophyll and photosynthesis process**
  - **may evaluate ripening of fruits etc.**

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UV Spectroscopy

Analytical Applications (Cont….)

- **Pharmaceutical manufacturer – metabolic pathways of drugs, nucleic acid analysis, kinetics and dissolution testing**
  - **Drug availability testing using dissolution apparatus**
  - **Drug availability in human body after infusion**
  - **Kinetic studies to monitor reaction speed and completion status**

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UV Spectroscopy

Analytical Applications (Cont….)

- Chemical manufacturer
  - Raw material quality assurance as per specs
  - Quality control testing of tablets and syrups etc.

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Instrumentation
UV Spectroscopy-XIV

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UV Spectroscopy

Applications:

1. **Structure Determination**
2. **Analytical applications**
3. **Kinetic measurements**
4. **Ionization constants**
5. **Stereochemical study**

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UV Spectroscopy

Kinetic Measurement (Biological Applications)

- **Protein Analysis**
- **DNA and RNA analysis**
- **Fermentation**
- **Kinetics – monitor rate of reaction and enzymatic reaction**
  - **Draw samples from body fluids and monitor using UV spectrophotometer at some specific wavelength.**
  - **For comparison of results draw graphs between time and concentrations etc.**

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UV Spectroscopy

Ionization Constant

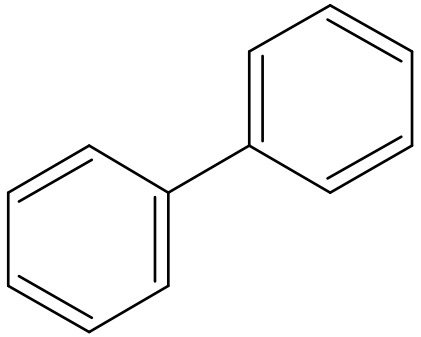
- For acid and bases
- End point determination
  - Dissociation of acid and bases leads to increase in concentration of specific moiety in the system, that can be quantitatively assessed using UV Spectrophotometer.
  - This may lead to conclusion that one specific substance is no more in the system, means end point of reaction or titration.

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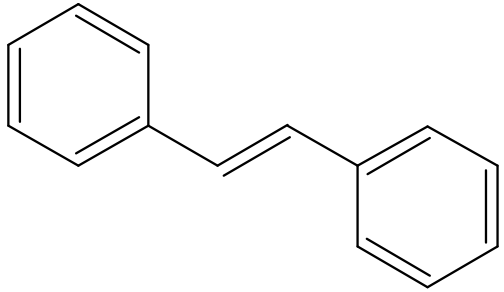
UV Spectroscopy

Stereochemical Study

- Symmetry of molecule
  - Stereochemical studies
    - Hindered rotation in ortho-substituted biphenyls can be studied easily



- Cis-trans distinguish
  - Geometrical isomerism
    - cis-stilbene 283 nm
    - trans-Stilbene 295 nm



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UV Spectroscopy

Academic and Government Application

- **Environmental testing – contaminants in streams and water sources**
- **Medical testing**
- **Academic - students**

END

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Instrumentation
IR Spectroscopy-I

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IR Spectroscopy

IR Spectroscopy

- UV ranges from 190 to 350 nm
- Visible region extended from 350 (Violet) to 760 nm (Red)
- IR region is up to 1 mm wavelength of electromagnetic radiation

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IR Spectroscopy

Wavenumber:

$$c = \lambda \upsilon \qquad \bar{\upsilon} = \frac{1}{\lambda}$$

$$E = h\nu = h\frac{c}{\lambda} = hc\bar{\upsilon}$$

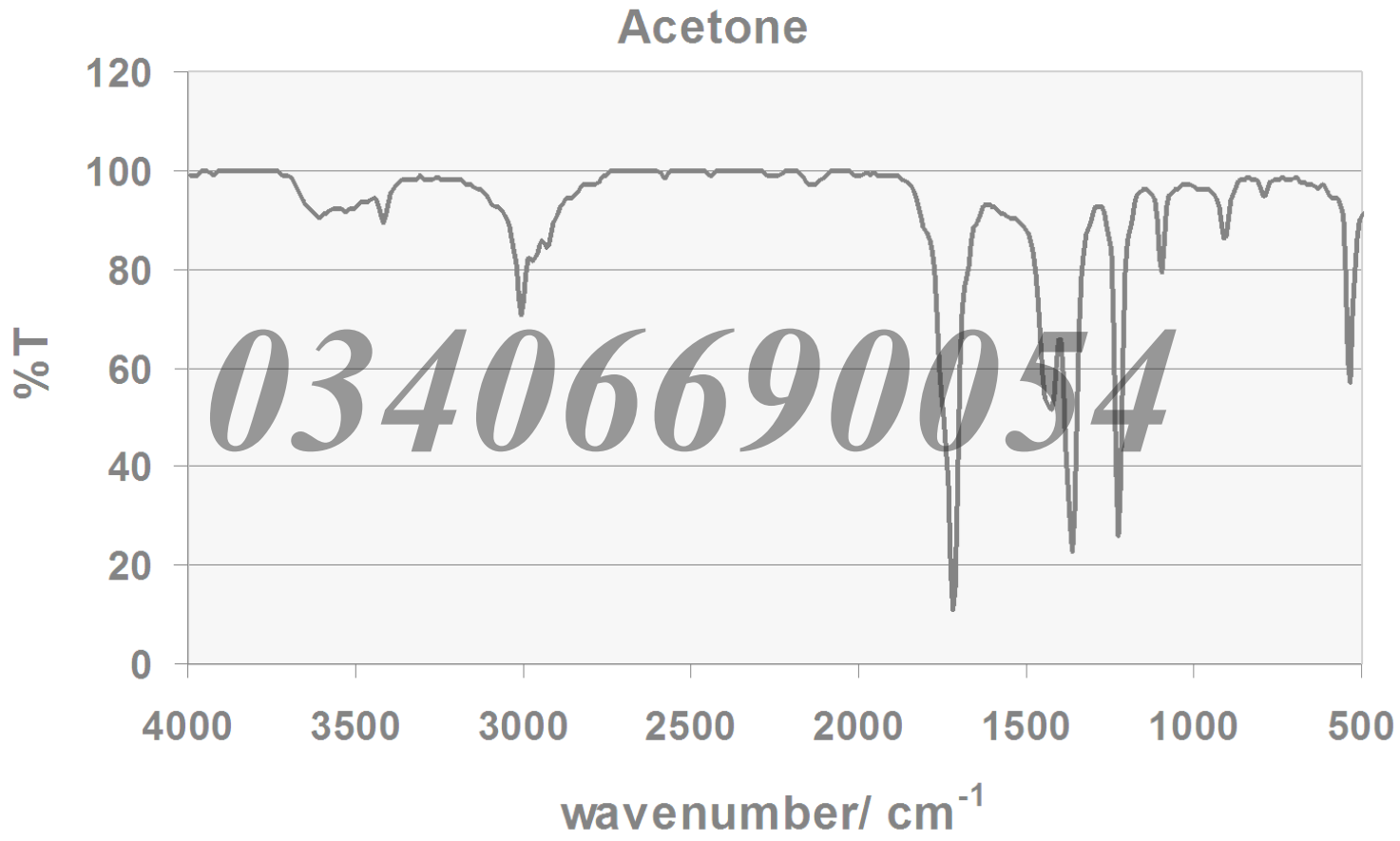
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- $\bar{\upsilon}$ (nu-bar) represents wavenumber; the number of wavelengths per cm (cm^{-1})
- Mid IR region represents 2.5 to 16 µm (4000 to 625 cm^{-1})

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IR Spectroscopy

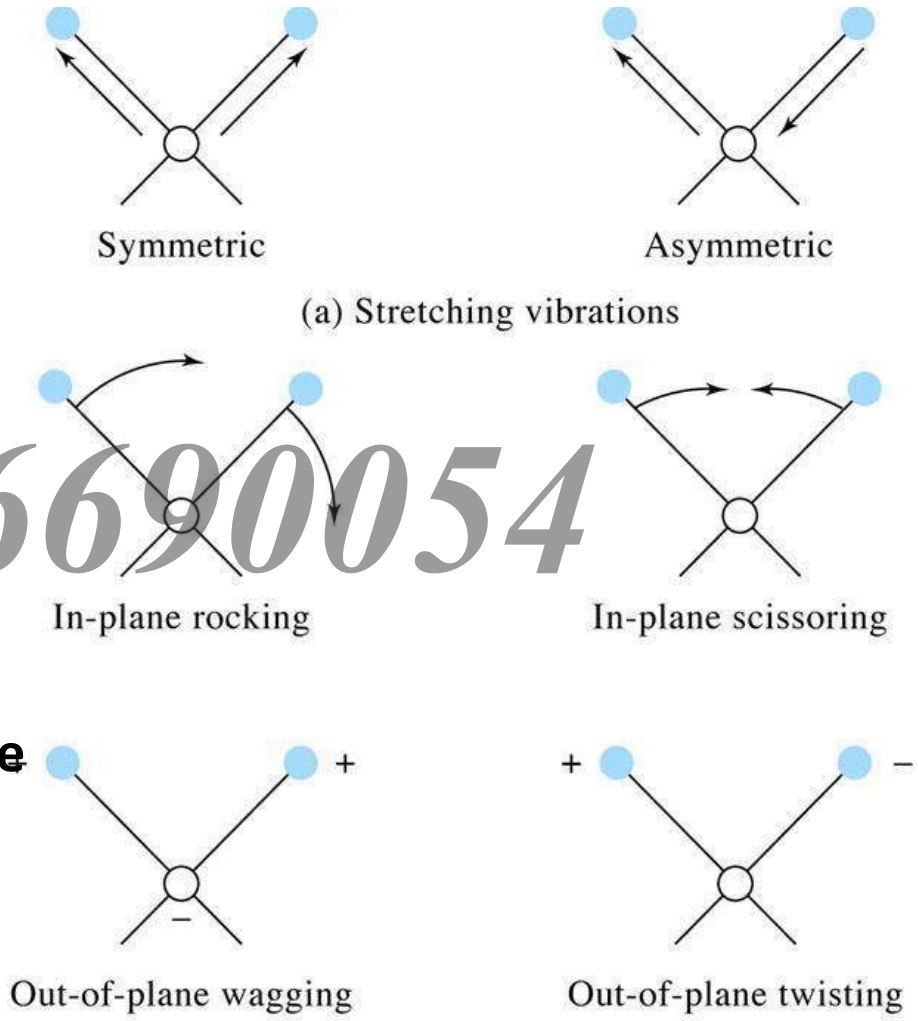


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IR Spectroscopy

Vibrational Modes

- The total number of vibrations in a molecule depends upon the number of atoms present in the molecule.
- Each atom has three degrees of freedom because three coordinates (x, y, z) are required to describe its position relative to other atoms in a molecule.
- Vibrational modes in AB2 system are represented here.



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IR Spectroscopy

- Only vibrations that cause a change in 'polarity' give rise to bands in IR spectra.
- Instead of symmetric stretch the unsymmetrical stretches appear in IR spectra.

END

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Instrumentation
IR Spectroscopy-II

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IR Spectroscopy

Vibrational Frequency

- We can calculate an approximation value for the stretching vibrational frequency of a bond by HOOK's LAW.

$$\bar{\nu} = \frac{1}{2\pi c}\sqrt{\frac{f}{\mu}}$$

$$\mu = \frac{m_a m_b}{m_a + m_b}$$

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IR Spectroscopy

Bond →	C-H	C-D	C-O	C-Cl
ν/cm^{-1} →	3000	2200	1100	700
Bond →	C≡O	C=O	C-O	
ν/cm^{-1} →	2143	1715	1100	***Due to increase in force constant***

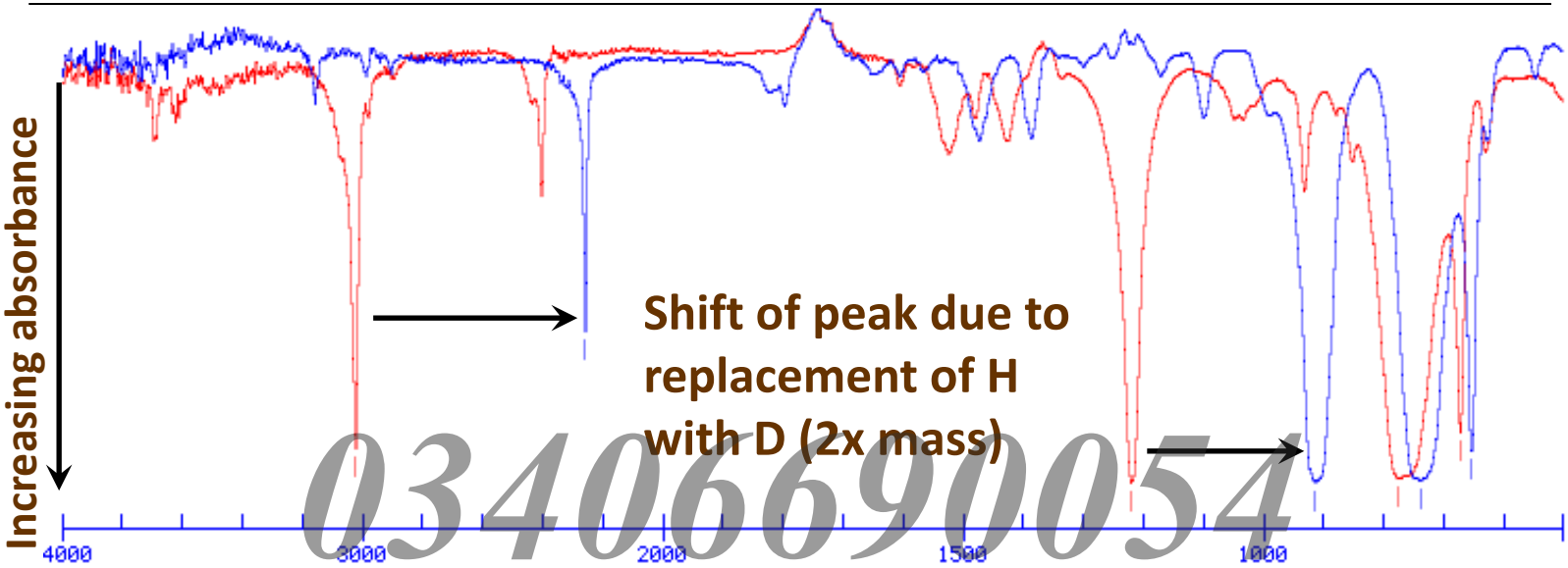
$$\bar{\nu}_m\,(cm^{-1}) = \frac{1}{2\pi c}\sqrt{\frac{f(m_a + m_b)}{m_a \bullet m_b}}$$

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IR Spectroscopy



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Mode of vibration	$CHCl_3$ Calculated*	$CHCl_3$ Measured	$CDCl_3$ Measured
C-H stretching	3002	3020	2256
C-Cl stretching	701	773	737

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IR Spectroscopy

- Lower masses appear in left most region of IR spectra (3700-2500 cm^{-1}).
- Less force constant will shift values of wavenumber to right region of IR spectra (1300-800 cm^{-1})

END

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Instrumentation
IR Spectroscopy-III

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IR Spectroscopy

- Lower masses appear in left most region of IR spectra (3700-2500 cm^{-1}).
- Less force constant will shift values of wavenumber to right region of IR spectra (1300-800 cm^{-1})

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IR Spectroscopy

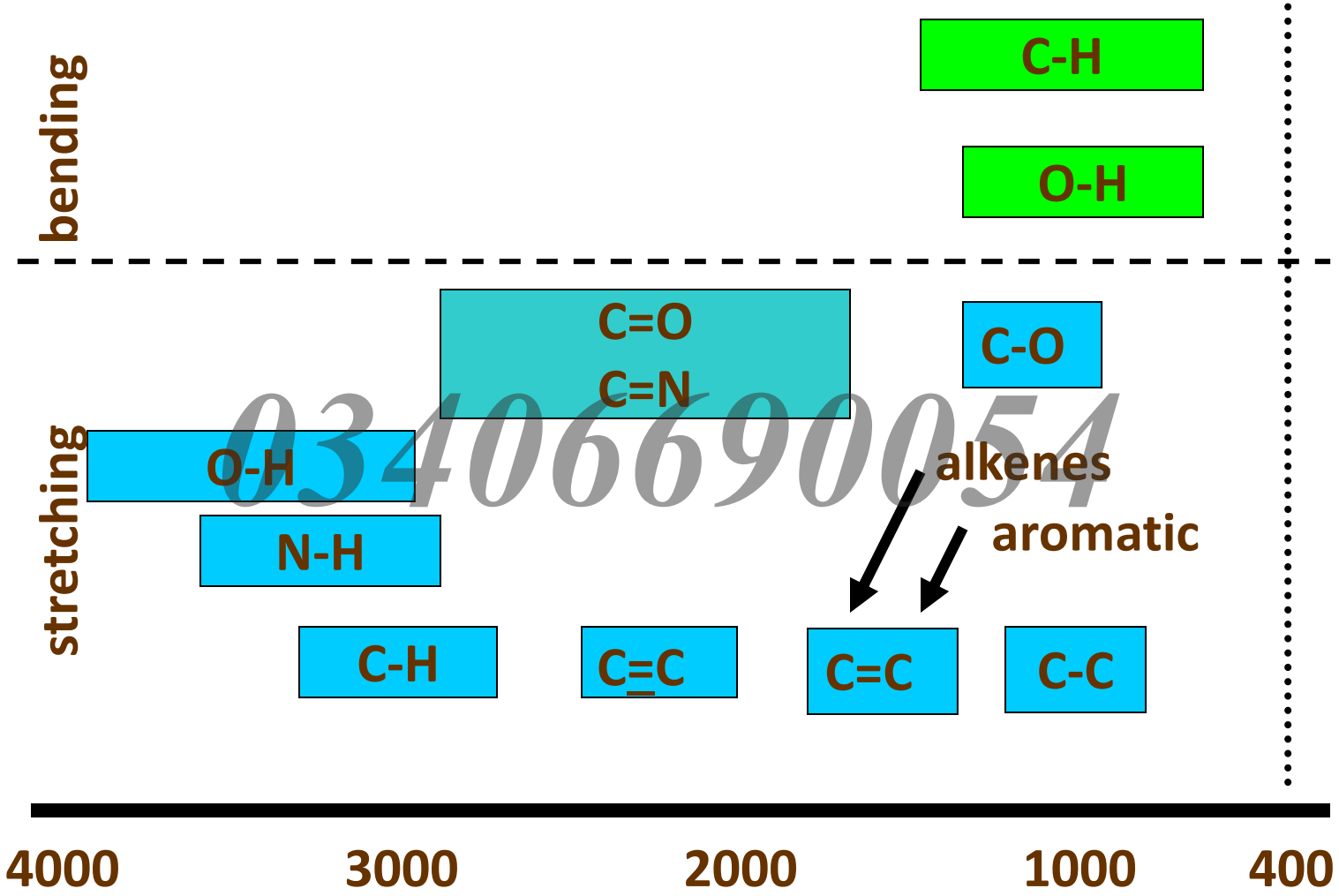
3700-2500 cm^{-1}	2300-2100 cm^{-1}	1900-1600 cm^{-1}	1300-800 cm^{-1}
O-H N-H C-H	C≡C C≡N	C=C C=O C=N	C-O C-C C-N
			Deformations

increasing energy / increasing wavenumber / increasing frequency

increasing wavelength

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IR Spectroscopy



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IR Spectroscopy

- We solve IR spectra for major peaks of functional groups.
- IR Spectra contains many tiny peaks due to overtones.
- Finger print region is very important in differentiating two compounds with similar functional groups.

END

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Instrumentation
IR Spectroscopy-IV

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IR Spectroscopy

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IR Spectra

- IR Spectra have three main features.

1. Position of peaks.
2. Strength of peaks
3. Width of peaks

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IR Spectroscopy

POSITION	REDUCED MASS	LIGHT ATOMS HIGH FREQUENCY
	BOND STRENGTH (STIFFNESS)	STRONG BONDS HIGH FREQUENCY
STRENGTH	CHANGE IN 'POLARITY'	STRONGLY POLAR BONDS GIVE INTENSE BANDS
WIDTH	HYDROGEN BONDING	STRONG HYDROGEN BONDING GIVES BROAD BANDS

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IR Spectroscopy

- In IR Spectra the C=O shows most intense peaks.
- Only OH peaks appear as wide peaks when OH involve in hydrogen bonding.
- NH appear as one peak while NH_2 as two peaks.

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Instrumentation
IR Spectroscopy-V

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IR Spectroscopy

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Factors influencing IR Absorptions

- **The most intense peak will be of C=O (carbonyl); we will discuss factors influencing absorption frequencies of carbonyl as a case study.**

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IR Spectroscopy

IR: Carbonyl

From 1850 – 1650 cm^{-1}

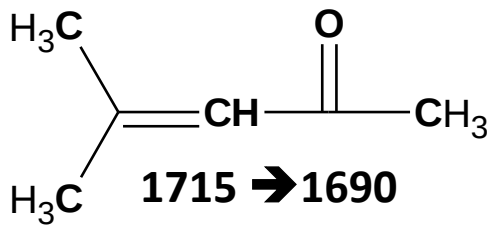
1810	1800	1735	1725	1715	1710	1690
Anhydr Band	Acid Chloride	Ester	Aldehyde	Ketone	Acid	Amide

Ketone 1715 cm^{-1} is used as reference point for comparisons

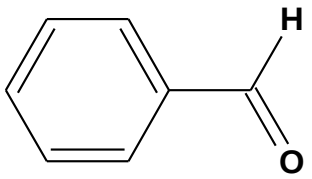
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Conjugation increase single bond character of C=O

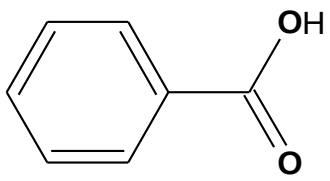
➔ Lower force constant ➔ lower frequency number



1715 ➔1690



1725 ➔ 1700



1710 ➔1680

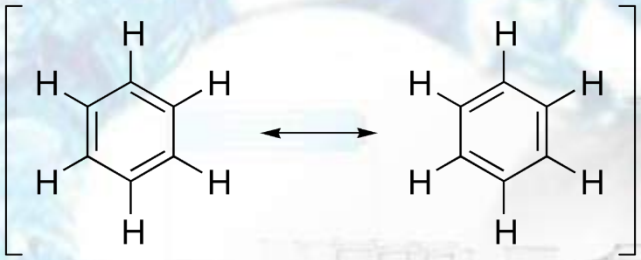
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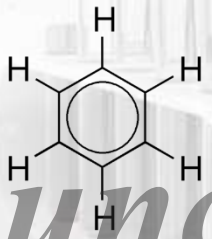
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IR Spectroscopy

- **Benzene ring due to conjugation shows multiple (4) peaks around 1400 cm^{-1}.**
- **C-C absorb around 1300-800 cm^{-1} while C=C absorb at 1900-1600 cm^{-1}.**

END





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Instrumentation
IR Spectroscopy-VI

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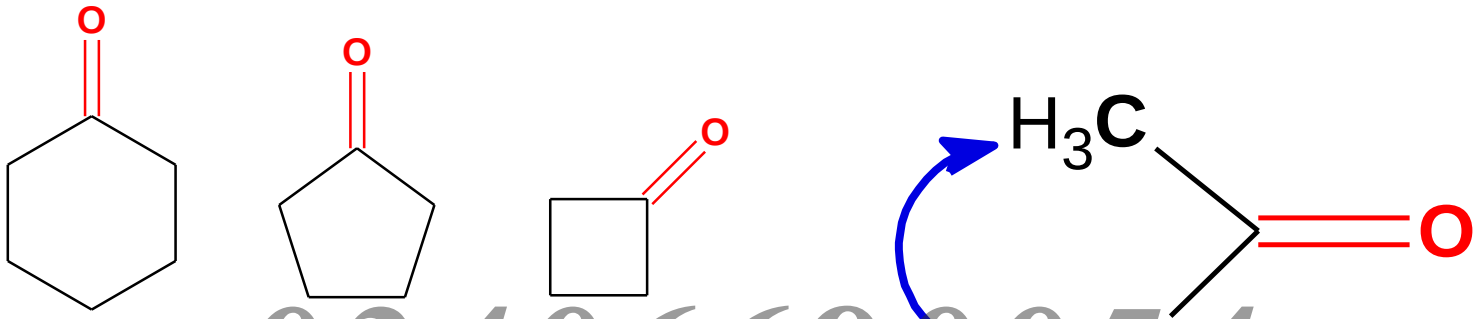
IR Spectroscopy

Factors influencing IR absorption

- **Additional to conjugation effect the other factors influence carbonyl absorption are**
  - **Ring strain**
  - **α substitution effect**
  - **Hydrogen bonding**

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IR Spectroscopy



1715 cm^{-1} 1740 cm^{-1} 1775 cm^{-1}

Angle ~ 120° ~ 109° ~90°

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Ring Strain: Increased double bond character: Higher

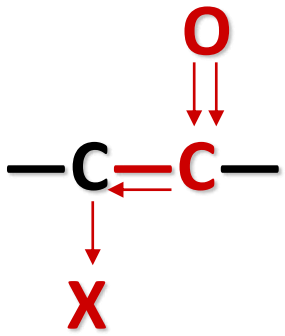
If the bond angle increases above 120°, the opposite effect is observed, as for di-*tert*-butyl ketone the Absorption frequency of carbonyl is 1695 cm^{-1}.

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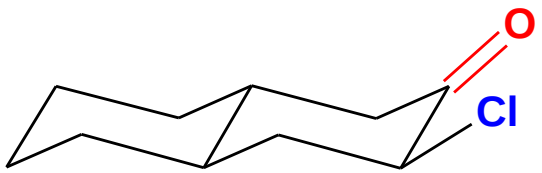
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IR Spectroscopy

α substitution effect (Chlorine or other halogens)



Result in stronger bound ➔ higher frequency

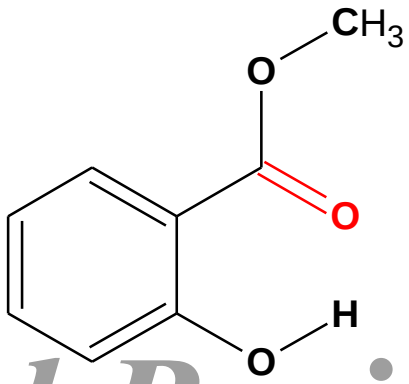


1750 cm^{-1}

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Hydrogen bonding

Decrease C=O strength → lower frequency



1680 cm^{-1}

2800 cm^{-1}

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IR Spectroscopy

- Strong hydrogen bonding in carboxylic acids leads to dimerization and due to this all three bonds i.e. OH, C=O and CO appear at lower frequency in IR spectra.

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Instrumentation
IR Spectroscopy-VII

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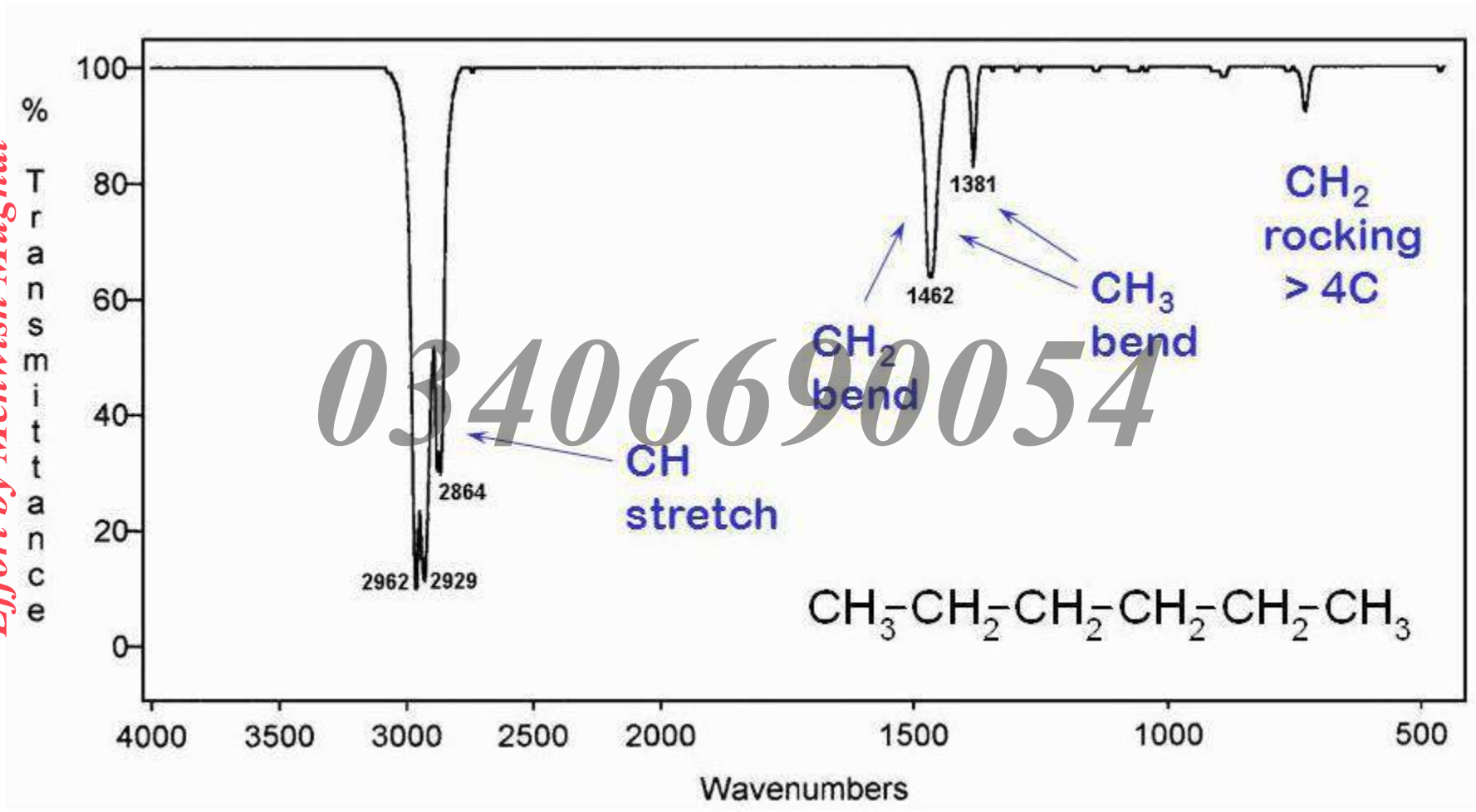
IR Spectroscopy

Interpretation of IR spectra

- The **most useful regions** are as follows:
  - 1680-1750 cm^{-1}:C=O stretches feature very strongly in IR specta
  - 2700-3100 cm^{-1}: different types of C-H stretching vibrations.
  - 3200-3700 cm^{-1}: various types of O-H and N-H stretching vibrations.

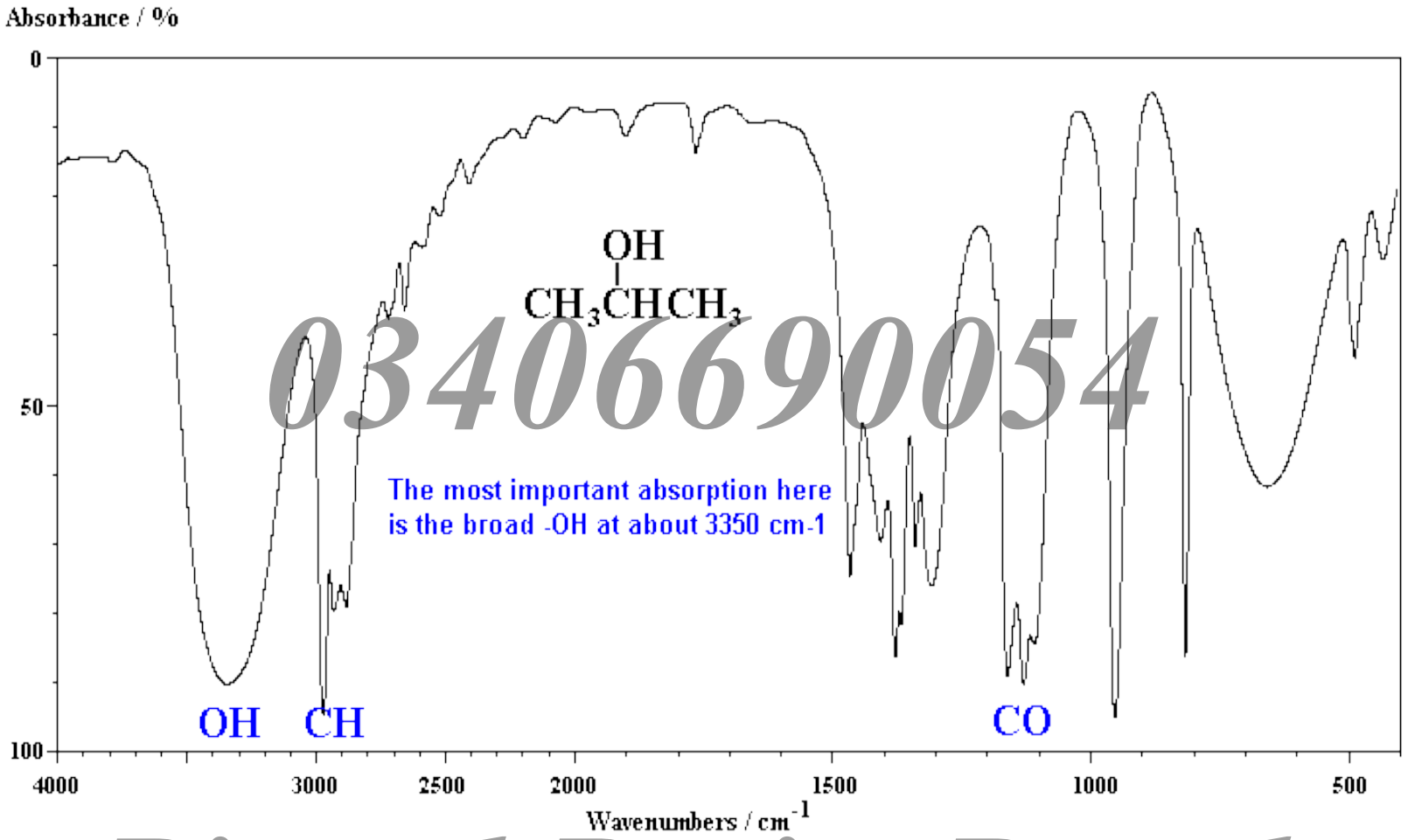
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IR Spectroscopy



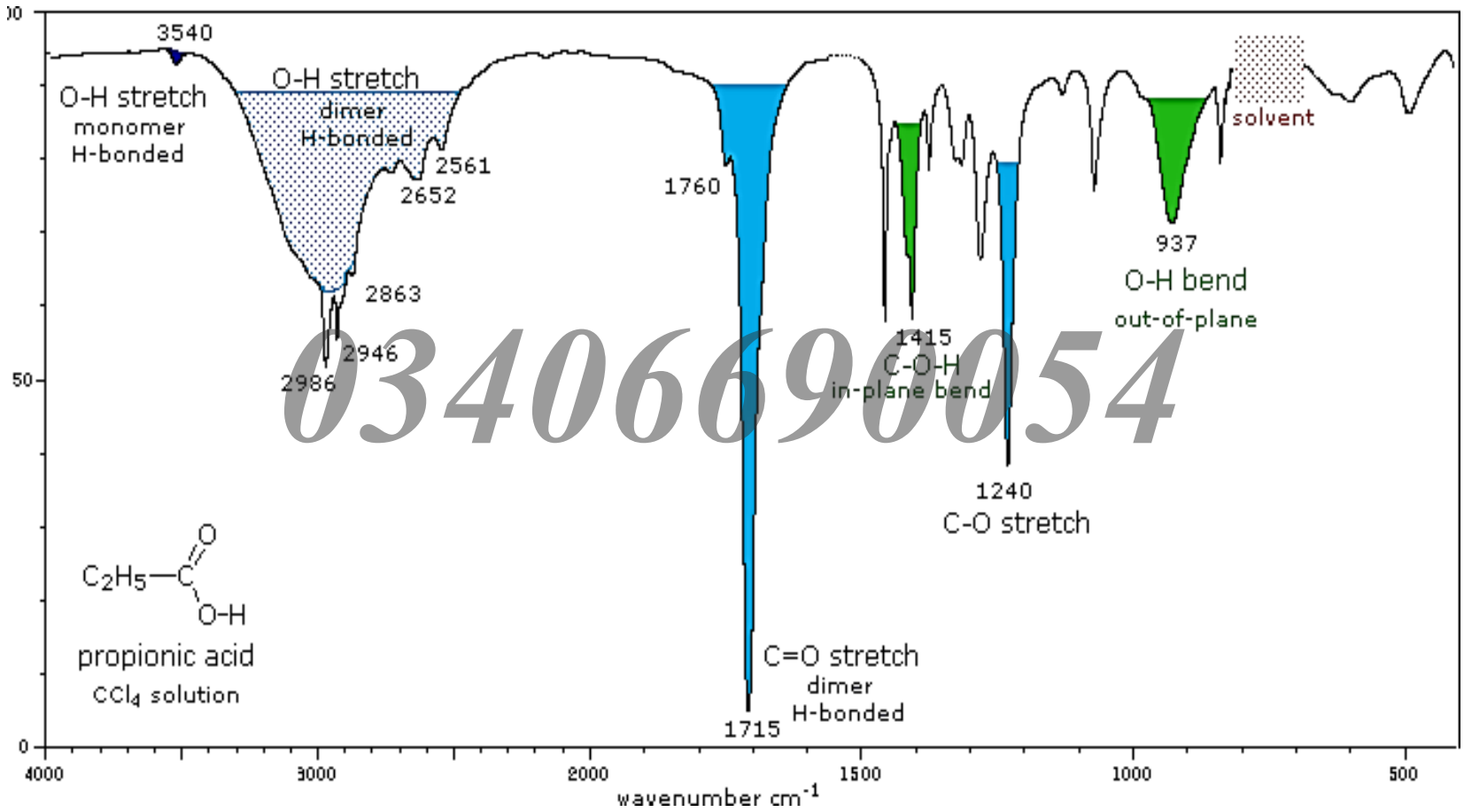
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IR Spectroscopy



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IR Spectroscopy



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IR Spectroscopy

- Too many bonds absorb in the region of 600-1600 cm^{-1} to allow confident assignment of individual bands.
- However, this region is useful as a fingerprint of a molecule, *i.e.* if the spectrum is almost identical to an authentic reference spectrum then the structure can be assigned with some confidence.

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Instrumentation
IR Spectroscopy-VIII

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IR Spectroscopy

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IR Spectrophotometer

- **Main components are**

1. **IR radiation source**
2. **Sample compartment**
3. **Monochromator**
4. **Read out device**

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IR Spectroscopy

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IR Spectroscopy



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IR Spectroscopy

Radiation Source:

- **Nernest filament is composed of a mixture of oxides of zirconium, thorium and yttrium, which is formed into a hollow rod about 2 mm in diameter and 2-5 cm in length.**
- **Globar is a bonded silicon carbide rod about 7 mm in diameter and 5 cm in length.**

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IR Spectroscopy

- Detector works on principle of a thermocouple, which converts the infrared (electromagnetic) energy into electrical energy.
- Sample are placed as drop of liquid and solids can be mixed with nujol.

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Instrumentation
IR Spectroscopy-IX

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IR Spectroscopy

Applications of IR Spectroscopy

1. **Structure Determination**
   - Search for benzene ring presence
   - Search for functional group regions (carbonyl, NH, OH, CO etc.)
   - Search for CH of different types

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IR Spectroscopy

Applications of IR Spectroscopy

2. Identification of compound

- Matching given structure (metal complex formation)
- Matching of standard through finger print region
- Differentiate if we have alcohol or ether

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IR Spectroscopy

Applications of IR Spectroscopy

3. Detection of impurities

- Cyclohexanone is readily detectable in cyclohexanol
- Based on functional groups or finger print region

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Instrumentation
IR Spectroscopy-X

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IR Spectroscopy

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Applications of IR Spectroscopy

4. Progress of reaction
   - Oxidation of secondary alcohol to ketone
   - Reduction of alcohol to alkene
   - Hydrolysis of sugars

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IR Spectroscopy

Applications of IR Spectroscopy

5. Presence of composition of mixture

- IR is a qualitative technique
- In rare cases it may be used as quantitative e.g. ortho, meta and para xylene

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END

- More experimental data is available in Introduction to spectroscopy by Pavia and in local published book of organic spectroscopy by Dr Younas.

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Instrumentation
Atomic Spectroscopy-I

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Atomic Spectroscopy

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Atomic Structure

- Atom is a divisible particle that represents the properties of matter.
- It has positive charge confined in the centre.
- The negative charge particles are found around the nucleus in different energy orbits.

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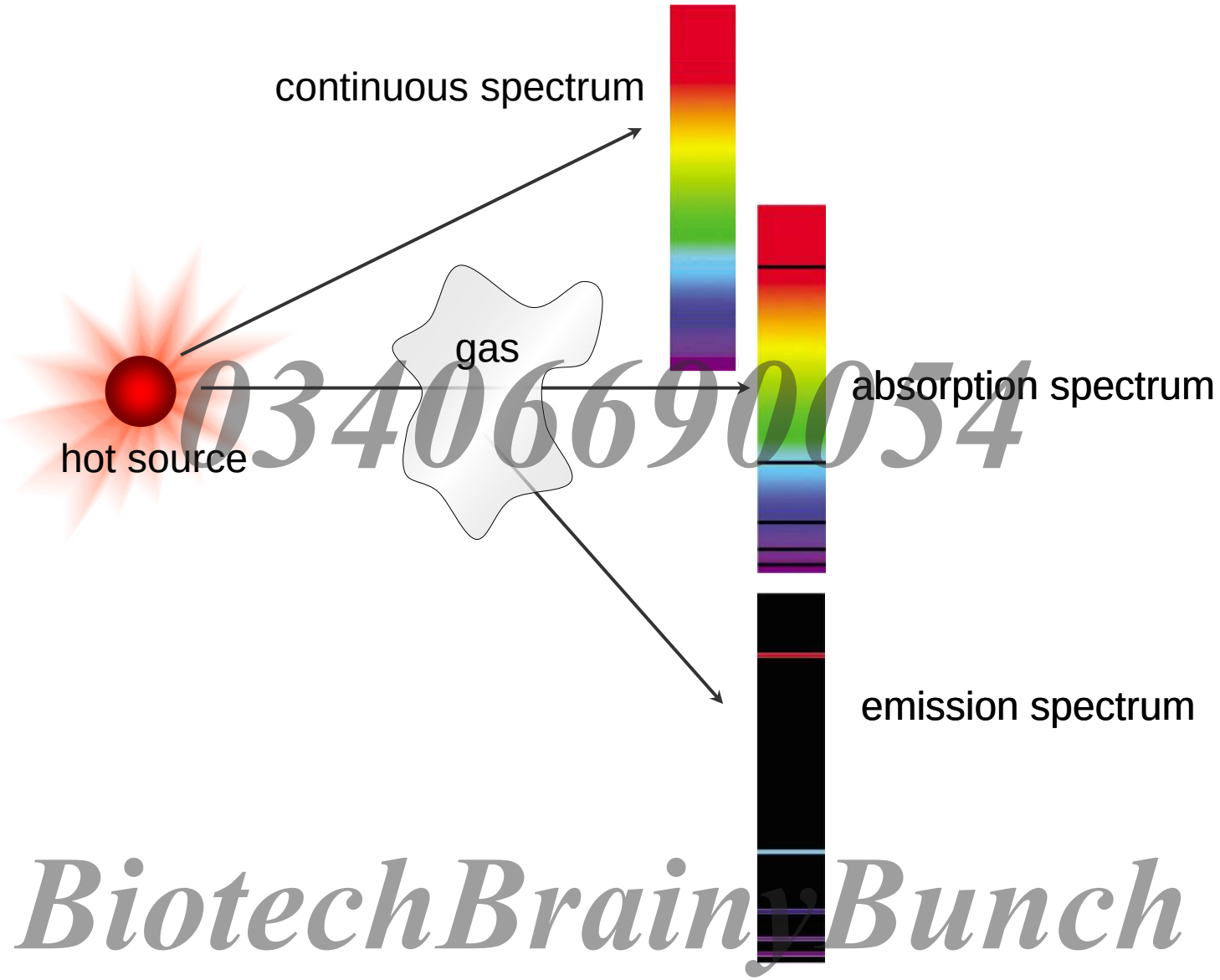
Atomic Spectroscopy

Atomic Spectrum

- When we give energy to the atom in any form, it could be electromagnetic radiation or thermal energy, electrons absorb that energy and jump in higher shell.
- They emit same amount of energy when come back to ground state.

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Atomic Spectroscopy



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Atomic Spectroscopy

Atomic Spectrum

- Bohr postulate that the electrons reside in different energy levels and they need a specific amount of energy to jump.
- When they come back to ground state same amount of energy emit.
- The colour in the spectrum correspond to that energy.

END

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Instrumentation Atomic Spectroscopy-II

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Atomic Spectroscopy

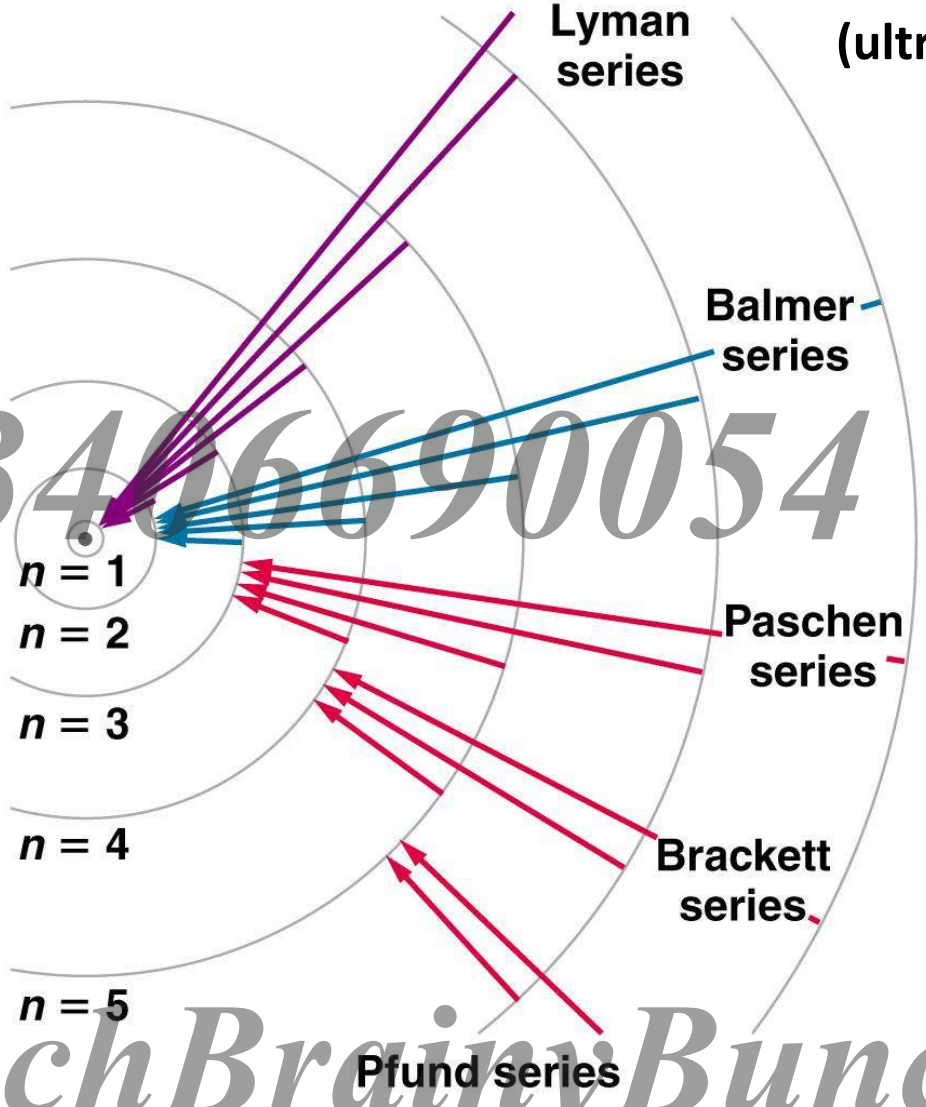
Hydrogen spectral lines

- Before Bohr different scientists have proved experimentally that hydrogen gas absorb or emit energy of different ranges i.e. UV range (Lymen series), Visible range (Balmer series), IR region (Paschen series).
- Bohr explained it.

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Atomic Spectroscopy

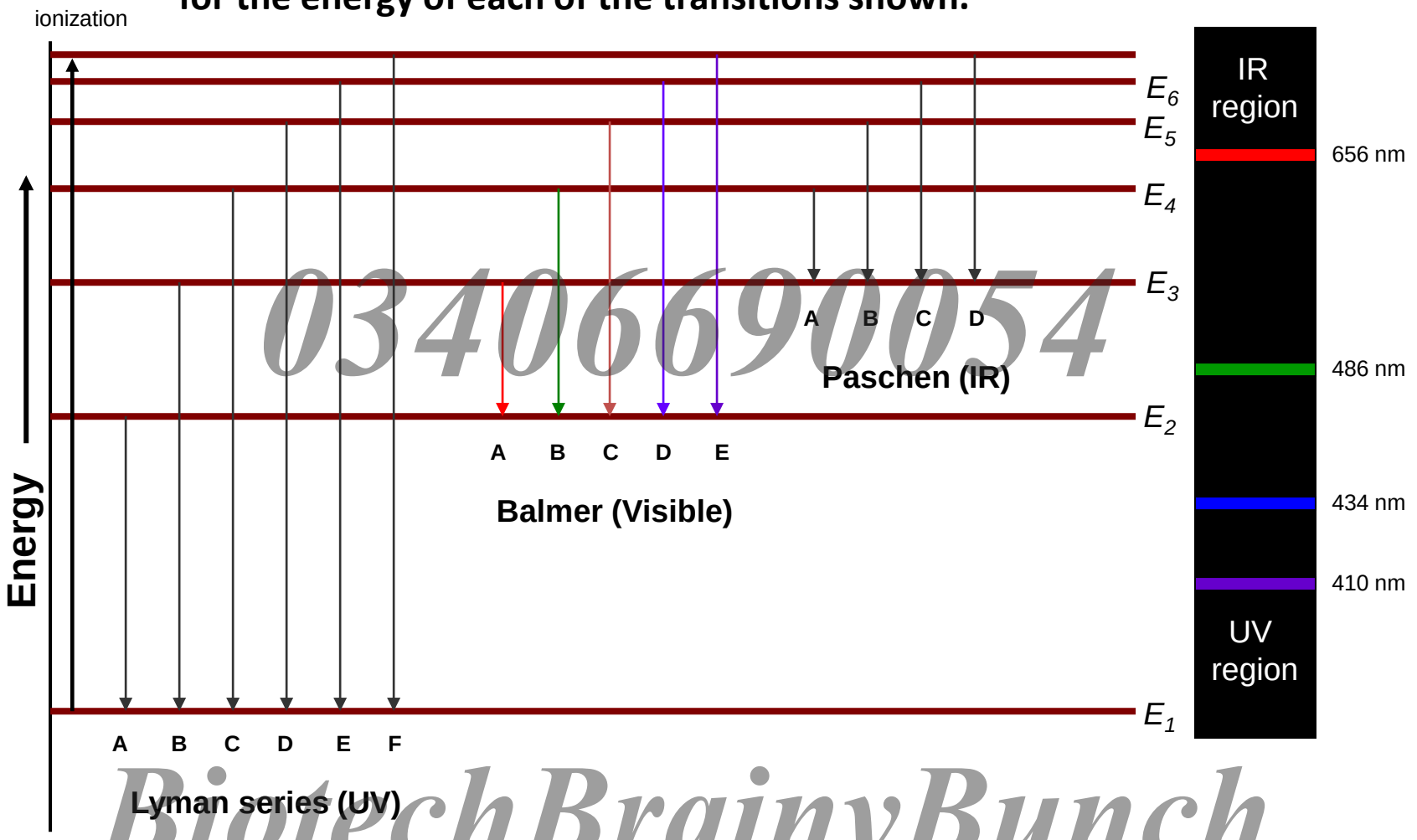
HYDROGEN SPECTRAL LINES



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Atomic Spectroscopy

Bohr's model of the atom accounted mathematically for the energy of each of the transitions shown.



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Atomic Spectroscopy

Absorption spectrum of different atoms

- These absorption or emission spectrum are different for different atoms.
- Some of the spectral lines may be common but some other may be characteristic of that atom.

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Instrumentation Atomic Spectroscopy-III

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Atomic Spectroscopy

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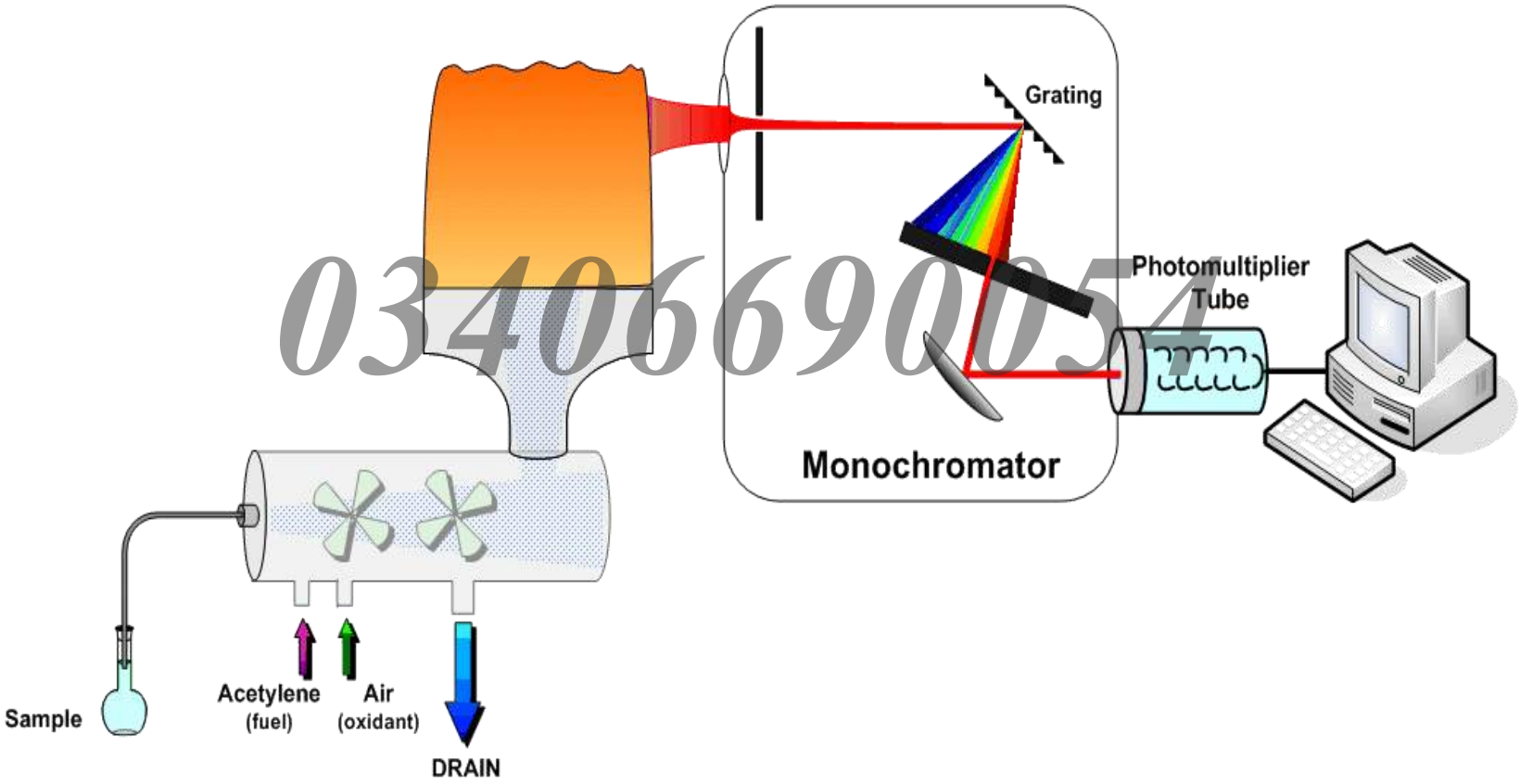
Atomic absorption spectroscopy

- Light source (HCL)
- Nebulizer
- Atomizers
- Monocromators
- Readout device
- Printer

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Atomic Spectroscopy

Atomic Emission Spectroscopy



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Atomic Spectroscopy

- Flame photometers can be used for very few metals that burn with coloured flames (Na, K, Li etc.).
- We will discuss the detail of these parts one after another.
- Some we have already discussed.

END

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Analytical Chemistry & Instrumentation

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Instrumentation
Atomic
Spectroscopy-IV

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Atomic Spectroscopy

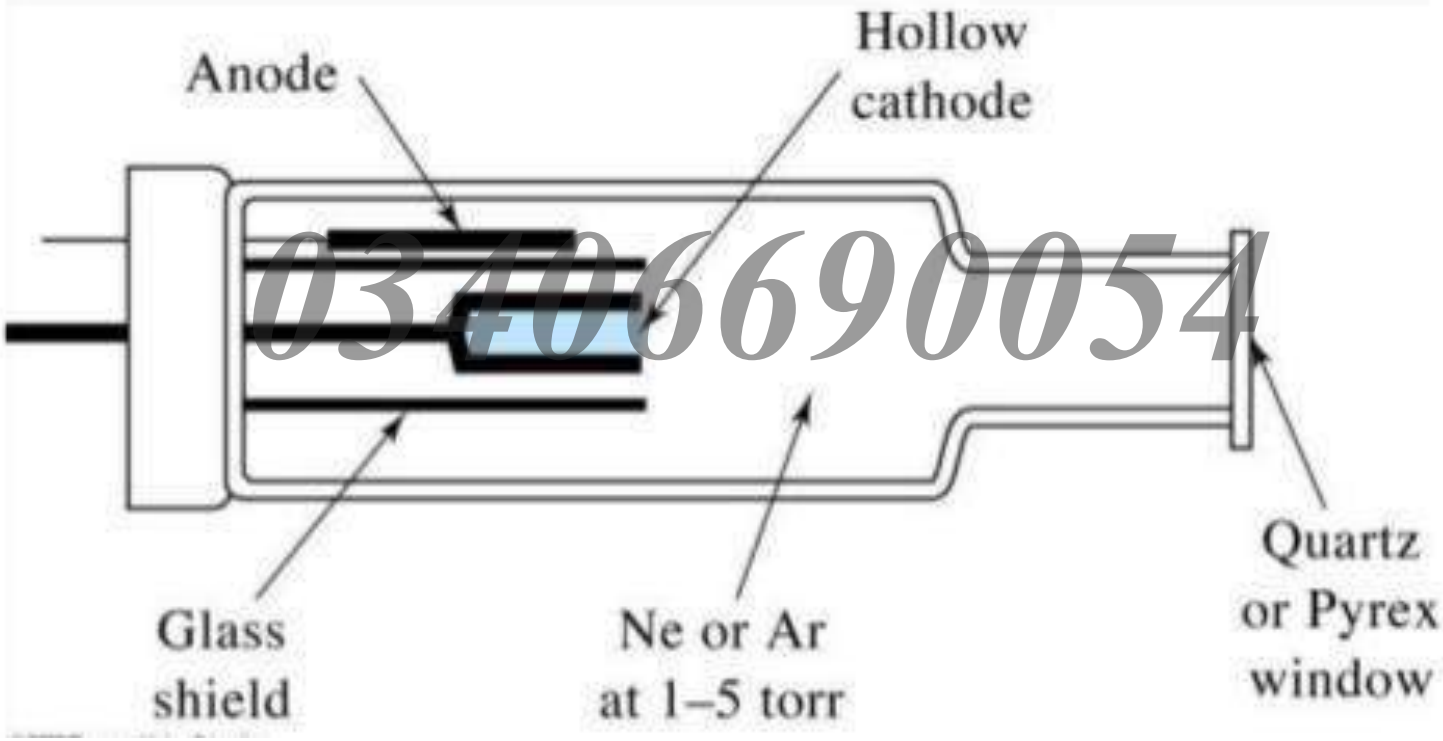
Hollow Cathode Lamp

- HCL is a vacuumed tube bearing cylindrical cathode and some noble gas at very low pressure.
- It generates photons of particular metal for which cathode is made.

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Atomic Spectroscopy

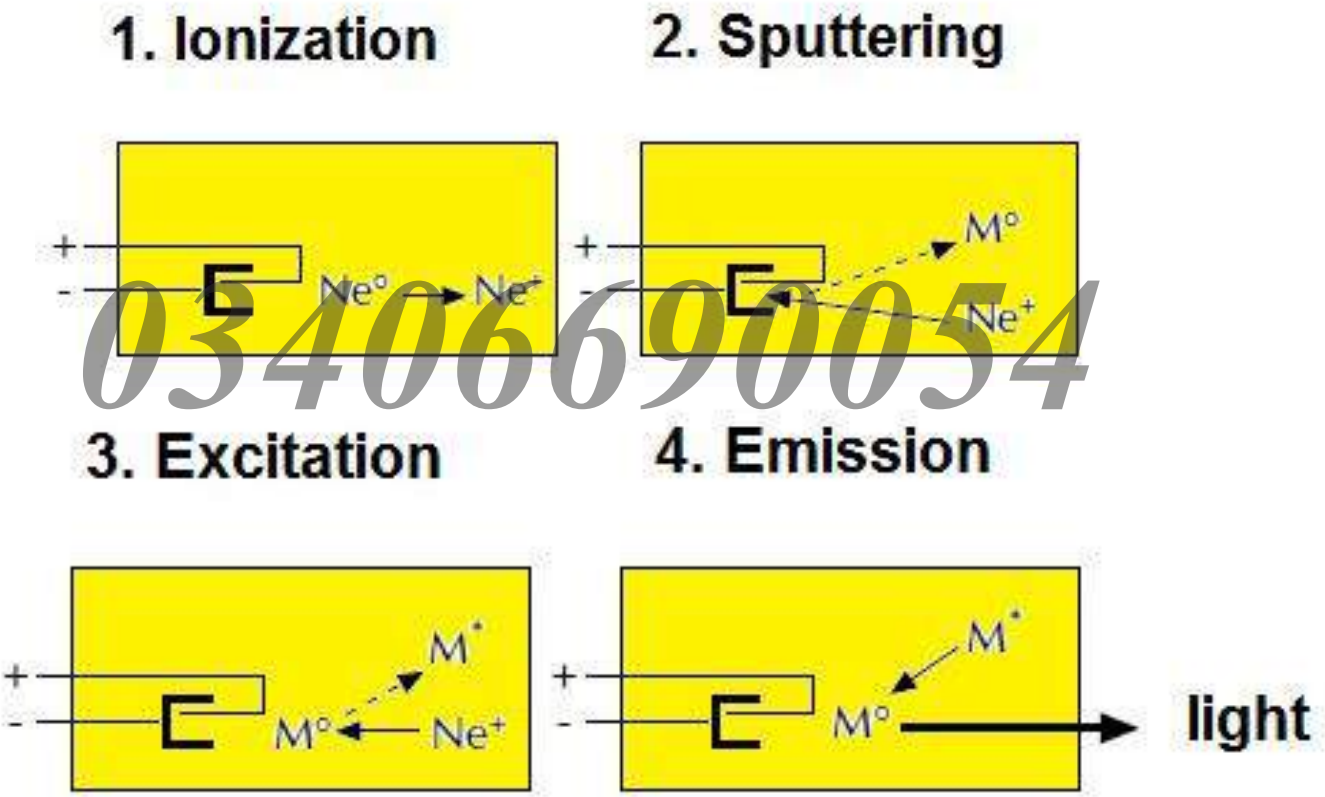
Hollow Cathode Lamp



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Atomic Spectroscopy

Generation of Photons



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Atomic Spectroscopy

- In cases where we have to analyze more volatile metals, the electrodeless discharge lamp (EDL) is used. A small amount of the metal or salt of the element is sealed inside a quartz bulb.
- They offer a better precision and lower detection limits.

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Instrumentation Atomic Spectroscopy-V

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Atomic Spectroscopy

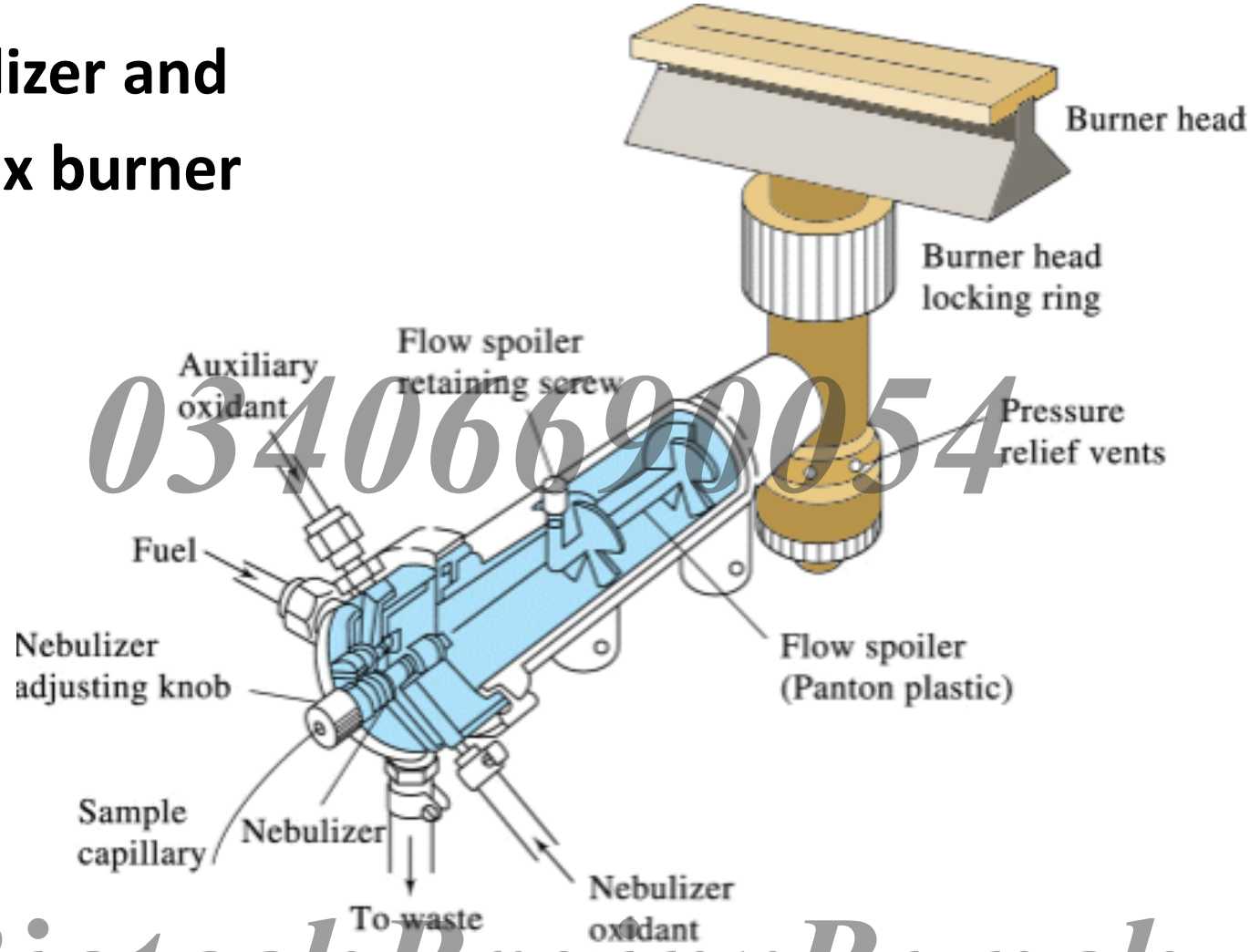
Nebulizers

- Nebulizers convert liquid into mist.
- Then burner remove heavy droplets and only allow the mist to pass through it.
- The remaining part move to waste while almost equal amount of sample reaches to flame.

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Atomic Spectroscopy

Nebulizer and Premix burner



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Atomic Spectroscopy

- These nebulizers are pneumatic and work with help of pressurized gases.
- These gases push the sample into burner which atomize them.

END

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Analytical Chemistry & Instrumentation

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Instrumentation
Atomic
Spectroscopy-VI

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Atomic Spectroscopy

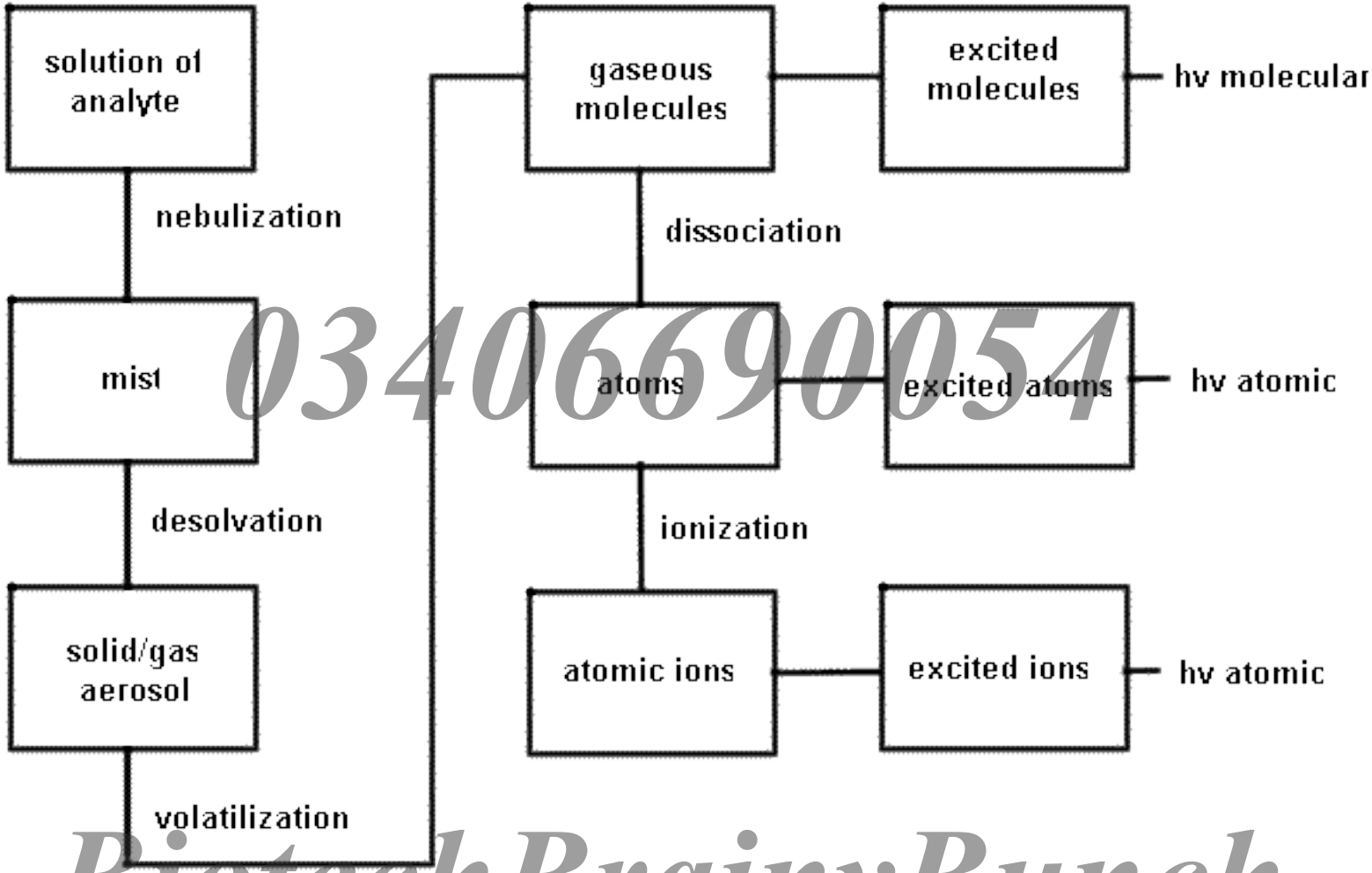
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Atomizers

- Atomizers are used to atomize the sample.
- The sample contain the metal in ionic form (salt).
- Atomizers / burners heat the sample and convert them to atomic form.

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Atomic Spectroscopy



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Atomic Spectroscopy

Use of Atomizer

- Metal like As and Hg are not atomized in this form so they are transformed to metal hydrides (gaseous form) that can be atomized in normal flame.
- Solid or biological samples have to digest properly to aspirate in flame.

END

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Analytical Chemistry & Instrumentation

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Instrumentation
Atomic Spectroscopy-VII

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Atomic Spectroscopy

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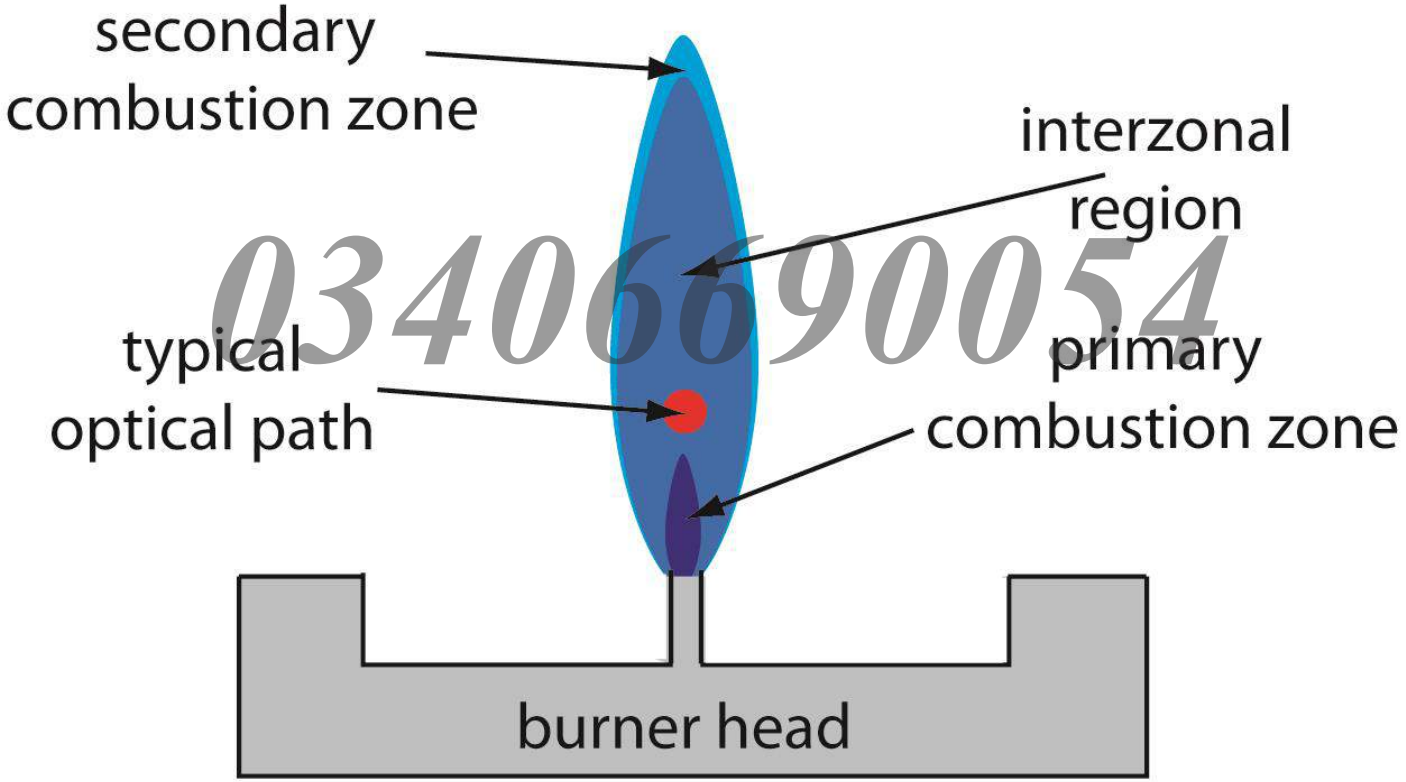
Flame Atomization

- Flame atomizers transformed sample into atomic form.
  - From their ionic solution
  - From hydrides gases

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Atomic Spectroscopy

Flame Structure



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Atomic Spectroscopy

- **Preheating zone-** Where combustion mixture is heated to the ignition temperature by thermal conduction from the primary reaction zone.
- **Primary reaction zone-** This zone is about 0.1 mm thick at atmospheric pressure
  - There is no thermodynamic equilibrium in this zone and the concentration of ions and free radicals is very high.
- **Interconal zone –** Extend up to considerable height. The maximum temperature is achieved just above the tip of the inner zone. This zone is used for flame photometry.
- **Secondary reaction zone -** In this zone, the products of the combustion processes are burnt to stable molecular species by the surrounding air.

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Atomic Spectroscopy

FULE	OXIDANT	TEMPERTURES ^{0}C
Natural Gas	Air	1700-1900
Natural Gas	Oxygen	2700-2800
Hydrogen	Air	2000-2100
Hydrogen	Oxygen	2550-2700
Acetylene	Air	2100-2400
Acetylene	Oxygen	3050-3150
Acetylene	Nitrous Oxide	2600-2800

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Atomic Spectroscopy

Use of Flame atomization

- **Most abundantly used flame is air-acetylene.**
- **Some of the metals like Al can not be atomized at this temperature so have to use nitrous oxide-acetylene flame.**

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Instrumentation
Atomic
Spectroscopy-VIII

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Atomic Spectroscopy

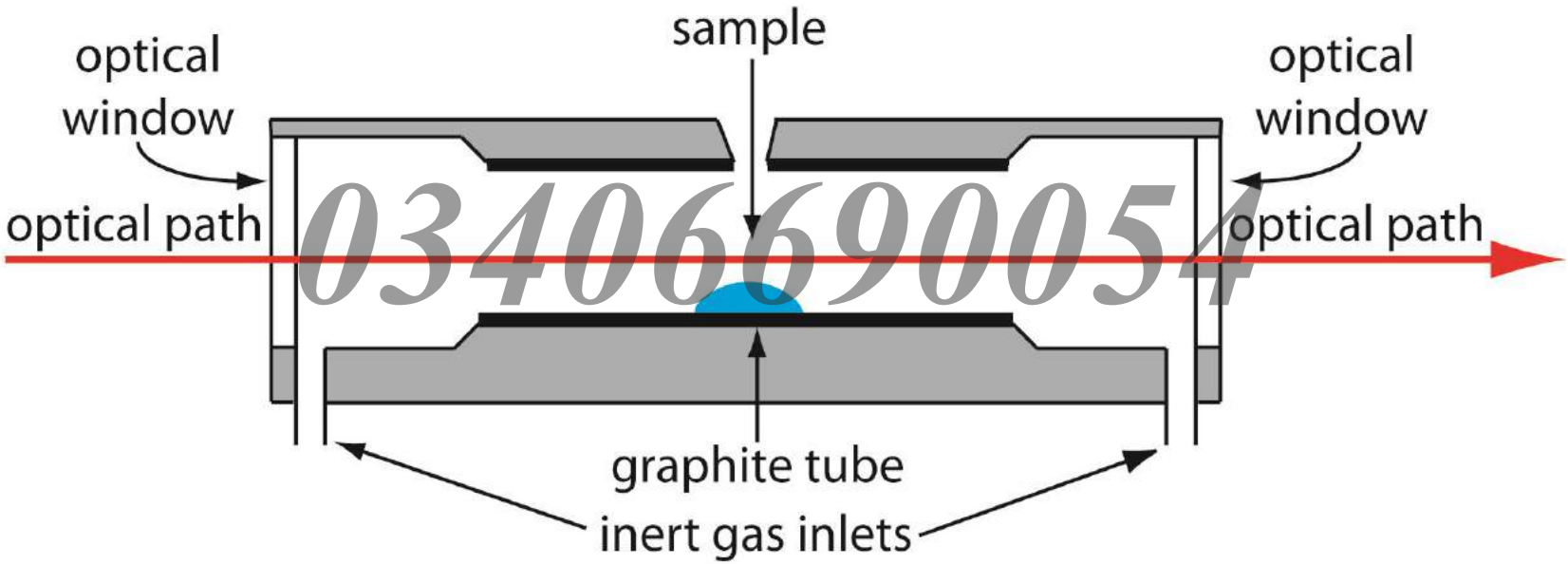
Graphite Furnace Atomizer

- Is a hollow graphite cylinder (5 cm length x 1 cm diameter).
- Tube is surrounded by a water circulation jacket separated by a gas space where an inert gas (Ar/ N) is circulated.
- A small amount of analyte sample solution (1-100 µL) is introduced in the cell.

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Atomic Spectroscopy

Graphite Tube Furnace



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Atomic Spectroscopy

Comparison of FAAS with GFAAS:

- In FAAS only liquid sample can be injected.
- In GFAAS both liquid and solid samples can be used.
- Sample quantity for GFAAS is minimized as compared to FAAS.
- Analytical precision is good in FAAS (10%) but has low sensitivity.

END

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Instrumentation
Atomic Spectroscopy-IX

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Atomic Spectroscopy

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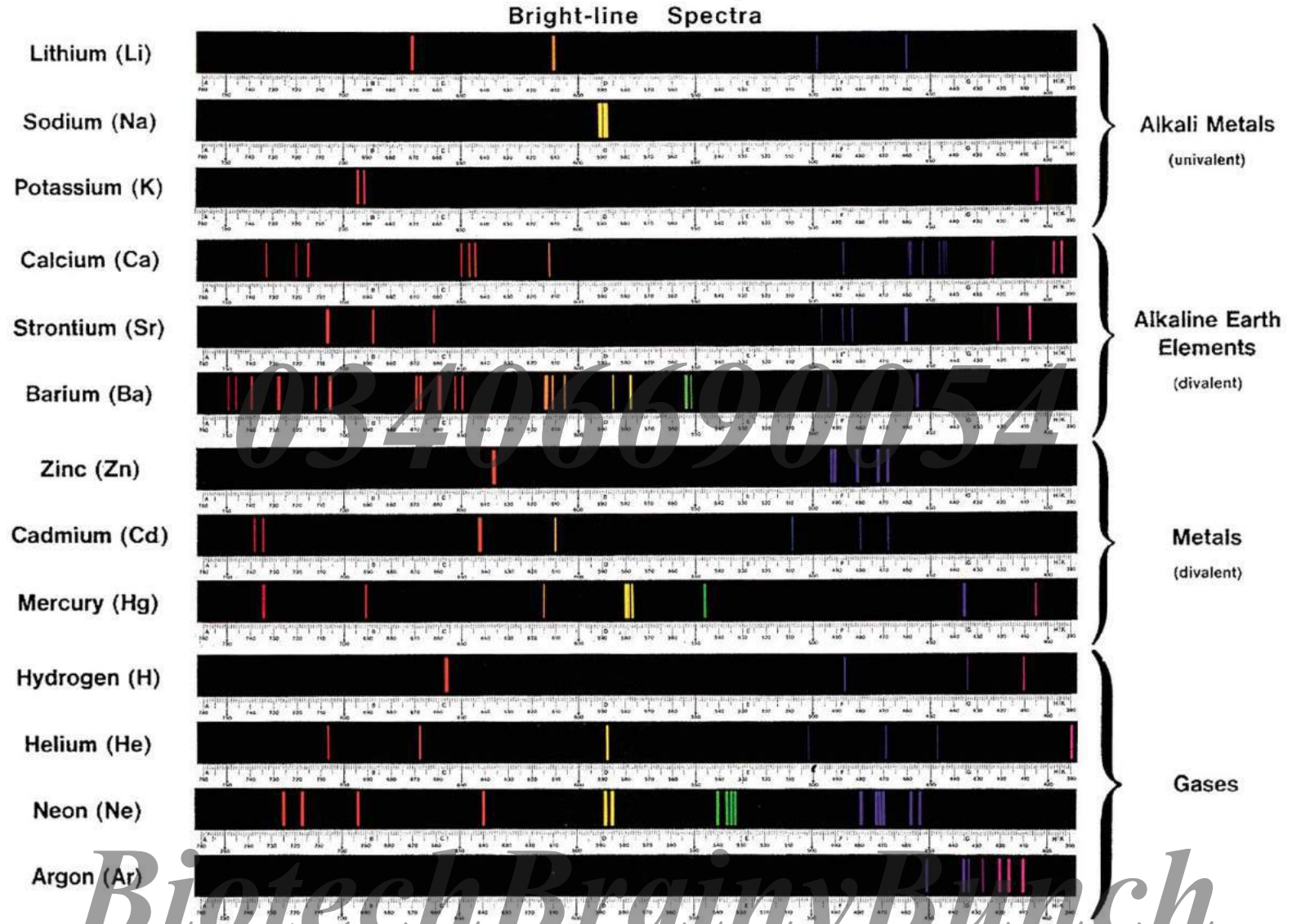
Interferences

- Spectral interferences
- Non-spectral interferences
  - Matrix interferences
  - Chemical interferences
  - Ionization interferences

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- If emission line of two overlap, a spectral interference is said to exist (a rare case for wavelength-specific atomic absorption spectroscopy, as the typical emission line width may be only 0.002 nanometers, actual overlap is extremely rare).
- The chances for spectral interference increase when multi-element lamps are used, where the source may contain close emission lines for several elements.
- Particular care should be exercised when secondary analytical wavelengths are being used in a multi-element lamp.
- Procedures for circumventing spectral interference include narrowing the monochromator slit width or using an alternate wavelength.

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Removal of spectral interferences

1. Continuous source background correction
   - BGC D2
2. Zeeman background correction
   - uses the principle that the electronic energy levels of an atom, placed in a strong magnetic field, are changed.

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Instrumentation

Atomic Spectroscopy-X

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Matrix interferences

- **Occur at the nebulization step, if the sample is more viscous**
- **The sample uptake rate or nebulization efficiency is different**

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Atomic Spectroscopy

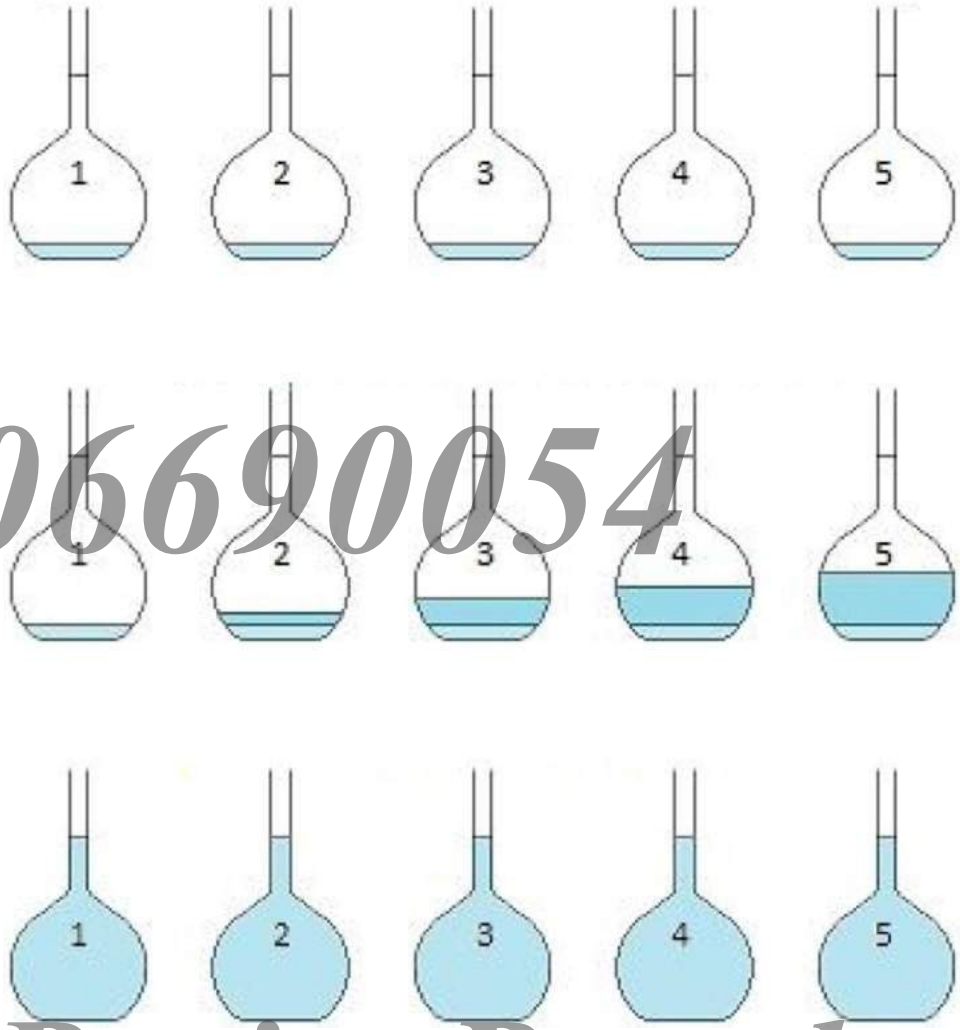
- An example of this type of interference is the effect of acid concentration on absorbance (phosphoric acid concentration increases, change the sample viscosity).
- Matrix interferences can also cause positive error. The presence of an organic solvent in a sample will produce an enhanced nebulization efficiency.
- One way of compensating for this type of interference is to match the matrix (by blank preparation)

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Method of Standard Additions

1. **Aliquots of a standard are added to portions of the sample**
2. **Add standard solution**
3. **Make up the volume**

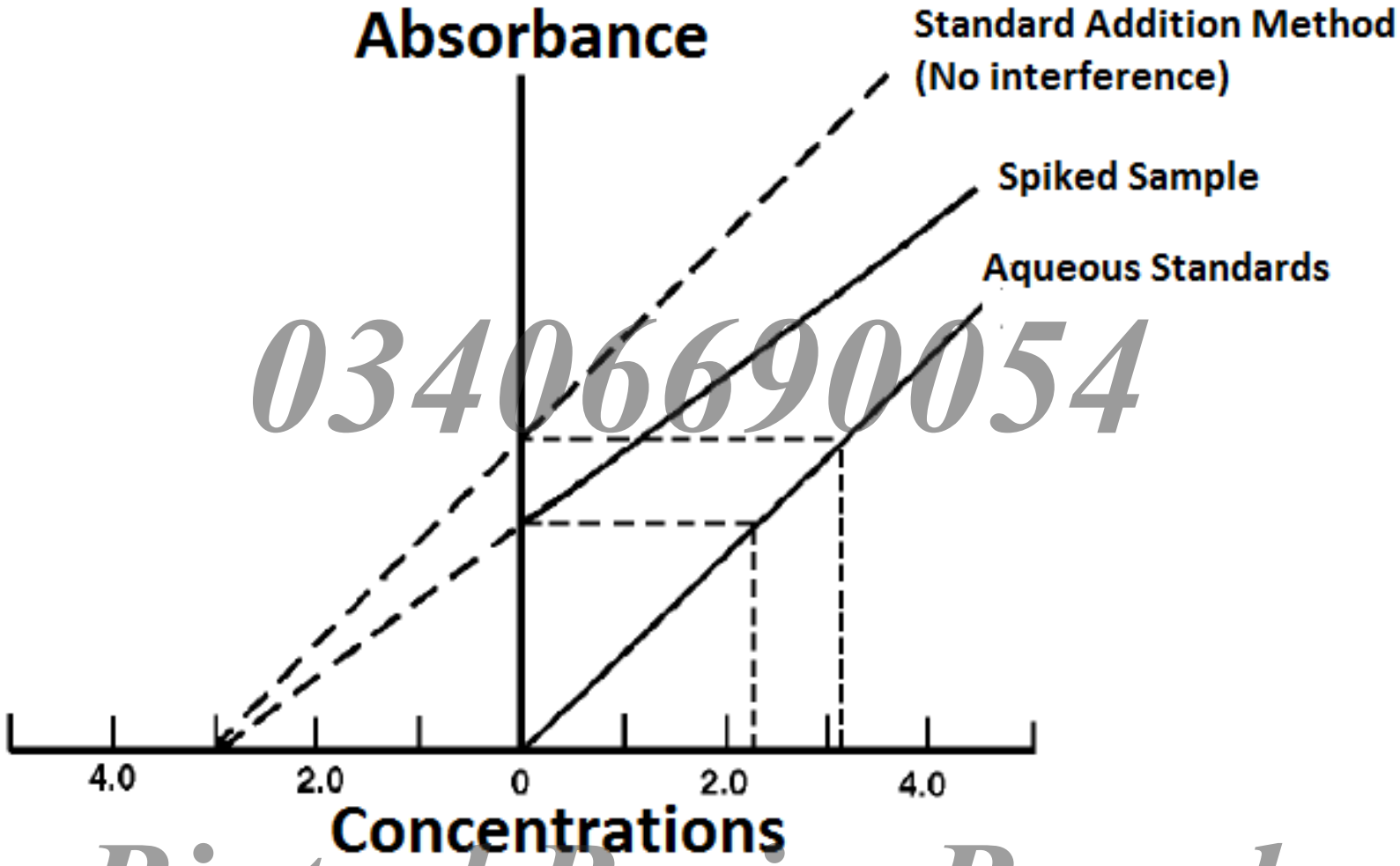


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The method of standard additions
will not compensate for background absorption or other types of spectral interference,
and normally will not compensate for chemical or ionization types of interference.

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Instrumentation
Atomic Spectroscopy-XI

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Atomic Spectroscopy

Chemical interferences

- In atomizer sufficient energy must be available to create free atoms.
- If the sample contains a component which forms a thermally stable compound with the analyte, will not decompose completely

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Atomic Spectroscopy

- The effect of phosphate on calcium is an example of a chemical interference.
- Calcium phosphate does not totally dissociate in an air acetylene flame.
- Therefore, as phosphate concentration is increased, the absorbance due to calcium atoms decreases.
- There are two means of dealing with this problem.

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- One is to eliminate the interference by adding an excess of another element or compound which will also form a thermally stable compound with the interferent. (In the case of calcium, lanthanum is added to tie up the phosphate and allow the calcium to be atomized, making the calcium absorbance independent of the amount of phosphate).
- Second way is using a hotter flame. Use of nitrous oxide-acetylene flame instead of air-acetylene and can often be used to minimize chemical interferences. (The phosphate interference on calcium is not observed with a nitrous oxide-acetylene flame)

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Atomic Spectroscopy

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Use of Graphite Furnace

- Graphite furnace can be used for higher atomization temperature.
- The sensitivity is also increased from mg/L to µg/L.

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Instrumentation
Atomic Spectroscopy-XII

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Atomic Spectroscopy

Ionization interferences

- In atomizer the dissociation process does not stop at the ground state atom.
- the ground state atom thermally raised to the excited state or an electron creating an ion.
- This electronic rearrangement deplete the number of ground state atoms

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Atomic Spectroscopy

- It was observed in case of alkali and alkaline earth metal (e.g. sodium).
- Ionization interference can be eliminated by adding an excess of an element which is very easily ionized, creating a large number of free electrons in the flame and suppressing the ionization of the analyte.
- Potassium, rubidium, and cesium salts are commonly used as ionization suppressants.

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Use of HVG

- Arsenic salts are not atomized in air-acetylene or nitrous oxide-acetylene flame.
- Hydride vapour generator is used to convert arsenic, or antimony to respective hydride by reacting with $NaBH_4$ and HCl.
- Mercury can be atomized by cold vapour mercury technique

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Instrumentation Atomic Spectroscopy-XIII

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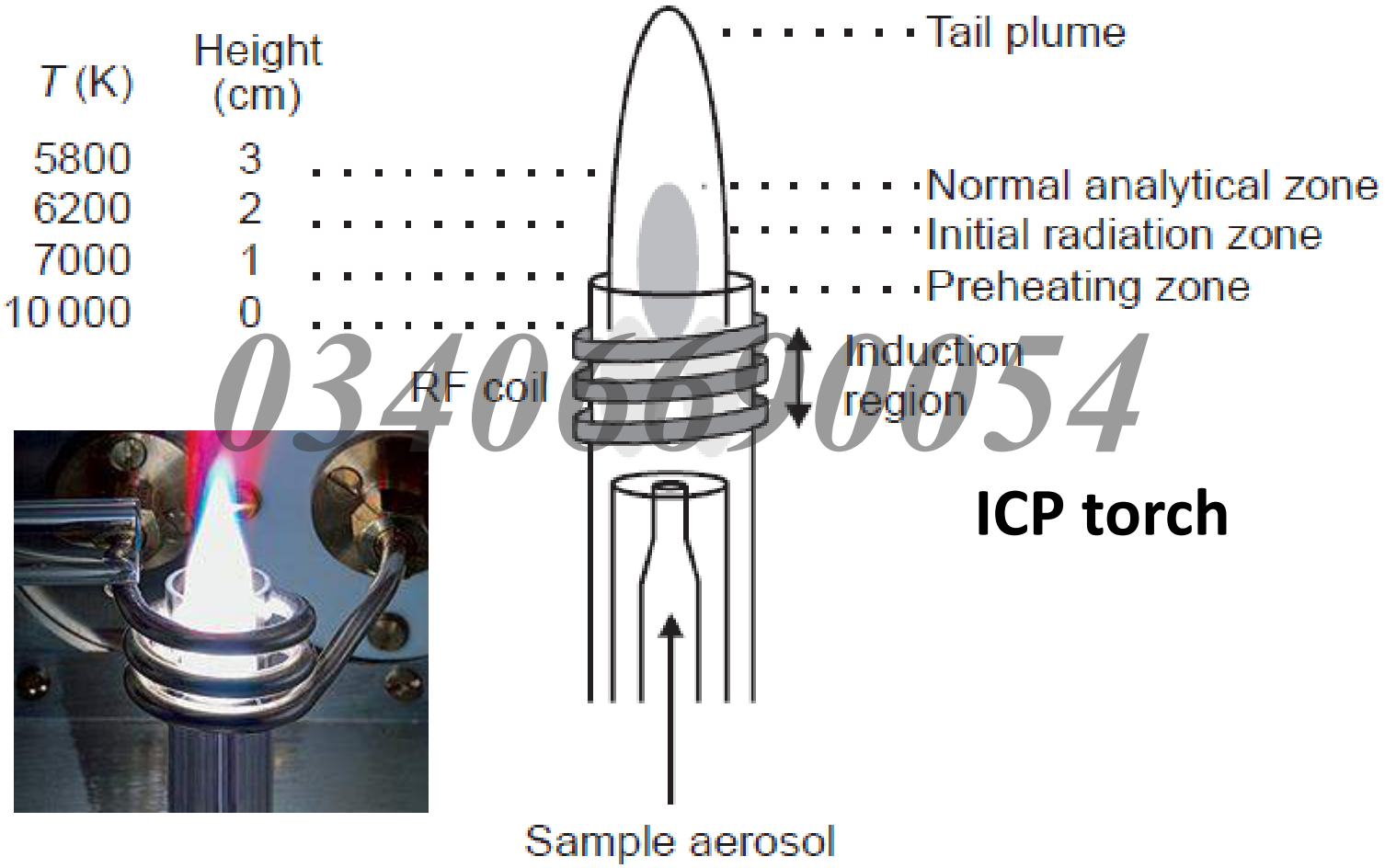
Atomic Spectroscopy

Atomic Emission Spectroscopy

- ICP/OES and ICP MS
- Atomization source changed to inductive coupled plasma
- Temperature of plasma can reach up to 10,000K
- 70 elements can be determined to ppb level
- Low chemical interferences

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Atomic Spectroscopy



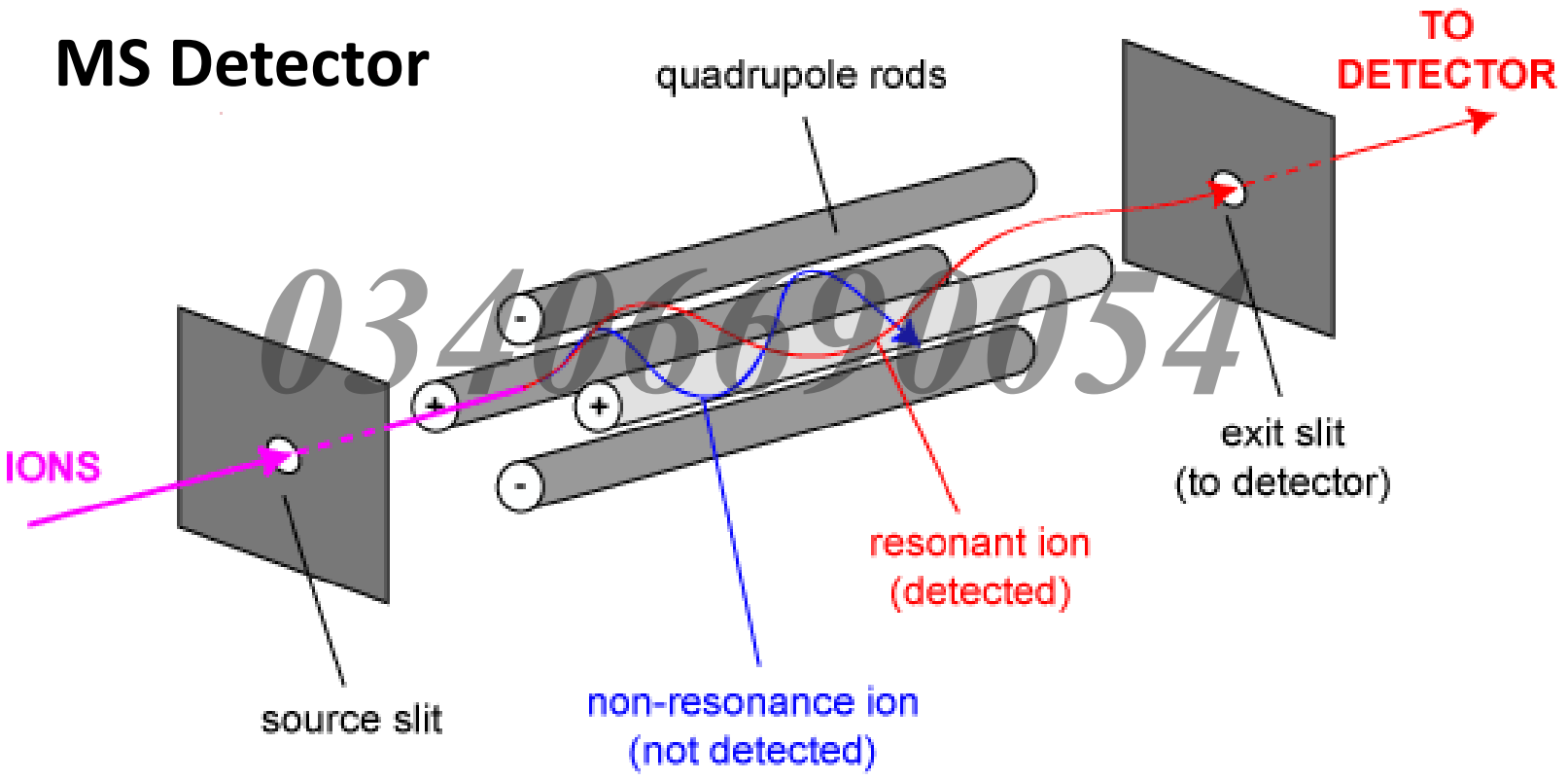
ICP torch

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Atomic Spectroscopy

Quadrupole

MS Detector



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Atomic Spectroscopy

Working of ICP

- The sample is prepared in liquid form as in case of FAAS.
- Sample is injected through nebulizer @ 0.5 to 1.5 mL/min.
- The emitted spectral lines pass through monochromator.
- The separation in MS is on basis of mass to charge ratio.

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Instrumentation
Atomic Spectroscopy-XIV

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Atomic Spectroscopy

Sample preparation techniques for atomic spectroscopy

- **Wet digestion**
- **Dry ashing**
- **Microwave dissolution**

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Atomic Spectroscopy

Wet Digestion

Biological samples → Treated with acids → Repeat till clear solution → Filter leftover



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Atomic Spectroscopy

Dry Ashing





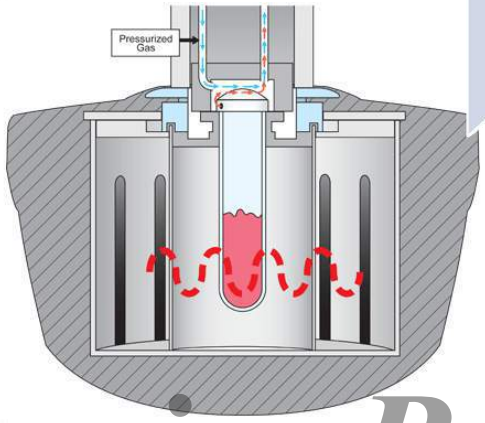
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Atomic Spectroscopy

Microwave dissolution

- Sealing of sample in microwave vessel
- Addition of suitable acids
- Keeping temperature to 200 C with increased pressure
- clear liquid sample of biological materials





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Atomic Spectroscopy

Sample preparation

- Technique can be optimized for better extraction
- Normally biological sample wet digested while plant materials digested through ashing
- Glass can be solubilize in HF while metal in aqua regia.
- Microwave can save time and sensitive material

END

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Instrumentation

Atomic Spectroscopy-XV

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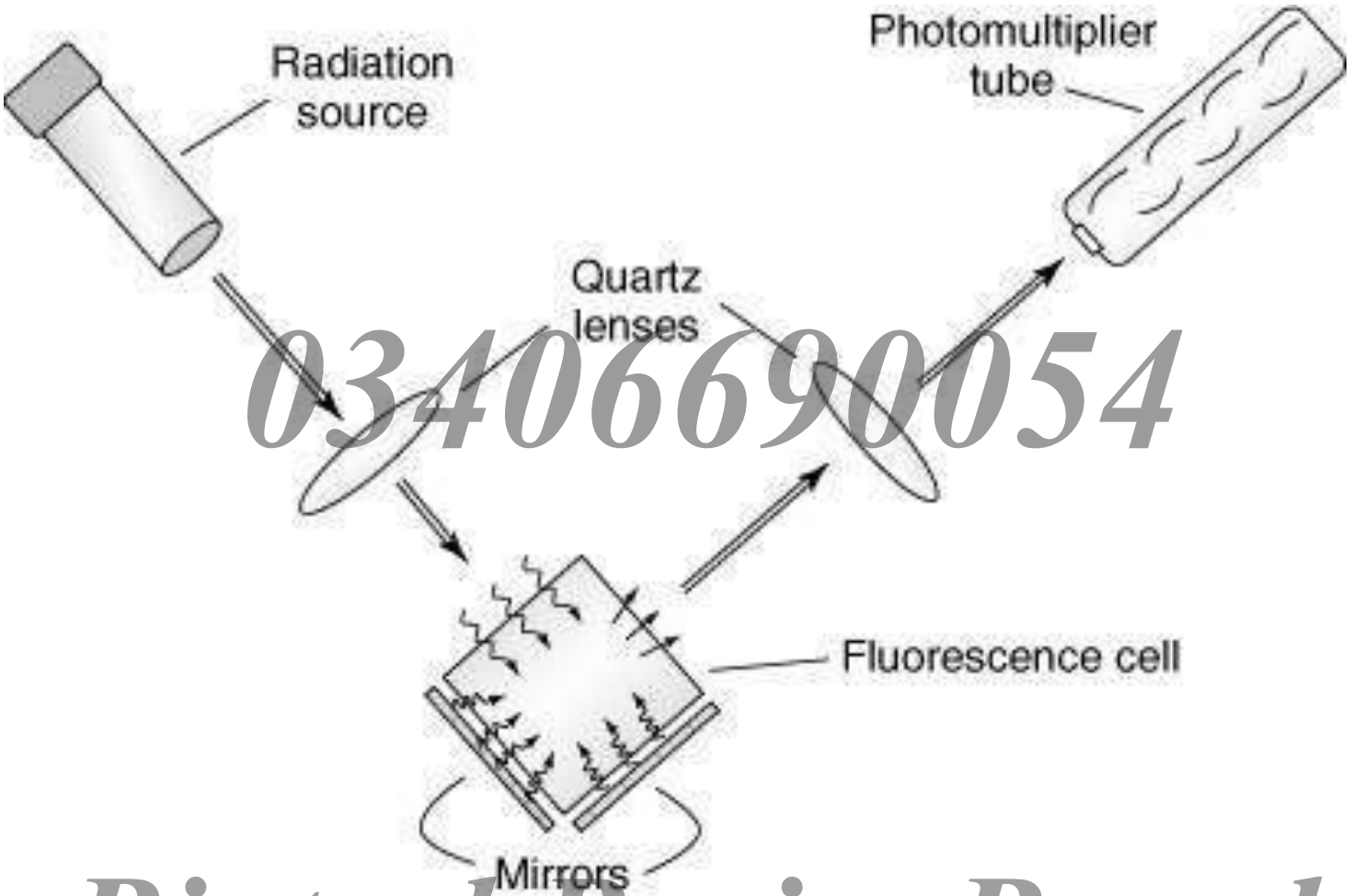
Atomic Spectroscopy

Atomic Florescence Spectroscopy

- Photoluminescence
  - Fluorescence
  - Phosphorescence
- Instrumentation

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Atomic Spectroscopy



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Atomic Spectroscopy

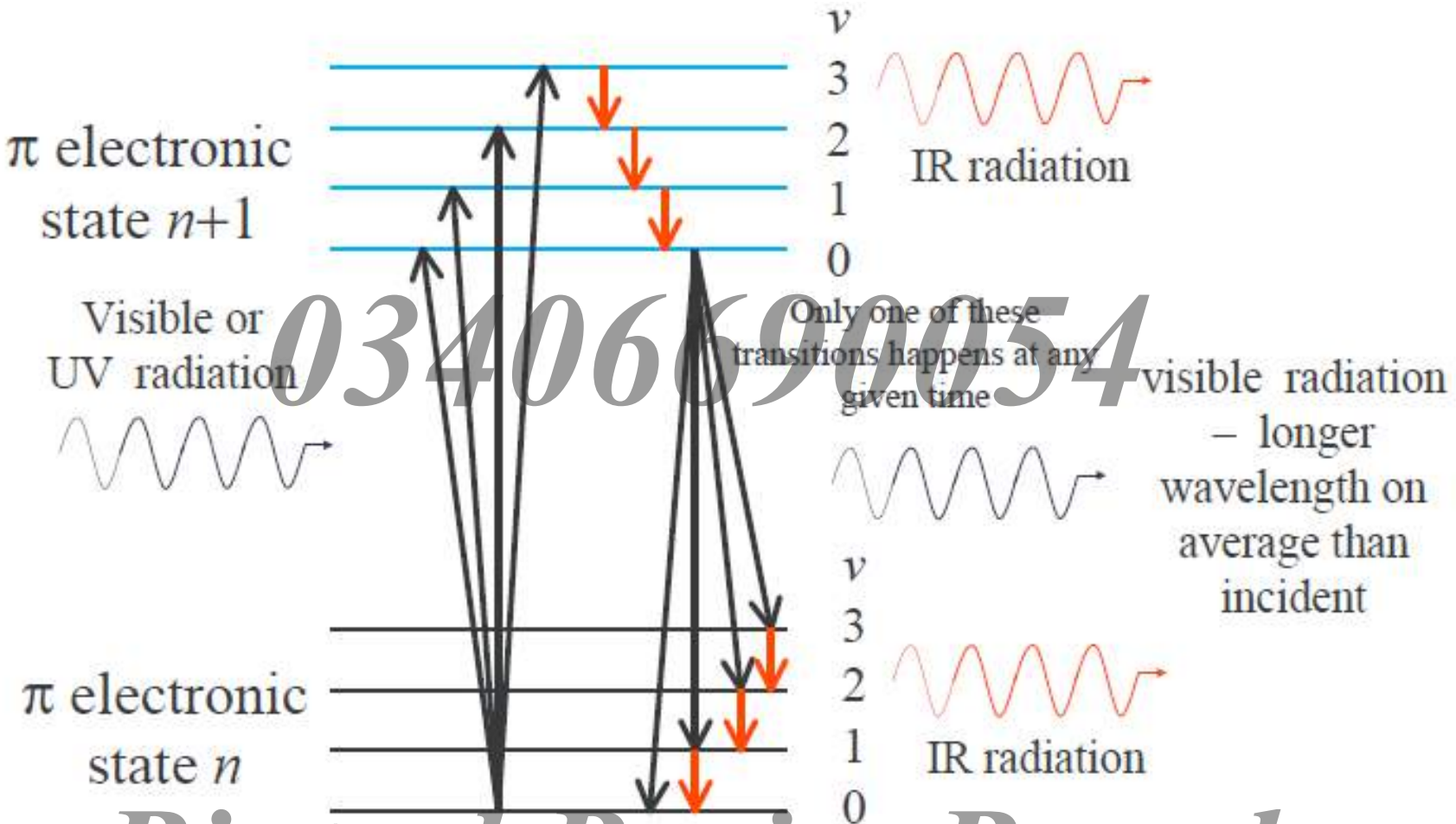
Photoluminescence

- Molecules that have an electronic excitation are *excited, with* vibrational excitation are *hot*
- When excited molecular states decay back to the ground state, resulting in the emission of light, undergoing a *luminescence* process
- If after de-excitation the emitting light is in visible range called *photoluminescence (Fluorescence* and *Phosphorescence)*

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Atomic Spectroscopy

Fluorescence



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Atomic Spectroscopy

- Absorption process occurs over short time interval (10^{-15} s).
- Vibrational relaxation (emission of IR while lowering vibrational state) occurs in ~10^{-12} s.
- De-excitation to electronic ground state with emission of IR occurs in 10^{-9} s.
- vibrational relaxation occurs ~1000 times faster

END

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Instrumentation
Atomic Spectroscopy-XVI

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Atomic Spectroscopy

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Phosphorescence

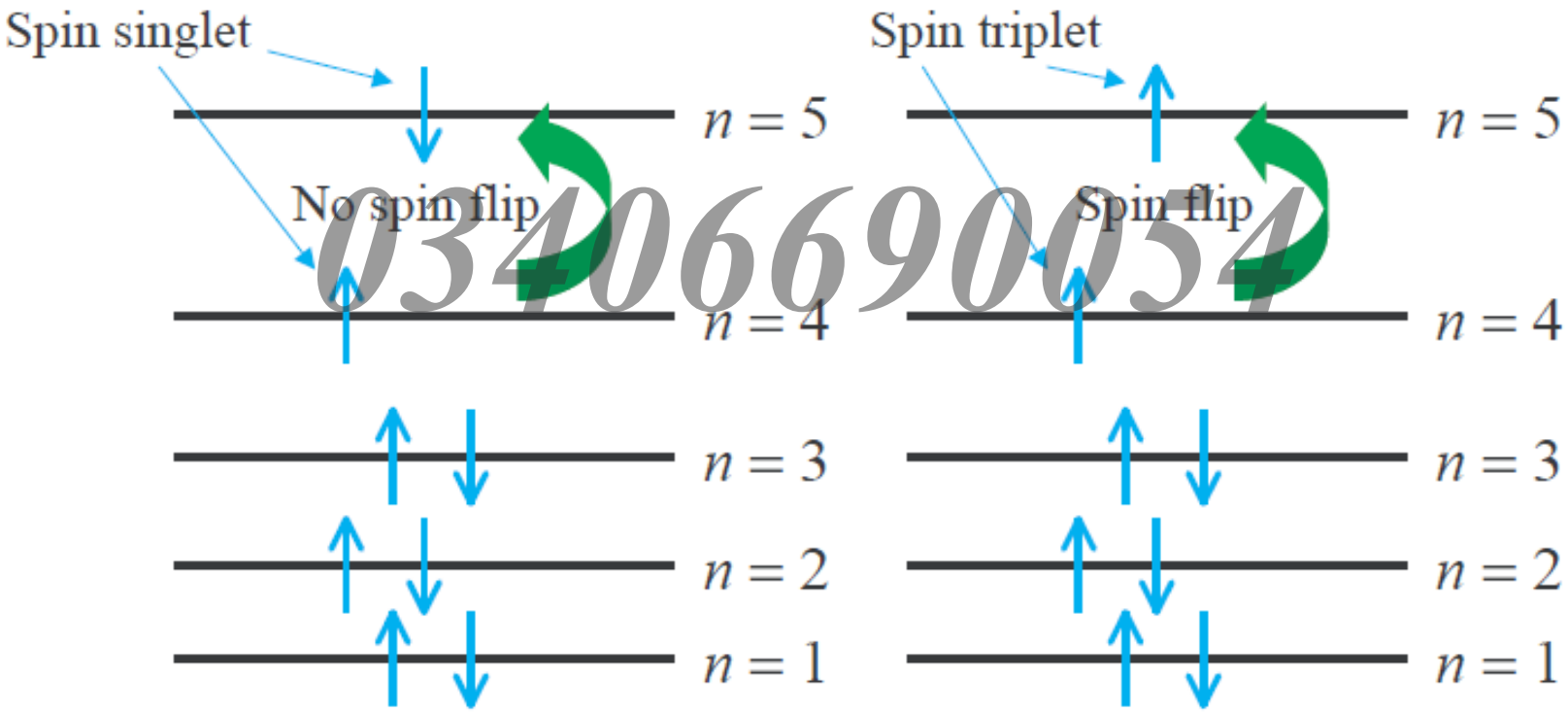
- In the fluorescence process, the electron did not change its spin direction
- But under the appropriate conditions, a spin-flip can occur (during absorption or afterward)

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Atomic Spectroscopy

Phosphorescence



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Atomic Spectroscopy

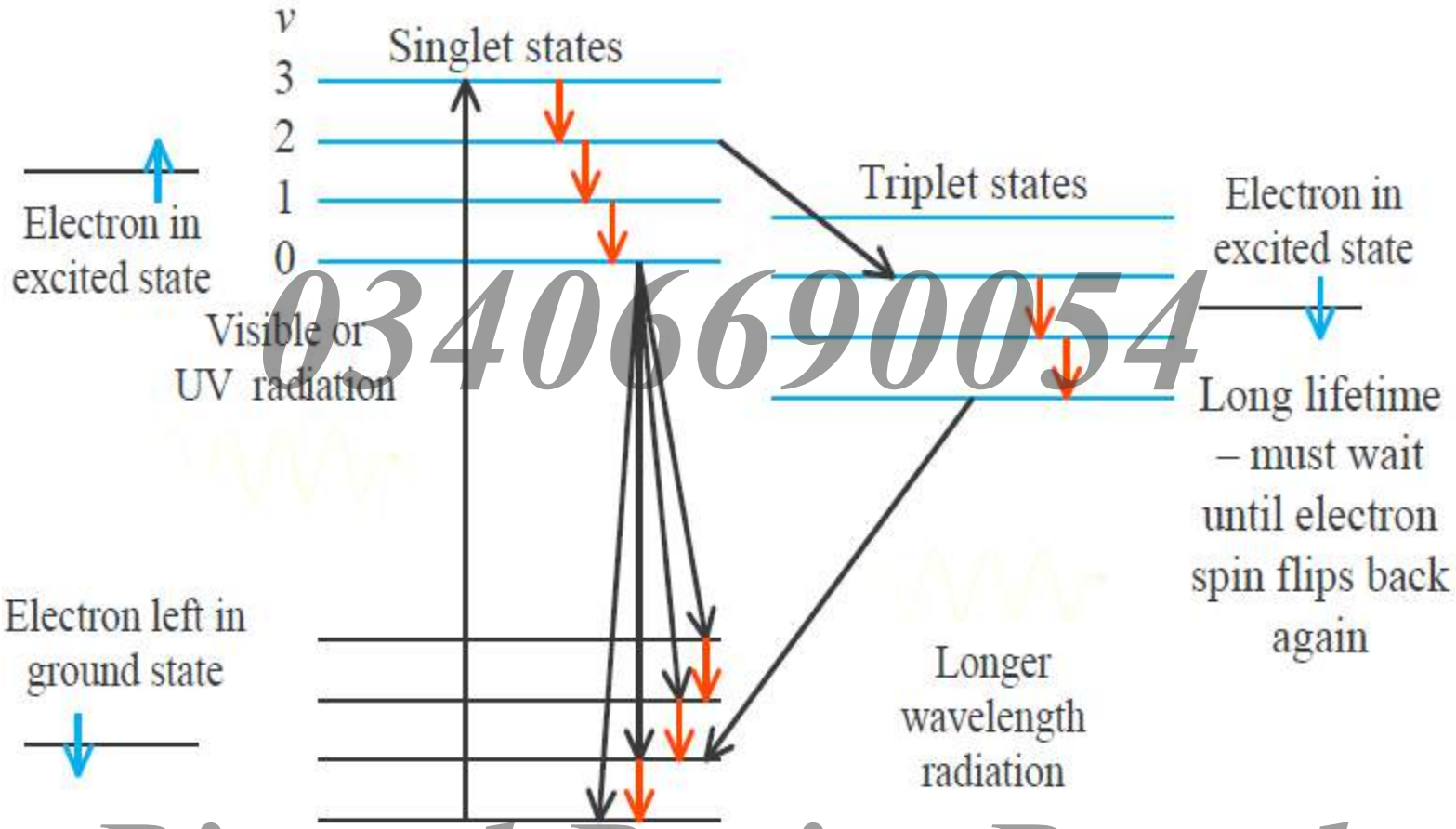
Phosphorescence

- The light emission process must wait until electron undergoes a spin-flip to revert back to it original state
  - May take 10^{-3} – 10^{2} s
  - Light emission is delayed long enough so that materials "glow in the dark" after exposure to light
  - Because of the lower energy of triplet state, lower energy photon emission than incident or fluorescence photons
- Because of the long lifetimes, molecular triplet states are more often involved in photo-chemical and photo-biological reactions than singlet states in molecules

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Atomic Spectroscopy

Phosphorescence



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Atomic Spectroscopy

Fluorescence spectrophotometer

- **Measure the intensity of fluorescence light as a function of wavelength**
- **Light source is perpendicular to detector**

END

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Instrumentation
Atomic Spectroscopy-XV

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Atomic Spectroscopy

Atomic Florescence Spectroscopy

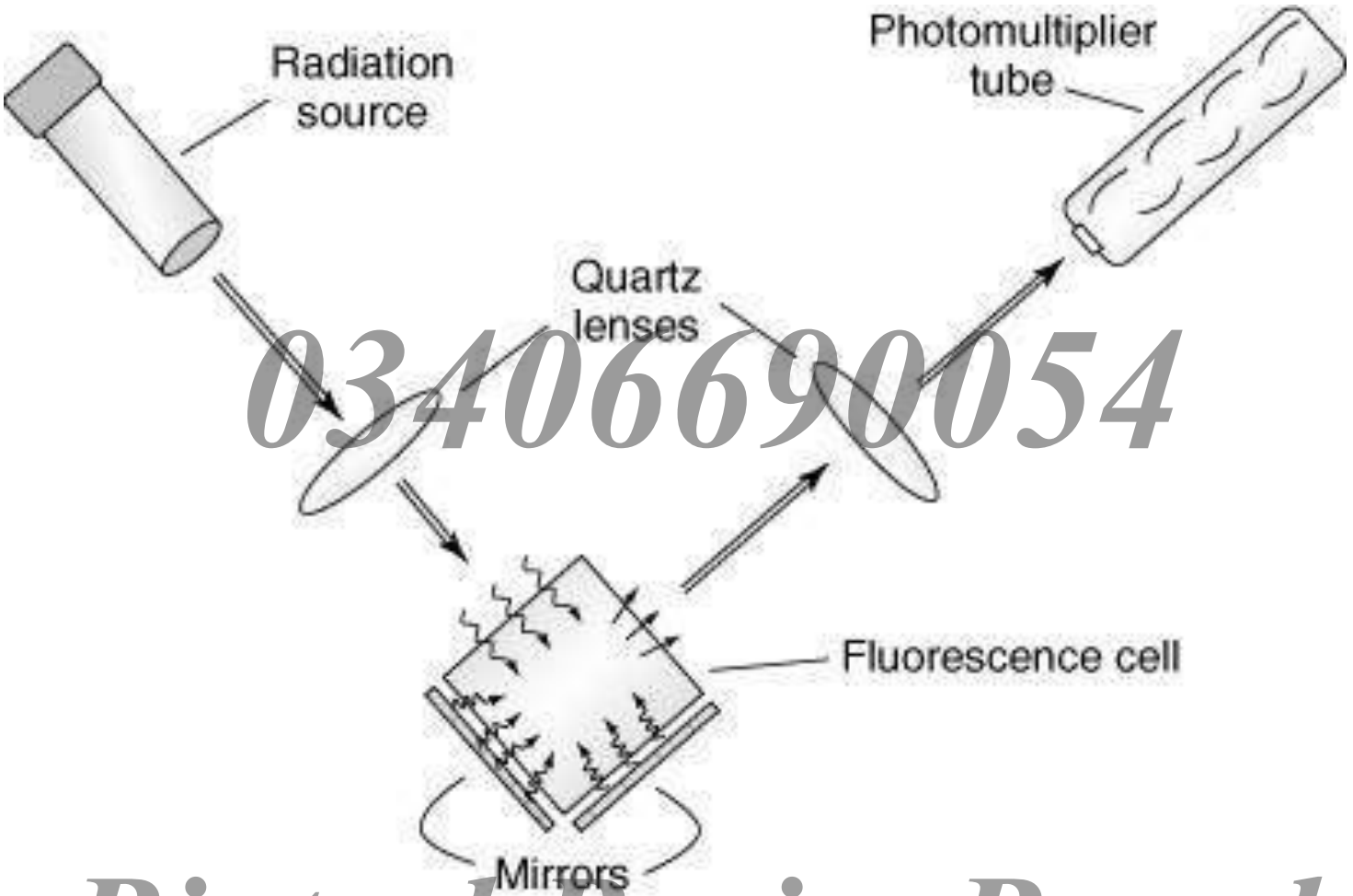
- Photoluminescence and its Types
  - Fluorescence
  - Phosphorescence
- Instrumentation

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Atomic Spectroscopy



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Atomic Spectroscopy

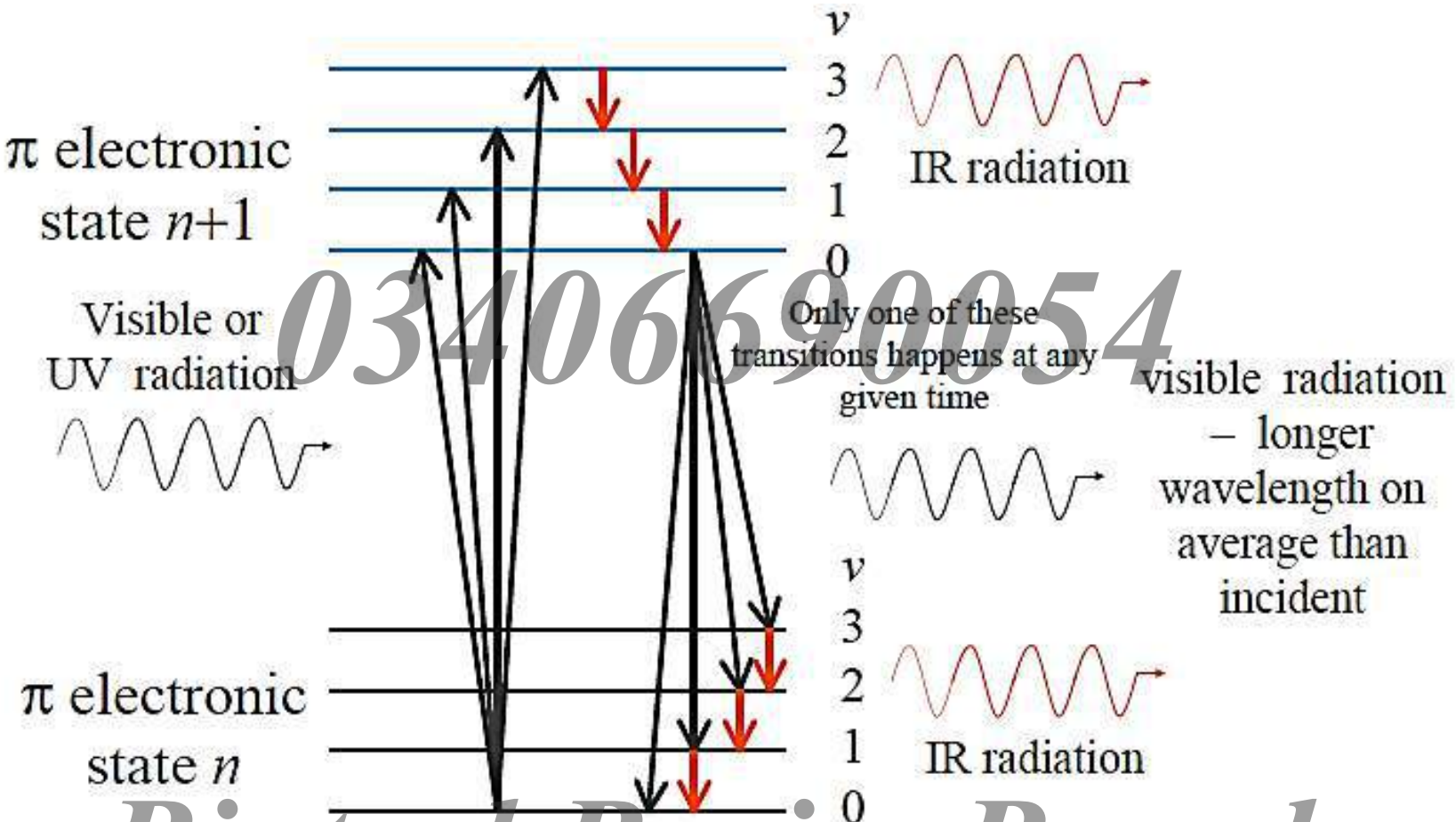
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Atomic Spectroscopy

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Instrumentation
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Atomic Spectroscopy

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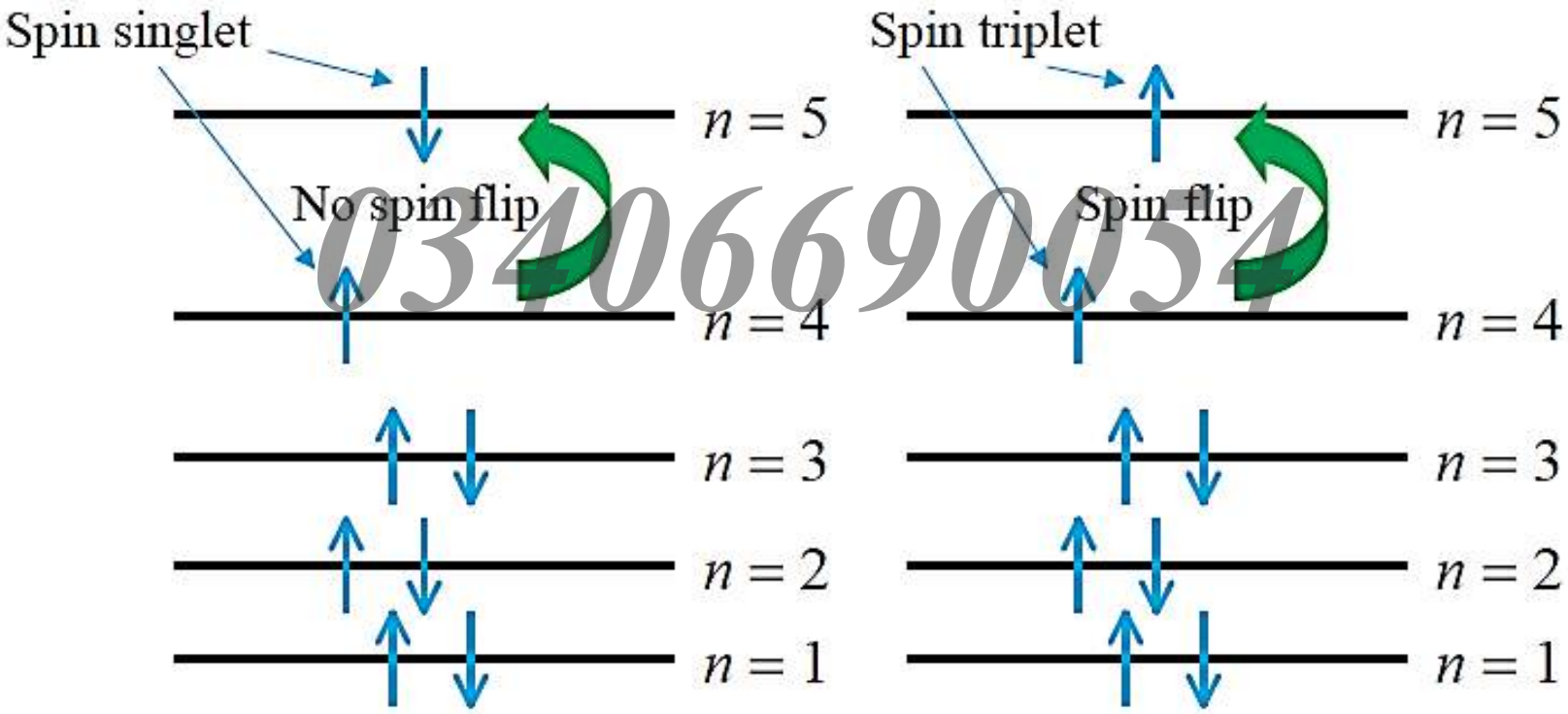
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Phosphorescence



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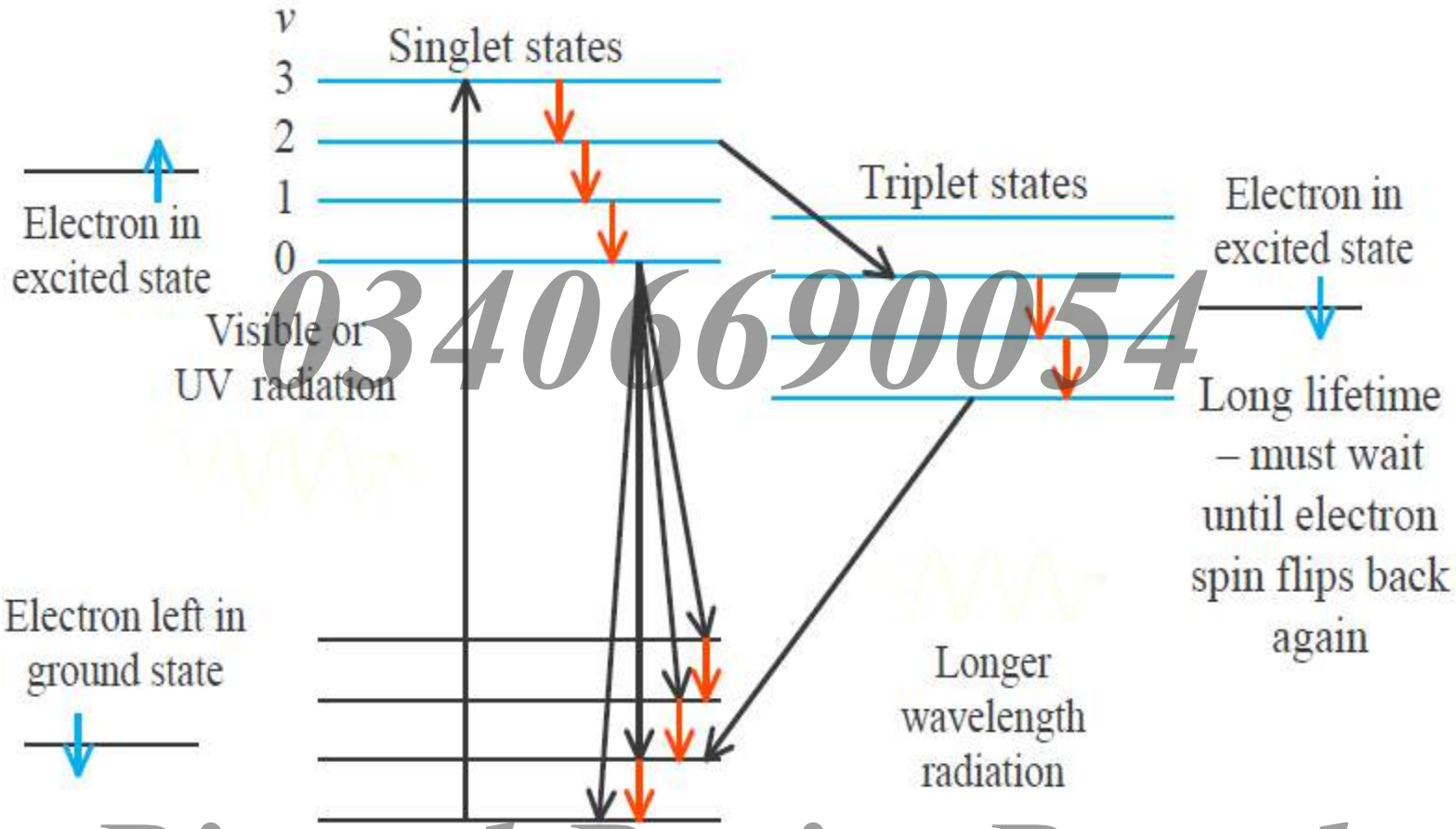
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Atomic Spectroscopy

Phosphorescence



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Atomic Spectroscopy

Fluorescence spectrophotometer

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END

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